

voloshin-hypergraph — formula report

2555 inline formulas (section omitted; all rendered), 266 display equations, 2 tables, 176 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0124	159	■ 0.310 ● N +0	$a_{ij} = \begin{cases} 1 & \text{if } \mathcal{D}(x_i) \cap \mathcal{D}(x_j) \neq \emptyset \\ 0 & \text{otherwise.} \end{cases}$	$a_{ij} = \begin{cases} 1 & \text{if } \mathcal{D}(x_i) \cap \mathcal{D}(x_j) \neq \emptyset \\ 0 & \text{otherwise.} \end{cases}$	$a_{ij} = \begin{cases} 1 & \text{if } \mathcal{D}(x_i) \cap \mathcal{D}(x_j) \neq \emptyset, \\ 0 & \text{otherwise.} \end{cases}$
voloshin-hypergraph_ EQ0133	170	■ 0.340 ● N +0	$(\mathcal{H})_2 = (X, \mathcal{E}) \text{ where } \{x_i, x_j\} \in \mathcal{E} \Leftrightarrow \mathcal{D}(x_i) \cap \mathcal{D}(x_j) \neq \emptyset.$	$(\mathcal{H})_2 = (X, \mathcal{E}) \text{ where } \{x_i, x_j\} \in \mathcal{E} \Leftrightarrow \mathcal{D}(x_i) \cap \mathcal{D}(x_j) \neq \emptyset.$	$(\mathcal{H})_2 = (X, \mathcal{E}) \text{ where } \{x_i, x_j\} \in \mathcal{E} \Leftrightarrow \mathcal{D}(x_i) \cap \mathcal{D}(x_j) \neq \emptyset.$
voloshin-hypergraph_ EQ0141 (8.1)	185	■ 0.473 ● N -1	$\tau(\mathcal{H}) = \nu(\mathcal{H}).$	$\tau(\mathcal{H}) = \nu(\mathcal{H}).$	$\tau(\mathcal{H}) = \nu(\mathcal{H}).$
voloshin-hypergraph_ EQ0197	220	■ 0.500 ● N -1	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0, \quad R(\mathcal{H}) = (0, 0, \dots, 0), \quad P(\mathcal{H}, \lambda) = 0.$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0, \quad R(\mathcal{H}) = (0, 0, \dots, 0), \quad P(\mathcal{H}, \lambda) = 0.$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 0, \quad R(\mathcal{H}) = (0, 0, \dots, 0), \quad P(\mathcal{H}, \lambda) = 0.$
voloshin-hypergraph_ EQ0244	258	■ 0.530 ● C -3	$\bar{\chi}(\mathcal{H}) = \alpha_C(\mathcal{H}) = X - \tau(\mathcal{H}_C) = X - \nu(\mathcal{H}_C).$	$\bar{\chi}(\mathcal{H}) = \alpha_C(\mathcal{H}) = X - \tau(\mathcal{H}_C) = X - \nu(\mathcal{H}_C).$	$\bar{\chi}(\mathcal{H}) = \alpha_C(\mathcal{H}) = X - \tau(\mathcal{H}_C) = X - \nu(\mathcal{H}_C).$
voloshin-hypergraph_ EQ0207	230	■ 0.531 ● K +0	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1), \text{ where }$	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1), \text{ where }$	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1), \text{ where }$
voloshin-hypergraph_ EQ0233	253	■ 0.543 ● N +0	$\alpha_c(\mathcal{H}) - \bar{\chi}(\mathcal{H}) > k.$	$\alpha_c(\mathcal{H}) - \bar{\chi}(\mathcal{H}) > k.$	$\alpha_c(\mathcal{H}) - \bar{\chi}(\mathcal{H}) > k.$
voloshin-hypergraph_ EQ0130 (7.3)	168	■ 0.556 ● N -1	$\tau(\mathcal{H}) = v(\mathcal{H}).$	$\tau(\mathcal{H}) = v(\mathcal{H}).$	$\tau(\mathcal{H}) = v(\mathcal{H}).$
voloshin-hypergraph_ EQ0190	216	■ 0.566 ● N +0	$2, 3, 1, 4, 5,$	$2, 3, 1, 4, 5,$	$2, 3, 1, 4, 5,$
voloshin-hypergraph_ EQ0029	085	■ 0.576 ● N +0	$n - m + f = 2.$	$n - m + f = 2.$	$n - m + f = 2.$
voloshin-hypergraph_ EQ0051	107	■ 0.608 ● N +1	$P(G \cdot e, \lambda) = \sum_{i=0}^{n-1} (-1)^i b_i \lambda^{n-1-i}.$	$P(G \cdot e, \lambda) = \sum_{i=0}^{n-1} (-1)^i b_i \lambda^{n-1-i}.$	$P(G \cdot e, \lambda) = \sum_{i=0}^{n-1} (-1)^i b_i \lambda^{n-1-i}.$
voloshin-hypergraph_ EQ0120	156	■ 0.613 ● W +1	$\mathcal{H} = (X, \mathcal{D}), \quad X = \{1, 2, 3, 4, 5, 6\}, \quad \mathcal{D} = \{D_1, D_2, D_3, D_4, D_5\},$ $D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{1, 2, 4\},$ $D_4 = \{2, 3, 5\}, \quad D_5 = \{3, 4, 5\}.$	$\mathcal{H} = (X, \mathcal{D}), \quad X = \{1, 2, 3, 4, 5, 6\}, \quad \mathcal{D} = \{D_1, D_2, D_3, D_4, D_5\},$ $D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{1, 2, 4\}$ $D_4 = \{2, 3, 5\}, \quad D_5 = \{3, 4, 5\}.$	$\mathcal{H} = (X, \mathcal{D}), \quad X = \{1, 2, 3, 4, 5, 6\}, \quad \mathcal{D} = \{D_1, D_2, D_3, D_4, D_5\},$ $D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{1, 2, 4\},$ $D_4 = \{2, 3, 5\}, \quad D_5 = \{3, 4, 5\}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ EQ0056	109	■ 0.653 ● W +1	$\begin{gathered} P(G, \lambda) = \frac{P(\text{left}(G_1), \lambda) P(\text{left}(G_2), \lambda)}{\lambda^2} = \\ \frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)] [\lambda(\lambda-1)(\lambda-2)]}{\lambda(\lambda-1)} = \\ \lambda(\lambda-1)(\lambda-2)(\lambda^2-3\lambda+3). \end{gathered}$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^2} =$ $\frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)] [\lambda(\lambda-1)(\lambda-2)]}{\lambda(\lambda-1)} =$ $\lambda(\lambda-1)(\lambda-2)(\lambda^2-3\lambda+3).$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^2} =$ $\frac{[\lambda(\lambda-1)(\lambda^2-3\lambda+3)] [\lambda(\lambda-1)(\lambda-2)]}{\lambda(\lambda-1)} =$ $\lambda(\lambda-1)(\lambda-2)(\lambda^2-3\lambda+3).$
voloshin-hypergraph_ EQ0125	160	■ 0.655 ● N +0	$ \text{left}(\mathcal{D}) \text{left}(K_n^r) \text{right} = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$	$ \mathcal{D}(K_n^r) = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$	$ \mathcal{D}(K_n^r) = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$
voloshin-hypergraph_ EQ0189	216	■ 0.679 ● K +0	$5, 4, 1, 3, 2.$	$5, 4, 1, 3, 2.$	$5, 4, 1, 3, 2.$
voloshin-hypergraph_ EQ0198 (10.3)	221	■ 0.699 ● N -2	$1 \leq \chi(\text{left}(\mathcal{H}) \text{right}) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_C) \leq n.$	$1 \leq \chi(\mathcal{H}_D) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_C) \leq n.$	$1 \leq \chi(\mathcal{H}_D) \leq \chi(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}) \leq \bar{\chi}(\mathcal{H}_C) \leq n.$
voloshin-hypergraph_ EQ0160	195	■ 0.709 ● N -1	$\wedge(\mathcal{H}) = \min_T \wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H}).$	$\wedge(\mathcal{H}) = \min_T \wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H}).$	$\wedge(\mathcal{H}) = \min_T \wedge(\mathcal{H}, T) = \sum_{D \in \mathcal{D}} (D - 1) - w(\mathcal{H}).$
voloshin-hypergraph_ EQ0040	102	■ 0.712 ● W +1	$\begin{gathered} P(G, \lambda) = P(\text{left}(K_3), \lambda) + P(\text{left}(K_2), \lambda) = \lambda^3 + \lambda^2 = \\ \lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) = \lambda(\lambda-1)^2. \end{gathered}$	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = \lambda^3 + \lambda^2 =$ $\lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) = \lambda(\lambda-1)^2$	$P(G, \lambda) = P(K_3, \lambda) + P(K_2, \lambda) = \lambda^3 + \lambda^2 =$ $\lambda(\lambda-1)(\lambda-2) + \lambda(\lambda-1) = \lambda(\lambda-1)^2.$
voloshin-hypergraph_ EQ0216	239	■ 0.715 ● K +0	$\mathcal{K}(n, l, m) = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{l}, \binom{X}{m} \right).$	$\mathcal{K}(n, l, m) = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{l}, \binom{X}{m} \right).$	$\mathcal{K}(n, l, m) = (X, \mathcal{C}, \mathcal{D}) = \left(X, \binom{X}{l}, \binom{X}{m} \right).$
voloshin-hypergraph_ EQ0266	287	■ 0.722 ● K +0	$\text{left}(G_1) = \begin{array}{llll} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}$	$I(G_1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$	$I(G_1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$
voloshin-hypergraph_ EQ0009	033	■ 0.738 ● K +0	$A(G) = \begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{array};$	$A(G) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix};$	$A(G) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix};$
voloshin-hypergraph_ EQ0137 (7.4)	173	■ 0.743 ● K +0	$\text{left}(D_1) \cap D_2 \text{right} \cup \text{left}(D_1) \cap D_3 \text{right} \subseteq D.$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ EQ0119	156	0.755 K +0	$\begin{gathered} \mathcal{Z}=\left\{X_{\{1\}}, X_{\{2\}}, \ldots, X_{\{n\}}\right\}, \quad X_{\{i\}}=\left\{d_{\{j\}}: x_{\{i\}} \in D_{\{j\}}\right\} \\ \text { in } \mathcal{H} \end{gathered}$	$\mathcal{Z}=\left\{X_1, X_2, \ldots, X_n\right\},$ $X_i=\left\{d_j: x_i \in D_j \text { in } \mathcal{H}\right\}.$	$\mathcal{Z}=\left\{X_1, X_2, \ldots, X_n\right\},$ $X_i=\left\{d_j: x_i \in D_j \text { in } \mathcal{H}\right\}.$
voloshin-hypergraph_ EQ0052	107	0.759 N +1	$\begin{gathered} P(G, \lambda)=P(G-e, \lambda)-P(G \cdot e, \lambda) \\ \backslash \backslash=\lambda^{\{n\}}-(m-1) \lambda^{\{n-1\}}+a_{\{2\}} \lambda^{\{n-2\}}-\cdots+(-1)^{\{i\}} a_{\{i\}} \lambda^{\{n-i\}} \cdots \\ \backslash \backslash-\left(\lambda^{\{n-1\}}-b_{\{1\}} \lambda^{\{n-2\}}+b_{\{2\}} \lambda^{\{n-2\}}-\cdots+(-1)^{\{i\}} b_{\{i-1\}} \lambda^{\{n-i\}} \cdots\right) \\ \backslash \backslash=\lambda^{\{n\}}-m(G) \lambda^{\{n-1\}}+\left(a_{\{2\}}+b_{\{1\}}\right) \lambda^{\{n-2\}} \cdots(-1)^{\{i\}}\left(a_{\{i\}}+b_{\{i-1\}}\right) \lambda^{\{n-i\}} \cdots \end{gathered}$	$P(G, \lambda)=P(G-e, \lambda)-P(G \cdot e, \lambda)$ $=\lambda^n-(m-1) \lambda^{n-1}+a_2 \lambda^{n-2}-\cdots+(-1)^i a_i \lambda^{n-i} \cdots$ $-\left(\lambda^{n-1}-b_1 \lambda^{n-2}+b_2 \lambda^{n-2}-\cdots+(-1)^i b_{i-1} \lambda^{n-i} \cdots\right)$ $=\lambda^n-m(G) \lambda^{n-1}+\left(a_2+b_1\right) \lambda^{n-2} \cdots(-1)^i\left(a_i+b_{i-1}\right) \lambda^{n-i} \cdots$	$P(G, \lambda)=P(G-e, \lambda)-P(G \cdot e, \lambda)$ $=\lambda^n-(m-1) \lambda^{n-1}+a_2 \lambda^{n-2}-\cdots+(-1)^i a_i \lambda^{n-i} \cdots$ $-\left(\lambda^{n-1}-b_1 \lambda^{n-2}+b_2 \lambda^{n-2}-\cdots+(-1)^i b_{i-1} \lambda^{n-i} \cdots\right)$ $=\lambda^n-m(G) \lambda^{n-1}+\left(a_2+b_1\right) \lambda^{n-2} \cdots(-1)^i\left(a_i+b_{i-1}\right) \lambda^{n-i} \cdots$
voloshin-hypergraph_ EQ0064	111	0.763 W +0	$\begin{gathered} P\left(G_{\{1\}}, \lambda\right)=\left(\lambda-k_{\{1\}}\right) P\left(G_{\{2\}}, \lambda\right)=\left(\lambda-k_{\{1\}}\right)\left(\lambda-k_{\{2\}}\right) P\left(G_{\{3\}}, \lambda\right) \cdots= \\ \left(\lambda-k_{\{1\}}\right)\left(\lambda-k_{\{2\}}\right)\left(\lambda-k_{\{3\}}\right) \cdots\left(\lambda-k_{\{n\}}\right) \end{gathered}$	$P\left(G_1, \lambda\right)=\left(\lambda-k_1\right) P\left(G_2, \lambda\right)=\left(\lambda-k_1\right)\left(\lambda-k_2\right) P\left(G_3, \lambda\right) \cdots=$ $\left(\lambda-k_1\right)\left(\lambda-k_2\right)\left(\lambda-k_3\right) \cdots\left(\lambda-k_n\right)$	$P\left(G_1, \lambda\right)=\left(\lambda-k_1\right) P\left(G_2, \lambda\right)=\left(\lambda-k_1\right)\left(\lambda-k_2\right) P\left(G_3, \lambda\right) \cdots=$ $\left(\lambda-k_1\right)\left(\lambda-k_2\right)\left(\lambda-k_3\right) \cdots\left(\lambda-k_n\right).$
voloshin-hypergraph_ EQ0206	230	0.771 W -1	$\begin{gathered} \mathcal{H}_{\{1\}}=\left(X, \mathcal{C}, \mathcal{D}_{\{1\}}\right), \mathcal{D}_{\{1\}}=\mathcal{D} \cup\left\{x_k, x_l\right\}, \\ \mathcal{H}_{\{2\}}=\left(X, \mathcal{C}_1, \mathcal{D}\right), \mathcal{C}_1=\mathcal{C} \cup\left\{x_k, x_l\right\} \text { (splitting). } \end{gathered}$	$\mathcal{H}_1=\left(X, \mathcal{C}, \mathcal{D}_1\right), \mathcal{D}_1=\mathcal{D} \cup\left\{x_k, x_l\right\},$ $\mathcal{H}_2=\left(X, \mathcal{C}_1, \mathcal{D}\right), \mathcal{C}_1=\mathcal{C} \cup\left\{x_k, x_l\right\} \text { (splitting).}$	$\mathcal{H}_1=\left(X, \mathcal{C}, \mathcal{D}_1\right), \mathcal{D}_1=\mathcal{D} \cup\left\{x_k, x_l\right\},$ $\mathcal{H}_2=\left(X, \mathcal{C}_1, \mathcal{D}\right), \mathcal{C}_1=\mathcal{C} \cup\left\{x_k, x_l\right\} \text { (splitting).}$
voloshin-hypergraph_ EQ0007	030	0.775 K +0	$\begin{aligned} & J(G)=\left\{\left\{x_1, x_2\right\},\left\{x_2, x_3\right\},\left\{x_3, x_4\right\},\right. \\ & \left.\left\{x_4, x_5\right\},\left\{x_1, x_5\right\},\left\{x_2, x_5\right\},\left\{x_2, x_4\right\}\right\} . \end{aligned}$	$J(G)=\left\{\left\{x_1, x_2\right\},\left\{x_2, x_3\right\},\left\{x_3, x_4\right\},\right.$ $\left.\left\{x_4, x_5\right\},\left\{x_1, x_5\right\},\left\{x_2, x_5\right\},\left\{x_2, x_4\right\}\right\} .$	$J(G)=\left\{\left\{x_1, x_2\right\},\left\{x_2, x_3\right\},\left\{x_3, x_4\right\},\right.$ $\left.\left\{x_4, x_5\right\},\left\{x_1, x_5\right\},\left\{x_2, x_5\right\},\left\{x_2, x_4\right\}\right\} .$
voloshin-hypergraph_ EQ0143 (8.3)	190	0.779 N +0	$\wedge\left(\mathcal{H}, T\right)=\sum_{D \in \mathcal{D}}(D -1)-w(T)+l\left(\mathcal{H}, T\right) .$	$\wedge(\mathcal{H}, T)=\sum_{D \in \mathcal{D}}(D -1)-w(T)+l(\mathcal{H}, T).$	$\Lambda(\mathcal{H}, T)=\sum_{D \in \mathcal{D}}(D -1)-w(T)+l(\mathcal{H}, T).$
voloshin-hypergraph_ EQ0245	258	0.790 C -3	$\nu\left(\mathcal{H}_C\right)=\tau\left(\mathcal{H}_C\right) .$	$\nu\left(\mathcal{H}_C\right)=\tau\left(\mathcal{H}_C\right) .$	$\mathfrak{v}\left(\mathcal{H}_C\right)=\mathfrak{t}\left(\mathcal{H}_C\right) .$
voloshin-hypergraph_ EQ0161	195	0.792 K +0	$\wedge\left(\mathcal{H}\right)=\wedge\left(\mathcal{H}_1\right) .$	$\wedge(\mathcal{H})=\wedge\left(\mathcal{H}_1\right) .$	$\Lambda(\mathcal{H})=\Lambda\left(\mathcal{H}_1\right) .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ EQ0118	155	■ 0.808 ● S +0	$(i, j) \text{ \texttt{-entry} } = \begin{cases} 1 & \text{if } x_i \in D_j, \\ 0 & \text{if } x_i \notin D_j. \end{cases}$	$(i, j)\text{-entry} = \begin{cases} 1 & \text{if } x_i \in D_j, \\ 0 & \text{if } x_i \notin D_j. \end{cases}$	$(i, j)\text{-entry} = \begin{cases} 1 & \text{if } x_i \in D_j, \\ 0 & \text{if } x_i \notin D_j. \end{cases}$
voloshin-hypergraph_ EQ0138	173	■ 0.808 ● K +0	$\left(D_1 \cap D_2\right) \cup \left(D_1 \cap D_3\right) \cup \left(D_2 \cap D_3\right) \subseteq D$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$	$(D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3) \subseteq D.$
voloshin-hypergraph_ EQ0162	195	■ 0.828 ● N +1	$\wedge(\mathcal{H}) = \wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1).$	$\wedge(\mathcal{H}) = \wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1).$	$\wedge(\mathcal{H}) = \wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1).$
voloshin-hypergraph_ EQ0030	085	■ 0.840 ● N +0	$m(G) \leq 3n(G) - 6.$	$m(G) \leq 3n(G) - 6.$	$m(G) \leq 3n(G) - 6.$
voloshin-hypergraph_ EQ0199	221	■ 0.852 ● N -2	$1 \leq \chi(\mathcal{H}) = 2 \leq \bar{\chi}(\mathcal{H}) = 4 \leq \bar{\chi}(\mathcal{H}_C) = 4 \leq n = 5.$	$1 \leq \chi(\mathcal{H}_D) = 2 \leq \chi(\mathcal{H}) = 2 \leq \bar{\chi}(\mathcal{H}) = 4 \leq \bar{\chi}(\mathcal{H}_C) = 4 \leq n = 5.$	$1 \leq \chi(\mathcal{H}_D) = 2 \leq \chi(\mathcal{H}) = 2 \leq \bar{\chi}(\mathcal{H}) = 4 \leq \bar{\chi}(\mathcal{H}_C) = 4 \leq n = 5.$
voloshin-hypergraph_ EQ0087	116	■ 0.863 ● N -1	$\begin{gathered} P(G, \lambda) = \lambda P(G - x, \lambda - 1) = \frac{P(G - x, \lambda)}{P(G - x, \lambda)} \lambda P(G - x, \lambda - 1) = \\ P(G - x, \lambda) W(G_{N(x)}, \lambda) \end{gathered}$	$P(G, \lambda) = \lambda P(G - x, \lambda - 1) = \frac{P(G - x, \lambda)}{P(G - x, \lambda)} \lambda P(G - x, \lambda - 1) = \\ P(G - x, \lambda) W(G_{N(x)}, \lambda)$	$P(G, \lambda) = \lambda P(G - x, \lambda - 1) = \frac{P(G - x, \lambda)}{P(G - x, \lambda)} \lambda P(G - x, \lambda - 1) = \\ P(G - x, \lambda) W(G_{N(x)}, \lambda).$
voloshin-hypergraph_ EQ0153	193	■ 0.873 ● N -1	$\wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1) \text{ if and only if } \mathcal{D}(x) = \mathcal{D}_T(x) + m_2.$	$\wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1) \text{ if and only if } \mathcal{D}(x) = \mathcal{D}_T(x) + m_2.$	$\wedge(\mathcal{H}, T) = \wedge(\mathcal{H}_1, T_1) \text{ if and only if } \mathcal{D}(x) = \mathcal{D}_T(x) + m_2.$
voloshin-hypergraph_ EQ0214	233	■ 0.876 ● C -5	$\begin{gathered} \chi(\mathcal{H}) = \max \left\{ \chi(\mathcal{H}_1), \chi(\mathcal{H}_2), \dots, \chi(\mathcal{H}_k) \right\}, \\ \bar{\chi}(\mathcal{H}) = \bar{\chi}(\mathcal{H}_1) + \bar{\chi}(\mathcal{H}_2) + \dots + \bar{\chi}(\mathcal{H}_k), \\ P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) P(\mathcal{H}_2, \lambda) \dots P(\mathcal{H}_k, \lambda). \end{gathered}$	$\begin{gathered} \chi(\mathcal{H}) = \max \left\{ \chi(\mathcal{H}_1), \chi(\mathcal{H}_2), \dots, \chi(\mathcal{H}_k) \right\}, \\ \bar{\chi}(\mathcal{H}) = \bar{\chi}(\mathcal{H}_1) + \bar{\chi}(\mathcal{H}_2) + \dots + \bar{\chi}(\mathcal{H}_k), \\ P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) P(\mathcal{H}_2, \lambda) \dots P(\mathcal{H}_k, \lambda). \end{gathered}$	$\begin{gathered} \chi(\mathcal{H}) = \max \left\{ \chi(\mathcal{H}_1), \chi(\mathcal{H}_2), \dots, \chi(\mathcal{H}_k) \right\}, \\ \bar{\chi}(\mathcal{H}) = \bar{\chi}(\mathcal{H}_1) + \bar{\chi}(\mathcal{H}_2) + \dots + \bar{\chi}(\mathcal{H}_k), \\ P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) P(\mathcal{H}_2, \lambda) \dots P(\mathcal{H}_k, \lambda). \end{gathered}$
voloshin-hypergraph_ EQ0099	129	■ 0.890 ● N +0	$\chi'(G) = \Delta(G), \text{ or } \chi'(G) = \Delta(G) + 1.$	$\chi'(G) = \Delta(G), \text{ or } \chi'(G) = \Delta(G) + 1.$	$\chi'(G) = \Delta(G), \text{ or } \chi'(G) = \Delta(G) + 1.$
voloshin-hypergraph_ EQ0235 (10.5)	253	■ 0.901 ● N +0	$\bar{\chi}(\mathcal{H}') = \alpha_c(\mathcal{H}').$	$\bar{\chi}(\mathcal{H}') = \alpha_c(\mathcal{H}').$	$\bar{\chi}(\mathcal{H}') = \alpha_c(\mathcal{H}').$
voloshin-hypergraph_ EQ0201	224	■ 0.913 ● N +1	$o(x, \mathcal{H}) = \mathcal{C}(x) - b(x, \mathcal{H}) \geq 0$	$o(x, \mathcal{H}) = \mathcal{C}(x) - b(x, \mathcal{H}) \geq 0$	$o(x, \mathcal{H}) = \mathcal{C}(x) - b(x, \mathcal{H}) \geq 0$
voloshin-hypergraph_ EQ0232 (10.4)	252	■ 0.932 ● K +0	$\bar{\chi}(\mathcal{H}) \leq \alpha_c(\mathcal{H}).$	$\bar{\chi}(\mathcal{H}) \leq \alpha_c(\mathcal{H}).$	$\bar{\chi}(\mathcal{H}) \leq \alpha_c(\mathcal{H}).$
voloshin-hypergraph_ EQ0259	273	■ 0.935 ● K +0	$P(\mathcal{H}, \lambda) = \sum_{i \geq 1} f_i(G^*) \lambda(\lambda - 1)^i.$	$P(\mathcal{H}, \lambda) = \sum_{i \geq 1} f_i(G^*) \lambda(\lambda - 1)^i.$	$P(\mathcal{H}, \lambda) = \sum_{i \geq 1} f_i(G^*) \lambda(\lambda - 1)^i.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0226	248	■0.938 ●K +0	$ C \geq n-\chi\ .$	$ C \geq n-\chi.$	$ C \geq n-\chi.$
voloshin-hypergraph_ EQ0231	250	■0.941 ●K +0	$\boldsymbol{\sigma}=\left(x_5,\ x_3,\ x_4,\ x_2,\ x_{\{1\}}\right)\ .$	$\boldsymbol{\sigma}=(x_5,x_3,x_4,x_2,x_1).$	$\boldsymbol{\sigma}=(x_5,x_3,x_4,x_2,x_1).$
voloshin-hypergraph_ EQ0149	192	■0.945 ●C -3	$\begin{gathered}\wedge(\mathcal{H},T)=\sum_D\\ \text{in } \mathcal{D}\}(D -1)-w(T)+l(\mathcal{H},T) \\ \backslash\backslash = \\ \text{\texttt{\{ ignore the singletons \}}}=\sum_{ D \geq 2}\\ (D -1)-w(T)+l(\mathcal{H},T) \\ \backslash\backslash = \\ \text{\texttt{\{ insert the weight of the forest \}}} \\ \backslash\backslash =\sum_{ D \geq 2}\\ (D -1)-\sum_{ D \geq 2} E_D +l(\mathcal{H},T) \\ \backslash\backslash = \\ \text{\texttt{\{ unite the sum \}}}=\sum_{ D \geq 2}\\ \left(D -\left E_D\right -1\right) +l(\mathcal{H},T) \\ \backslash\backslash = \\ \text{\texttt{\{ use (8.5) \}}}=\sum_{ D \geq 2}\\ (c(T,D)-1)+l(\mathcal{H},T) \\ \backslash\backslash = \\ \text{\texttt{\{ bring in singletons back \}}} \\ =\sum_D\\ \text{in } \mathcal{D}\}(c(T,D)-1)+l(\mathcal{H},T) \\ .\end{gathered}$	$\begin{aligned}\Lambda(\mathcal{H},T)&= \sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T) \\ &= \{ \textbf{ignore the singletons} \}= \sum_{ D \geq 2}(D -1)-w(T)+l(\mathcal{H},T) \\ &= \{ \textbf{insert the weight of the forest} \} \\ &= \sum_{ D \geq 2}(D -1)-\sum_{ D \geq 2} E_D +l(\mathcal{H},T) \\ &= \{ \textbf{unite the sum} \}= \sum_{ D \geq 2}(D - E_D -1)+l(\mathcal{H},T) \\ &= \{ \textbf{use (8.5)} \}= \sum_{ D \geq 2}(c(T,D)-1)+l(\mathcal{H},T) \\ &= \{ \textbf{bring in singletons back} \}= \sum_{D\in\mathcal{D}}(c(T,D)-1)+l(\mathcal{H},T).\end{aligned}$	$\begin{aligned}\Lambda(\mathcal{H},T)&= \sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T) \\ &= \{ \textbf{ignore the singletons} \}= \sum_{ D \geq 2}(D -1)-w(T)+l(\mathcal{H},T) \\ &= \{ \textbf{insert the weight of the forest} \} \\ &= \sum_{ D \geq 2}(D -1)-\sum_{ D \geq 2} E_D +l(\mathcal{H},T) \\ &= \{ \textbf{unite the sum} \}= \sum_{ D \geq 2}(D - E_D -1)+l(\mathcal{H},T) \\ &= \{ \textbf{use (8.5)} \}= \sum_{ D \geq 2}(c(T,D)-1)+l(\mathcal{H},T) \\ &= \{ \textbf{bring in singletons back} \}= \sum_{D\in\mathcal{D}}(c(T,D)-1)+l(\mathcal{H},T).\end{aligned}$
voloshin-hypergraph_ EQ0144	191	■0.946 ●N +0	$\begin{gathered}\wedge(\mathcal{H},T)=\\ \sum_{ D =2}\\ (D -1)-w(T)+l(\mathcal{H},T) \\ \backslash\backslash = \\ \mathcal{D} -(X -1)=m-n+1\ .\end{gathered}$	$\begin{aligned}\Lambda(\mathcal{H},T)&= \sum_{ D =2}(D -1)-w(T)+l(\mathcal{H},T) \\ &= \mathcal{D} -(X -1)=m-n+1.\end{aligned}$	$\begin{aligned}\Lambda(\mathcal{H},T)&= \sum_{ D =2}(D -1)-w(T)+l(\mathcal{H},T) \\ &= \mathcal{D} -(X -1)=m-n+1.\end{aligned}$
voloshin-hypergraph_ EQ0202	224	■0.952 ●N -1	$O(\mathcal{H})=\max_Y\min_{x\in Y}o(x,\mathcal{H}_Y)$	$O(\mathcal{H})=\max_{Y\subseteq X}\min_{x\in Y}o(x,\mathcal{H}_Y)$	$O(\mathcal{H})=\max_{Y\subseteq X}\min_{x\in Y}o(x,\mathcal{H}_Y)$
voloshin-hypergraph_ EQ0145	191	■0.952 ●S +0	$\begin{gathered}\Lambda(\mathcal{H},T)=\sum_D\\ \text{in } \mathcal{D}\}(D -1)-w(T)+l(\mathcal{H},T) \\ \backslash\backslash =2+2+0+0-3+1=2\ .\end{gathered}$	$\begin{aligned}\Lambda(\mathcal{H},T)&= \sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T) \\ &= 2+2+0+0-3+1=2.\end{aligned}$	$\begin{aligned}\Lambda(\mathcal{H},T)&= \sum_{D\in\mathcal{D}}(D -1)-w(T)+l(\mathcal{H},T) \\ &= 2+2+0+0-3+1=2.\end{aligned}$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0041 (5.4)	103	0.953 ● W -1	$\begin{gathered} P\left(C_4, \lambda\right)=0 \cdot P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ \lambda^2+2 \lambda^3+\lambda^4=\lambda(\lambda-1)+2 \lambda(\lambda-1)(\lambda-2)+\lambda(\lambda-1)(\lambda-2)(\lambda-3)= \\ \lambda(\lambda-1)[1+2(\lambda-2)+(\lambda-2)(\lambda-3)]=\lambda(\lambda-1)\left(\lambda^2-3 \lambda+3\right)= \\ \lambda^4-4 \lambda^3+6 \lambda^2-3 \lambda . \end{gathered}$	$P\left(C_4, \lambda\right)=0 \cdot P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ \lambda^2+2 \lambda^3+\lambda^4=\lambda(\lambda-1)+2 \lambda(\lambda-1)(\lambda-2)+\lambda(\lambda-1)(\lambda-2)(\lambda-3)= \\ \lambda(\lambda-1)[1+2(\lambda-2)+(\lambda-2)(\lambda-3)]=\lambda(\lambda-1)\left(\lambda^2-3 \lambda+3\right)= \\ \lambda^4-4 \lambda^3+6 \lambda^2-3 \lambda .$	$P\left(C_4, \lambda\right)=0 \cdot P\left(K_1, \lambda\right)+1 \cdot P\left(K_2, \lambda\right)+2 \cdot P\left(K_3, \lambda\right)+1 \cdot P\left(K_4, \lambda\right)= \\ \lambda^2+2 \lambda^3+\lambda^4=\lambda(\lambda-1)+2 \lambda(\lambda-1)(\lambda-2)+\lambda(\lambda-1)(\lambda-2)(\lambda-3)= \\ \lambda(\lambda-1)[1+2(\lambda-2)+(\lambda-2)(\lambda-3)]=\lambda(\lambda-1)\left(\lambda^2-3 \lambda+3\right)= \\ \lambda^4-4 \lambda^3+6 \lambda^2-3 \lambda .$
voloshin-hypergraph_ EQ0265	286	0.955 ● N +1	$A\left(G_1\right)=\left(\begin{array}{llll} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right)$	$A\left(G_1\right)=\left(\begin{array}{llll} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right)$	$A\left(G_1\right)=\left(\begin{array}{llll} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array}\right)$
voloshin-hypergraph_ EQ0136	171	0.957 ● N +0	$\nu\left(\mathcal{H}\right)=\alpha\left(L\left(\mathcal{H}\right)\right) \text { and } \nu\left(\mathcal{H}^*\right)=\alpha\left(\left(\mathcal{H}\right)_2\right) .$	$\nu\left(\mathcal{H}\right)=\alpha\left(L\left(\mathcal{H}\right)\right) \text { and } \nu\left(\mathcal{H}^*\right)=\alpha\left(\left(\mathcal{H}\right)_2\right) .$	$\nu\left(\mathcal{H}\right)=\alpha\left(L\left(\mathcal{H}\right)\right) \text { and } \nu\left(\mathcal{H}^*\right)=\alpha\left(\left(\mathcal{H}\right)_2\right) .$
voloshin-hypergraph_ EQ0152 (8.6)	193	0.970 ● N -1	$\wedge\left(\mathcal{H}, T\right)=\wedge\left(\mathcal{H}_1, T_1\right)+\left \mathcal{D}(x)\right -\left \mathcal{D}_T(x)\right -m_2 .$	$\wedge\left(\mathcal{H}, T\right)=\wedge\left(\mathcal{H}_1, T_1\right)+\left \mathcal{D}(x)\right -\left \mathcal{D}_T(x)\right -m_2 .$	$\wedge\left(\mathcal{H}, T\right)=\wedge\left(\mathcal{H}_1, T_1\right)+\left \mathcal{D}(x)\right -\left \mathcal{D}_T(x)\right -m_2 .$
voloshin-hypergraph_ EQ0218	240	0.971 ● K +0	$(l-1)(m-1) \leq n .$	$(l-1)(m-1) \leq n .$	$(l-1)(m-1) \leq n .$
voloshin-hypergraph_ EQ0243	257	0.973 ● K +0	$\bigcap\left\{C \in \mathcal{C}_1\right\} C \neq \emptyset .$	$\bigcap\left\{C \in \mathcal{C}_1\right\} C \neq \emptyset .$	$\bigcap\left\{C \in \mathcal{C}_1\right\} C \neq \emptyset .$
voloshin-hypergraph_ EQ0242	256	0.975 ● N -1	$\bar{\chi}\left(\mathcal{H}^r\right)=2 r-4<2 r-3=\alpha_C\left(\mathcal{H}^r\right) .$	$\bar{\chi}\left(\mathcal{H}^r\right)=2 r-4<2 r-3=\alpha_C\left(\mathcal{H}^r\right) .$	$\bar{\chi}\left(\mathcal{H}^r\right)=2 r-4<2 r-3=\alpha_C\left(\mathcal{H}^r\right) .$
voloshin-hypergraph_ EQ0072	114	0.983 ● K +0	$P\left(G', \lambda\right)=\lambda^{s_0'}(\lambda-1)^{s_1'} \ldots(\lambda-\chi'+1)^{s_{\chi'-1}'}$	$P\left(G', \lambda\right)=\lambda^{s_0'}(\lambda-1)^{s_1'} \ldots(\lambda-\chi'+1)^{s_{\chi'-1}'}$	$P\left(G', \lambda\right)=\lambda^{s_0'}(\lambda-1)^{s_1'} \ldots(\lambda-\chi'+1)^{s_{\chi'-1}'},$
voloshin-hypergraph_ EQ0015	048	0.988 ● N -1	$\tau(G) \geq v(G) .$	$\tau(G) \geq v(G) .$	$\tau(G) \geq v(G) .$
voloshin-hypergraph_ EQ0096	118	0.988 ● N +1	$\begin{gathered} P\left(G_1, \lambda\right) W\left(G_{N(4)}, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) \frac{(\lambda-2)^2}{\lambda-1}= \\ \lambda(\lambda-1)(\lambda-2)^5(\lambda-3)=P(G, \lambda) \end{gathered}$	$P\left(G_1, \lambda\right) W\left(G_{N(4)}, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) \frac{(\lambda-2)^2}{\lambda-1}= \\ \lambda(\lambda-1)(\lambda-2)^5(\lambda-3)=P(G, \lambda)$	$P\left(G_1, \lambda\right) W\left(G_{N(4)}, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) \frac{(\lambda-2)^2}{\lambda-1}= \\ \lambda(\lambda-1)(\lambda-2)^5(\lambda-3)=P(G, \lambda),$
voloshin-hypergraph_ EQ0228	248	0.988 ● K +0	$\left X_1\right -1+\left X_2\right -1+\ldots+\left X_{\chi}\right -1=\left X\right -\chi .$	$\left X_1\right -1+\left X_2\right -1+\ldots+\left X_{\chi}\right -1=\left X\right -\chi .$	$\left X_1\right -1+\left X_2\right -1+\ldots+\left X_{\chi}\right -1=\left X\right -\chi .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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voloshin-hypergraph_ EQ0107	136	■ 0.989 ● N -1	$\bar{\chi}^{\prime}(G)=c+m-n+p=1+5-4+1=3,$	$\bar{\chi}'(G) = c + m - n + p = 1 + 5 - 4 + 1 = 3,$	$\overline{\chi}(G) = c + m - n + p = 1 + 5 - 4 + 1 = 3,$
voloshin-hypergraph_ EQ0186	215	■ 0.990 ● C -3	$M(\mathrm{mathcal{H}})=m\left(y,\mathrm{mathcal{H}}_Y\right)\leq m\left(x_k,\mathrm{mathcal{H}}_k\right)\leq t.$	$M(\mathcal{H}) = m(y, \mathcal{H}_Y) \leq m(x_k, \mathcal{H}_k) \leq t.$	$M(\mathcal{H}) = m(y, \mathcal{H}_Y) \leq m(x_k, \mathcal{H}_k) \leq t.$
voloshin-hypergraph_ EQ0254	266	■ 0.990 ● N +1	$R\left(\mathrm{mathcal{H}}^{\prime}\right)=\left(0,r_2\left(\mathrm{mathcal{H}}^{\prime}\right),0,\ldots,0,r_{\chi},\ldots,r_{\bar{\chi}},0,\ldots,0\right).$	$R(\mathcal{H}') = (0, r_2(\mathcal{H}'), 0, \dots, 0, r_{\chi}, \dots, r_{\bar{\chi}}, 0, \dots, 0).$	$R(\mathcal{H}') = (0, r_2(\mathcal{H}'), 0, \dots, 0, r_{\chi}, \dots, r_{\bar{\chi}}, 0, \dots, 0).$
voloshin-hypergraph_ EQ0054	108	■ 0.991 ● N +0	$\begin{gathered} P(G,\lambda)=\frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}\lambda^{(k)}=\\ \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}. \end{gathered}$	$P(G,\lambda) = \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}\lambda^{(k)} = \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}.$	$P(G,\lambda) = \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}\lambda^{(k)} = \frac{P\left(G_{X_1\cup S},\lambda\right)P\left(G_{X_2\cup S},\lambda\right)}{\lambda^{(k)}}.$
voloshin-hypergraph_ EQ0132	170	■ 0.993 ● K +0	$L(\mathrm{mathcal{H}})=(\mathrm{mathcal{D}},\mathrm{mathcal{E}}),\text{ where }\left\{D_i,D_j\right\}\in\mathcal{E}\Leftrightarrow D_i\cap D_j\neq\emptyset.$	$L(\mathcal{H}) = (\mathcal{D}, \mathcal{E}), \text{ where } (D_i, D_j) \in \mathcal{E} \Leftrightarrow D_i \cap D_j \neq \emptyset.$	$L(\mathcal{H}) = (\mathcal{D}, \mathcal{E}), \text{ where } (D_i, D_j) \in \mathcal{E} \Leftrightarrow D_i \cap D_j \neq \emptyset.$
voloshin-hypergraph_ EQ0150	192	■ 0.994 ● K +0	$\wedge(\mathrm{mathcal{H}},T)=(2-1)+(1-1)+(1-1)+1=2.$	$\wedge(\mathcal{H}, T) = (2 - 1) + (1 - 1) + (1 - 1) + 1 = 2.$	$\wedge(\mathcal{H}, T) = (2 - 1) + (1 - 1) + (1 - 1) + 1 = 2.$
voloshin-hypergraph_ EQ0065 (S.11)	112	■ 0.994 ● K +0	$P(G,\lambda)=\lambda^{S_0}(\lambda-1)^{S_1}(\lambda-2)^{S_2}\cdots(\lambda-\chi(G)+1)^{S_{\chi-1}}$	$P(G, \lambda) = \lambda^{S_0}(\lambda - 1)^{S_1}(\lambda - 2)^{S_2} \cdots (\lambda - \chi(G) + 1)^{S_{\chi-1}}$	$P(G, \lambda) = \lambda^{S_0}(\lambda - 1)^{S_1}(\lambda - 2)^{S_2} \cdots (\lambda - \chi(G) + 1)^{S_{\chi-1}}$
voloshin-hypergraph_ EQ0223	243	■ 0.994 ● K +0	$P(\mathrm{mathcal{H}},\lambda)=\lambda(\lambda-1)(\lambda-2)\cdots(\lambda-\chi+1)=\lambda^{(\chi)}.$	$P(\mathcal{H}, \lambda) = \lambda(\lambda - 1)(\lambda - 2) \cdots (\lambda - \chi + 1) = \lambda^{(\chi)}.$	$P(\mathcal{H}, \lambda) = \lambda(\lambda - 1)(\lambda - 2) \cdots (\lambda - \chi + 1) = \lambda^{(\chi)}.$
voloshin-hypergraph_ EQ0182	212	■ 0.995 ● K +0	$\chi(\mathrm{mathcal{H}})\leq\wedge(\mathrm{mathcal{H}})+2.$	$\chi(\mathcal{H}) \leq \wedge(\mathcal{H}) + 2.$	$\chi(\mathcal{H}) \leq \wedge(\mathcal{H}) + 2.$
voloshin-hypergraph_ EQ0257	268	■ 0.996 ● C -4	$\begin{array}{lll} 123 & \& 341 & \& 561 \\ 124 & \& 342 & \& 562 \\ 125 & \& 345 & \& 563 \\ 126 & \& 346 & \& 564 \end{array}.$	$\begin{array}{lll} 123 & 341 & 561 \\ 124 & 342 & 562 \\ 125 & 345 & 563 \\ 126 & 346 & 564. \end{array}$	$\begin{array}{lll} 123 & 341 & 561 \\ 124 & 342 & 562 \\ 125 & 345 & 563 \\ 126 & 346 & 564. \end{array}$
voloshin-hypergraph_ EQ0176	211	■ 0.996 ● K +0	$\left\lfloor\frac{ D }{\lambda}\right\rfloor\leq D\cap S_i \leq\left\lceil\frac{ D }{\lambda}\right\rceil,$	$\left\lfloor\frac{ D }{\lambda}\right\rfloor\leq D\cap S_i \leq\left\lceil\frac{ D }{\lambda}\right\rceil,$	$\left\lfloor\frac{ D }{\lambda}\right\rfloor\leq D\cap S_i \leq\left\lceil\frac{ D }{\lambda}\right\rceil,$
voloshin-hypergraph_ EQ0085	116	■ 0.996 ● N -1	$W(G,\lambda)=\frac{\lambda P(G,\lambda-1)}{P(G,\lambda)}.$	$W(G, \lambda) = \frac{\lambda P(G, \lambda - 1)}{P(G, \lambda)}.$	$W(G, \lambda) = \frac{\lambda P(G, \lambda - 1)}{P(G, \lambda)}.$
voloshin-hypergraph_ EQ0156	193	■ 0.996 ● N -1	$\wedge(\mathrm{mathcal{H}}-x,T-x)+ \mathcal{D}(x) - \mathcal{D}_T(x) -m_2=2+3-1-1=3=\wedge(\mathrm{mathcal{H}},T).$	$\wedge(\mathcal{H} - x, T - x) + \mathcal{D}(x) - \mathcal{D}_T(x) - m_2 = 2 + 3 - 1 - 1 = 3 = \wedge(\mathcal{H}, T).$	$\wedge(\mathcal{H} - x, T - x) + \mathcal{D}(x) - \mathcal{D}_T(x) - m_2 = 2 + 3 - 1 - 1 = 3 = \wedge(\mathcal{H}, T).$
voloshin-hypergraph_ EQ0258	273	■ 0.996 ● N +0	$r_k(G)=\sum_{i\geq 1}S(i,k-1)f_i(G^*),1\leq k\leq n(G),$	$r_k(G) = \sum_{i \geq 1} S(i, k - 1) f_i(G^*), 1 \leq k \leq n(G),$	$r_k(G) = \sum_{i \geq 1} S(i, k - 1) f_i(G^*), 1 \leq k \leq n(G),$
voloshin-hypergraph_ EQ0208	230	■ 0.996 ● C -3	$\mathcal{C}_1=\mathcal{C}-\mathcal{C}_t\text{ (contraction).}$	$\mathcal{C}_1 = \mathcal{C} - \mathcal{C}_t \text{ (contraction).}$	$\mathcal{C}_1 = \mathcal{C} - \mathcal{C}_t \text{ (contraction).}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0246	259	■ 0.997 ● C -3	$\begin{aligned} & \& V_{\{1\}}=(\{1,2,3,4\},\{\{1,2,3\}, \\ & \{1,3,4\},\{1,2,4\}\}) \text{ (monostar)}; \} \& \& \\ & V_{\{2\}}=(\{1,2,3,4,5\},\{\{1,2,3\},\{1,4,5\}\}) \text{ (monostar)}; \} \& \& \\ & V_{\{3\}}=(\{1,2,3,4,5\},\{\{1,2,3\}, \\ & \{1,3,4\},\{1,4,5\}\}) \text{ (monostar)}; \} \& \& \\ & V_{\{4\}}=(\\ & \{1,2,3,4,5\},\{\{1,2,3\},\{1,3,4\},\{1,4,5\},\{1,2,5\} \\ & \}) \text{ (monostar)}; \} \& \& \\ & \left.V_{\{5\}}=(\{1,2,3,4,5 \\ & \},\{\{1,2,3\},\{2,3,4\},\{3,4,5\},\{4,5,1\},\{5,1,2\} \\ & \}) \text{ (cycloid } C_5^3\text{)}\right) ; \& \& \\ & K_{\{1\}}=(\\ & \{1,2,3,4,5,6\},\{\{1,2,4\},\{2,3,5\},\{3,4,6\}, \\ & \quad \{4,5,1\}, \quad \quad \{5,6, \quad 2\}, \quad \quad \{6,1,3\}, \\ & \quad \quad \quad \{3,5\},\{2,4,6\}\}) \text{ (Král's example)}. \end{aligned}$	$\begin{aligned} & V_1 = (\{1, 2, 3, 4\}, \{\{1, 2, 3\}, \{1, 3, 4\}, \{1, 2, 4\}\}) \text{ (monostar)}; \\ & V_2 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{1, 4, 5\}\}) \text{ (monostar)}; \\ & V_3 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{1, 3, 4\}, \{1, 4, 5\}\}) \text{ (monostar)}; \\ & V_4 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{1, 3, 4\}, \{1, 4, 5\}, \{1, 2, 5\}\}) \text{ (monostar)}; \\ & V_5 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{2, 3, 4\}, \{3, 4, 5\}, \{4, 5, 1\}, \{5, 1, 2\}\}) \text{ (cycloid } C_5^3); \\ & K_1 = (\{1, 2, 3, 4, 5, 6\}, \{\{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{4, 5, 1\}, \{5, 6, 2\}, \{6, 1, 3\}, \\ & 3, 5\}, \{2, 4, 6\}\}) \text{ (Král's example)}. \end{aligned}$	$\begin{aligned} & V_1 = (\{1, 2, 3, 4\}, \{\{1, 2, 3\}, \{1, 3, 4\}, \{1, 2, 4\}\}) \text{ (monostar)}; \\ & V_2 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{1, 4, 5\}\}) \text{ (monostar)}; \\ & V_3 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{1, 3, 4\}, \{1, 4, 5\}\}) \text{ (monostar)}; \\ & V_4 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{1, 3, 4\}, \{1, 4, 5\}, \{1, 2, 5\}\}) \text{ (monostar)}; \\ & V_5 = (\{1, 2, 3, 4, 5\}, \{\{1, 2, 3\}, \{2, 3, 4\}, \{3, 4, 5\}, \{4, 5, 1\}, \{5, 1, 2\}\}) \text{ (cycloid } C_5^3); \\ & K_1 = (\{1, 2, 3, 4, 5, 6\}, \{\{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 6\}, \{4, 5, 1\}, \{5, 6, 2\}, \{6, 1, 3\}, \\ & 3, 5\}, \{2, 4, 6\}\}) \text{ (Král's example)}. \end{aligned}$
voloshin-hypergraph_ EQ0195 (10.2)	220	■ 0.997 ● N +1	$P(\mathcal{H}, \lambda) = \sum_{i=\chi(\mathcal{H})}^{\bar{\chi}(\mathcal{H})} r_i(\mathcal{H}) \lambda^{(i)}.$	$P(\mathcal{H}, \lambda) = \sum_{i=\chi(\mathcal{H})}^{\bar{\chi}(\mathcal{H})} r_i(\mathcal{H}) \lambda^{(i)}.$	$P(\mathcal{H}, \lambda) = \sum_{i=\chi(\mathcal{H})}^{\bar{\chi}(\mathcal{H})} r_i(\mathcal{H}) \lambda^{(i)}.$
voloshin-hypergraph_ EQ0261	274	■ 0.997 ● N +0	$X=\{1,2,3,4\}, \mathcal{C}=\mathcal{D}=\{\{1,2,3\}, \{2,3,4\}, \{3,4,1\}, \{4,1,2\}\};$	$X = \{1, 2, 3, 4\}, \mathcal{C} = \mathcal{D} = \{\{1, 2, 3\}, \{2, 3, 4\}, \{3, 4, 1\}, \{4, 1, 2\}\};$	$X = \{1, 2, 3, 4\}, \mathcal{C} = \mathcal{D} = \{\{1, 2, 3\}, \{2, 3, 4\}, \{3, 4, 1\}, \{4, 1, 2\}\};$
voloshin-hypergraph_ EQ0026	073	■ 0.997 ● W +0	$M(G) = \max_{X' \subseteq X} \min_{x \in G'} d(x).$	$M(G) = \max_{X' \subseteq X} \min_{x \in G'} d(x).$	$M(G) = \max_{X' \subseteq X} \min_{x \in G'} d(x).$
voloshin-hypergraph_ EQ0018	063	■ 0.998 ● K +0	$\left N_G(S)\right \geq S .$	$ N_G(S) \geq S .$	$ N_G(S) \geq S .$
voloshin-hypergraph_ EQ0071	113	■ 0.998 ● W +1	$\begin{aligned} & \lambda(\lambda-1)^{n-1}-\left[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)\right]= \\ & (\lambda-1)^n+(-1)^n(\lambda-1) \end{aligned}$	$\lambda(\lambda-1)^{n-1}-\left[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)\right]=$ $(\lambda-1)^n+(-1)^n(\lambda-1).$	$\lambda(\lambda-1)^{n-1}-\left[(\lambda-1)^{n-1}+(-1)^{n-1}(\lambda-1)\right]=$ $(\lambda-1)^n+(-1)^n(\lambda-1).$
voloshin-hypergraph_ EQ0154	193	■ 0.999 ● K +0	$\wedge(\mathcal{H}, T)=2+2+1+0+0-3+1=3.$	$\wedge(\mathcal{H}, T)=2+2+1+0+0-3+1=3.$	$\wedge(\mathcal{H}, T)=2+2+1+0+0-3+1=3.$
voloshin-hypergraph_ EQ0155	193	■ 0.999 ● K +0	$\wedge(\mathcal{H}-x, T-x)=2+1+0+0-2+1=2.$	$\wedge(\mathcal{H}-x, T-x)=2+1+0+0-2+1=2.$	$\wedge(\mathcal{H}-x, T-x)=2+1+0+0-2+1=2.$
voloshin-hypergraph_ EQ0185	215	■ 0.999 ● C -3	$m\left(y, \mathcal{H}_Y\right)=\min _{z} m\left(z, \mathcal{H}_Y\right)=M(\mathcal{H}).$	$m\left(y, \mathcal{H}_Y\right)=\min _{z} m\left(z, \mathcal{H}_Y\right)=M(\mathcal{H}).$	$m(y, \mathcal{H}_Y) = \min_z m(z, \mathcal{H}_Y) = M(\mathcal{H}).$
voloshin-hypergraph_ EQ0203	227	■ 0.999 ● N -2	$o\left(y, \mathcal{H}^{\prime}\right)=\min _{z} o\left(z, \mathcal{H}^{\prime}\right)=O(\mathcal{H}) \geq t+1.$	$o\left(y, \mathcal{H}^{\prime}\right)=\min _{z} o\left(z, \mathcal{H}^{\prime}\right)=O(\mathcal{H}) \geq t+1.$	$o(y, \mathcal{H}') = \min_z o(z, \mathcal{H}') = O(\mathcal{H}) \geq t+1.$
voloshin-hypergraph_ EQ0025	064	■ 0.999 ● N -1	$\tau(G)=\mathcal{V}(G).$	$\tau(G) = \mathcal{V}(G).$	$\tau(G) = \mathcal{V}(G).$
voloshin-hypergraph_ EQ0013	040	■ 0.999 ● K +0	$\sigma=\left(\begin{array}{llllll} 1 & 2 & 3 & 4 & 5 & 6 \\ a & b & c & d & e & f \end{array}\right).$	$\sigma=\left(\begin{array}{llllll} 1 & 2 & 3 & 4 & 5 & 6 \\ a & b & c & d & e & f \end{array}\right).$	$\sigma=\left(\begin{array}{llllll} 1 & 2 & 3 & 4 & 5 & 6 \\ a & b & c & d & e & f \end{array}\right).$
voloshin-hypergraph_ EQ0224	244	■ 0.999 ● K +0	$x_{\{j\}} \neq x_{\{j+1\}} \neq x_{\{j+2\}} \neq x_{\{j\}} \rightarrow \left\{x_{\{j\}}, x_{\{j+1\}}, x_{\{j+2\}}\right\} \in \mathcal{C}, j=0,1, \ldots, k-2.$	$x_j \neq x_{j+1} \neq x_{j+2} \neq x_j \Rightarrow \{x_j, x_{j+1}, x_{j+2}\} \in \mathcal{C}, j = 0, 1, \dots, k-2.$	$x_j \neq x_{j+1} \neq x_{j+2} \neq x_j \Rightarrow \{x_j, x_{j+1}, x_{j+2}\} \in \mathcal{C}, j = 0, 1, \dots, k-2.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0248	260	■ 0.999 ● K +0	1122,1112,1121,1222,2122,1212,1221.	1122, 1112, 1121, 1222, 2122, 1212, 1221.	1122, 1112, 1121, 1222, 2122, 1212, 1221.
voloshin-hypergraph_ EQ0179	211	■ 0.999 ● N +1	$\left\lfloor\frac{n}{\lambda}\right\rfloor\leq\left S_i\right \leq\left\lceil\frac{n}{\lambda}\right\rceil$.	$\left\lfloor\frac{n}{\lambda}\right\rfloor\leq S_i \leq\left\lceil\frac{n}{\lambda}\right\rceil$.	$\left\lfloor\frac{n}{\lambda}\right\rfloor\leq S_i \leq\left\lceil\frac{n}{\lambda}\right\rceil$.
voloshin-hypergraph_ EQ0004	029	■ 1.000 ● S +0	$\begin{aligned} L(G)= & \left\{\left\{x_2, x_5\right\},\left\{x_1, x_3, x_4, x_5\right\},\left\{x_2, x_4\right\},\right. \\ & \left.\left\{x_2, x_3, x_5\right\},\left\{x_1, x_2, x_4\right\}\right\}. \end{aligned}$	$L(G)=\left\{\left\{x_2, x_5\right\},\left\{x_1, x_3, x_4, x_5\right\},\left\{x_2, x_4\right\},\right. \\ \left.\left\{x_2, x_3, x_5\right\},\left\{x_1, x_2, x_4\right\}\right\}.$	$L(G)=\left\{\left\{x_2, x_5\right\},\left\{x_1, x_3, x_4, x_5\right\},\left\{x_2, x_4\right\},\right. \\ \left.\left\{x_2, x_3, x_5\right\},\left\{x_1, x_2, x_4\right\}\right\}.$
voloshin-hypergraph_ EQ0006	030	■ 1.000 ● K +0	$I(G)=\left(\begin{array}{llllll} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right).$	$I(G)=\left(\begin{array}{llllll} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right).$	$I(G)=\left(\begin{array}{llllll} 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right).$
voloshin-hypergraph_ EQ0019 (2.1)	063	■ 1.000 ● K +0	$ S <\left N_{\{G\}}(S)\right $.	$ S < N_G(S) $.	$ S < N_G(S) $.
voloshin-hypergraph_ EQ0039 (5.3)	102	■ 1.000 ● N +1	$P(G, \lambda)=\sum_{i=\chi}^n r_i(G) \lambda^{\{i\}}$	$P(G, \lambda)=\sum_{i=\chi}^n r_i(G) \lambda^{(i)}$	$P(G, \lambda)=\sum_{i=\chi}^n r_i(G) \lambda^{(i)}.$
voloshin-hypergraph_ EQ0050	106	■ 1.000 ● N +0	$P(G-e, \lambda)=\sum_{i=0}^n(-1)^i a_i \lambda^{n-i}$	$P(G-e, \lambda)=\sum_{i=0}^n(-1)^i a_i \lambda^{n-i}$	$P(G-e, \lambda)=\sum_{i=0}^n(-1)^i a_i \lambda^{n-i}$
voloshin-hypergraph_ EQ0066 (5.12)	112	■ 1.000 ● W +0	$\begin{gathered} P\left(G_{\{1\}}, \lambda\right)= \\ \left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-3\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)= \\ \left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)=\lambda\left(\lambda-1\right)\left(\lambda-2\right)^5\left(\lambda-3\right) . \end{gathered}$	$P\left(G_1, \lambda\right)=\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-3\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)= \\ \lambda\left(\lambda-1\right)\left(\lambda-2\right)^5\left(\lambda-3\right) .$	$P\left(G_1, \lambda\right)=\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-3\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-2\right)\left(\lambda-1\right)\left(\lambda-0\right)= \\ \lambda\left(\lambda-1\right)\left(\lambda-2\right)^5\left(\lambda-3\right) .$
voloshin-hypergraph_ EQ0067	113	■ 1.000 ● K +0	$P\left(G_{\{1\}}, 4\right)=4\left(4-1\right)\left(4-2\right)^5\left(4-3\right)=384$.	$P\left(G_1, 4\right)=4\left(4-1\right)\left(4-2\right)^5\left(4-3\right)=384$.	$P\left(G_1, 4\right)=4\left(4-1\right)\left(4-2\right)^5\left(4-3\right)=384$.
voloshin-hypergraph_ EQ0079	115	■ 1.000 ● K +0	$\left\{x_0\right\} \cup N\left(x_0\right) \subseteq X_1 \cup A, N\left(x_0\right) \cap A \subseteq X_i \cup A, 1 \leq i \leq k$.	$\left\{x_0\right\} \cup N\left(x_0\right) \subseteq X_1 \cup A, N\left(x_0\right) \cap A \subseteq X_i \cup A, 1 \leq i \leq k$.	$\left\{x_0\right\} \cup N\left(x_0\right) \subseteq X_1 \cup A, N\left(x_0\right) \cap A \subseteq X_i \cup A, 1 \leq i \leq k$.
voloshin-hypergraph_ EQ0084	116	■ 1.000 ● K +0	$P\left(C_{\{k\}}, \lambda\right)=\lambda\left(\lambda-1\right)^2 \lambda\left(\lambda-1\right)^{k-2} \lambda^{-2}=\left(\lambda-1\right)^k \neq P\left(C_k, \lambda\right),$	$P\left(C_k, \lambda\right)=\lambda\left(\lambda-1\right)^2 \lambda\left(\lambda-1\right)^{k-2} \lambda^{-2}=\left(\lambda-1\right)^k \neq P\left(C_k, \lambda\right),$	$P\left(C_k, \lambda\right)=\lambda\left(\lambda-1\right)^2 \lambda\left(\lambda-1\right)^{k-2} \lambda^{-2}=\left(\lambda-1\right)^k \neq P\left(C_k, \lambda\right),$
voloshin-hypergraph_ EQ0090	117	■ 1.000 ● K +0	$P(C, \lambda)=P(C-x, \lambda) W\left(C_{N(x)}, \lambda\right)=\left(\lambda-1\right)^k \neq P(C, \lambda),$	$P(C, \lambda)=P(C-x, \lambda) W\left(C_{N(x)}, \lambda\right)=\left(\lambda-1\right)^k \neq P(C, \lambda),$	$P(C, \lambda)=P(C-x, \lambda) W\left(C_{N(x)}, \lambda\right)=\left(\lambda-1\right)^k \neq P(C, \lambda),$
voloshin-hypergraph_ EQ0106	135	■ 1.000 ● N -1	$\bar{\chi}^{\prime}\left(A_1\right) \geq \bar{\chi}^{\prime}\left(A_0\right)+1 \geq c+m\left(A_1\right)-n\left(A_1\right)+p\left(A_1\right),$	$\bar{\chi}^{\prime}\left(A_1\right) \geq \bar{\chi}^{\prime}\left(A_0\right)+1 \geq c+m\left(A_1\right)-n\left(A_1\right)+p\left(A_1\right),$	$\bar{\chi}^{\prime}\left(A_1\right) \geq \bar{\chi}^{\prime}\left(A_0\right)+1 \geq c+m\left(A_1\right)-n\left(A_1\right)+p\left(A_1\right),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ EQ0146 (8.4)	192	■ 1.000 ● N +0	$\wedge(\mathcal{H}, T)=\sum_{D \in \mathcal{D}}(c(T, D)-1)+l(\mathcal{H}, T)$.	$\wedge(\mathcal{H}, T)=\sum_{D \in \mathcal{D}}(c(T, D)-1)+l(\mathcal{H}, T)$.	$\wedge(\mathcal{H}, T)=\sum_{D \in \mathcal{D}}(c(T, D)-1)+l(\mathcal{H}, T)$.
voloshin-hypergraph_ EQ0180	211	■ 1.000 ● N +0	$0 \leq a_j \leq b_j \leq D_j $.	$0 \leq a_j \leq b_j \leq D_j $.	$0 \leq a_j \leq b_j \leq D_j $.
voloshin-hypergraph_ EQ0194	220	■ 1.000 ● N +0	$R(\mathcal{H})=(0,\ldots,0,r_\chi,\ldots,r_{\bar{\chi}},0,\ldots,0)$.	$R(\mathcal{H})=(0,\ldots,0,r_\chi,\ldots,r_{\bar{\chi}},0,\ldots,0)$.	$R(\mathcal{H})=(0,\ldots,0,r_\chi,\ldots,r_{\bar{\chi}},0,\ldots,0)$.
voloshin-hypergraph_ EQ0200	223	■ 1.000 ● N -1	$b(x,\mathcal{H})=\max_{y\in X} \mathcal{C}(x)\cap\mathcal{C}(y) $.	$b(x,\mathcal{H})=\max_y \mathcal{C}(x)\cap\mathcal{C}(y) :y\in X,y\neq x$	$b(x,\mathcal{H})=\max_y \mathcal{C}(x)\cap\mathcal{C}(y) :y\in X,y\neq x$
voloshin-hypergraph_ EQ0204	227	■ 1.000 ● W -2	$o(x_i,\mathcal{H}_i)\leq O(\mathcal{H})=\max_{Y\subseteq X}\min_{x\in Y}o(x,\mathcal{H}/Y)$.	$o(x_i,\mathcal{H}_i)\leq O(\mathcal{H})=\max_{Y\subseteq X}\min_{x\in Y}o(x,\mathcal{H}/Y)$.	$o(x_i,\mathcal{H}_i)\leq O(\mathcal{H})=\max_{Y\subseteq X}\min_{x\in Y}o(x,\mathcal{H}/Y)$.
voloshin-hypergraph_ EQ0229	248	■ 1.000 ● N +0	$M=\sum_{\substack{x\in T\\d(x)\geq 2}}\binom{d(x)}{2}$.	$M=\sum_{\substack{x\in T\\d(x)\geq 2}}\binom{d(x)}{2}$.	$M=\sum_{\substack{x\in T\\d(x)\geq 2}}\binom{d(x)}{2}$.
voloshin-hypergraph_ EQ0237	255	■ 1.000 ● N +0	$n_{12}+1+n_{23}<r, n_{23}+1+n_{31}<r$.	$n_{12}+1+n_{23}<r, n_{23}+1+n_{31}<r$.	$n_{12}+1+n_{23}<r, n_{23}+1+n_{31}<r$.
voloshin-hypergraph_ EQ0001	022	■ 1.000 ● K +0	$G=(X,E)$.	$G=(X,E)$.	$G=(X,E)$.
voloshin-hypergraph_ EQ0002 (1.1)	023	■ 1.000 ● N +0	$\sum_{i=1}^n d(x_i)=2m$.	$\sum_{i=1}^n d(x_i)=2m$.	$\sum_{i=1}^n d(x_i)=2m$.
voloshin-hypergraph_ EQ0003	023	■ 1.000 ● K +0	$2+4+2+3+3=14=2\times 7$.	$2+4+2+3+3=14=2\times 7$.	$2+4+2+3+3=14=2\times 7$.
voloshin-hypergraph_ EQ0005	029	■ 1.000 ● K +0	$A(G)=\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}$.	$A(G)=\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}$.	$A(G)=\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}$.
voloshin-hypergraph_ EQ0008	032	■ 1.000 ● K +0	$L(G)=\{\emptyset,\{2\},\{2,4\},\{1,1,3\}\}$.	$L(G)=\{\emptyset,\{2\},\{2,4\},\{1,1,3\}\}$.	$L(G)=\{\emptyset,\{2\},\{2,4\},\{1,1,3\}\}$.
voloshin-hypergraph_ EQ0010	033	■ 1.000 ● N +0	$I(G)=\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 & -1 & 1 \end{pmatrix}$.	$I(G)=\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 & -1 & 1 \end{pmatrix}$.	$I(G)=\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \\ -1 & -1 & 0 & 0 & -1 & 1 \end{pmatrix}$.
voloshin-hypergraph_ EQ0011	033	■ 1.000 ● K +0	$J(G)=\{(4,1),(4,1),(3,2),(2,2),(4,3),(3,4)\}$.	$J(G)=\{(4,1),(4,1),(3,2),(2,2),(4,3),(3,4)\}$.	$J(G)=\{(4,1),(4,1),(3,2),(2,2),(4,3),(3,4)\}$.
voloshin-hypergraph_ EQ0012	036	■ 1.000 ● N +0	$m=\binom{n}{2}=\frac{n(n-1)}{2}$.	$m=\binom{n}{2}=\frac{n(n-1)}{2}$.	$m=\binom{n}{2}=\frac{n(n-1)}{2}$.

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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voloshin-hypergraph_ EQ0014	047	■ 1.000 ● N -1	$\alpha(G)+\tau(G)= X $.	$\alpha(G) + \tau(G) = X .$	$\alpha(G) + \tau(G) = X .$
voloshin-hypergraph_ EQ0016	061	■ 1.000 ● K +0	$X=N_{\{0\}} \cup N_{\{1\}} \cup N_{\{2\}} \cup N_{\{3\}} \cup \ldots \cup N_{\{k\}}$.	$X = N_0 \cup N_1 \cup N_2 \cup N_3 \cup \ldots \cup N_k.$	$X = N_0 \cup N_1 \cup N_2 \cup N_3 \cup \ldots \cup N_k.$
voloshin-hypergraph_ EQ0017	062	■ 1.000 ● K +0	$N(S)=\cup_{x \in S} N(x) \backslash S$.	$N(S) = \cup_{x \in S} N(x) \backslash S.$	$N(S) = \cup_{x \in S} N(x) \backslash S.$
voloshin-hypergraph_ EQ0020	063	■ 1.000 ● K +0	$\left S^{\prime}\right \leq\left N_{G}\left(S^{\prime}\right)\right $.	$ S' \leq N_G(S') .$	$ S' \leq N_G(S') .$
voloshin-hypergraph_ EQ0021 (2.2)	063	■ 1.000 ● K +0	$\left S_{\{0\}}\right =\left N_{\{G\}}\left(S_{\{0\}}\right)\right $.	$ S_0 = N_G(S_0) .$	$ S_0 = N_G(S_0) .$
voloshin-hypergraph_ EQ0022	063	■ 1.000 ● N -1	$A=S_{\{0\}} \cup N_{\{G\}}\left(S_{\{0\}}\right) \text { and } B=\left(X \cup Y\right) \backslash A$	$A = S_0 \cup N_G(S_0) \text { and } B = (X \cup Y) \backslash A$	$A = S_0 \cup N_G(S_0) \text { and } B = (X \cup Y) \backslash A$
voloshin-hypergraph_ EQ0023	063	■ 1.000 ● K +0	$N_{\{G_{\{B\}}\}}(S)=N_{\{G\}}(S) \backslash N_{\{G\}}\left(S_{\{0\}}\right),$	$N_{G_B}(S) = N_G(S) \backslash N_G(S_0),$	$N_{G_B}(S) = N_G(S) \backslash N_G(S_0),$
voloshin-hypergraph_ EQ0024	063	■ 1.000 ● K +0	$\left S_{\{0\}}\right + S =\left S_{\{0\}} \cup S\right \leq\left N_{\{G\}}\left(S_{\{0\}} \cup S\right)\right =\left N_{\{G\}}\left(S_{\{0\}}\right)\right +\left N_{\{G_{\{B\}}\}}(S)\right $.	$ S_0 + S = S_0 \cup S \leq N_G(S_0 \cup S) = N_G(S_0) + N_{G_B}(S) .$	$ S_0 + S = S_0 \cup S \leq N_G(S_0 \cup S) = N_G(S_0) + N_{G_B}(S) .$
voloshin-hypergraph_ EQ0027	074	■ 1.000 ● N -1	$M(G)-(\omega(G)-1)$.	$M(G) - (\omega(G) - 1).$	$M(G) - (\omega(G) - 1).$
voloshin-hypergraph_ EQ0028	075	■ 1.000 ● K +0	$X=N_{\{0\}} \cup N_{\{1\}} \cup N_{\{2\}} \cup \cdots \cup N_{\{k-1\}} \cup N_{\{k\}}$	$X = N_0 \cup N_1 \cup N_2 \cup \cdots \cup N_{k-1} \cup N_k$	$X = N_0 \cup N_1 \cup N_2 \cup \cdots \cup N_{k-1} \cup N_k$
voloshin-hypergraph_ EQ0031	097	■ 1.000 ● N +0	$R(G)=\left(r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}\right)$	$R(G) = (r_1, r_2, \ldots, r_n)$	$R(G) = (r_1, r_2, \ldots, r_n)$
voloshin-hypergraph_ EQ0032	097	■ 1.000 ● N +0	$R(G)=\left(0,0, \ldots, 0, r_{\{\chi\}}, r_{\{\chi+1\}}, \ldots, r_{\{n\}}\right)$.	$R(G) = (0, 0, \ldots, 0, r_{\chi}, r_{\chi+1}, \ldots, r_n).$	$R(G) = (0, 0, \ldots, 0, r_{\chi}, r_{\chi+1}, \ldots, r_n).$
voloshin-hypergraph_ EQ0033	098	■ 1.000 ● K +0	$R(G)=(0,1,1)$.	$R(G) = (0, 1, 1).$	$R(G) = (0, 1, 1).$
voloshin-hypergraph_ EQ0034	099	■ 1.000 ● N +0	$P\left(K_{\{n\}}, \lambda\right)=\lambda(\lambda-1)(\lambda-2) \cdots(\lambda-n+1)$.	$P\left(K_n, \lambda\right) = \lambda(\lambda-1)(\lambda-2) \cdots(\lambda-n+1).$	$P(K_n, \lambda) = \lambda(\lambda-1)(\lambda-2) \cdots(\lambda-n+1).$
voloshin-hypergraph_ EQ0035 (5.1)	100	■ 1.000 ● K +0	$P(G, \lambda)=P\left(G_{\{1\}}, \lambda\right)+P\left(G_{\{2\}}, \lambda\right)$.	$P(G, \lambda) = P\left(G_1, \lambda\right) + P\left(G_2, \lambda\right).$	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda).$
voloshin-hypergraph_ EQ0036	101	■ 1.000 ● N +2	$P(G, \lambda)=P\left(K_{\{i_1\}}, \lambda\right)+P\left(K_{\{i_2\}}, \lambda\right)+\cdots+P\left(K_{\{i_s\}}, \lambda\right)$	$P(G, \lambda) = P\left(K_{i_1}, \lambda\right) + P\left(K_{i_2}, \lambda\right) + \cdots + P\left(K_{i_s}, \lambda\right)$	$P(G, \lambda) = P(K_{i_1}, \lambda) + P(K_{i_2}, \lambda) + \cdots + P(K_{i_s}, \lambda)$
voloshin-hypergraph_ EQ0037	101	■ 1.000 ● W +0	$\begin{gathered} P(G, \lambda)=P\left(K_{\{i_1\}}, \lambda\right)+P\left(K_{\{i_2\}}, \lambda\right)+\cdots+P\left(K_{\{i_s\}}, \lambda\right)=\sum_{i=1}^s r_i(G) P\left(K_{\{i\}}, \lambda\right)+r_{\{2\}}(G) P\left(K_{\{2\}}, \lambda\right)+\cdots+r_{\{n\}}(G) P\left(K_{\{n\}}, \lambda\right)=\end{gathered}$	$P(G, \lambda) = P\left(K_{i_1}, \lambda\right) + P\left(K_{i_2}, \lambda\right) + \cdots + P\left(K_{i_s}, \lambda\right) = r_1(G) P\left(K_1, \lambda\right) + r_2(G) P\left(K_2, \lambda\right) + \cdots + r_n(G) P\left(K_n, \lambda\right) =$	$P(G, \lambda) = P\left(K_{i_1}, \lambda\right) + P\left(K_{i_2}, \lambda\right) + \cdots + P\left(K_{i_s}, \lambda\right) = r_1(G) P\left(K_1, \lambda\right) + r_2(G) P\left(K_2, \lambda\right) + \cdots + r_n(G) P\left(K_n, \lambda\right) =$
voloshin-hypergraph_ EQ0038 (5.2)	102	■ 1.000 ● K +0	$\sum_{i=1}^n r_{\{i\}}(G) P\left(K_{\{i\}}, \lambda\right)$.	$\sum_{i=1}^n r_i(G) P\left(K_i, \lambda\right).$	$\sum_{i=1}^n r_i(G) P\left(K_i, \lambda\right).$
voloshin-hypergraph_ EQ0042	103	■ 1.000 ● K +0	$R\left(C_{\{4\}}\right)=(0,1,2,1)$.	$R\left(C_4\right) = (0, 1, 2, 1).$	$R(C_4) = (0, 1, 2, 1).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0043 (5.5)	105	■ 1.000 ● K +0	$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_k, \lambda).$	$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_k, \lambda).$	$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_k, \lambda).$
voloshin-hypergraph_ EQ0044	105	■ 1.000 ● K +0	$P(E_n, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n.$	$P(E_n, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n.$	$P(E_n, \lambda) = \lambda \cdot \lambda \cdot \dots \cdot \lambda = \lambda^n.$
voloshin-hypergraph_ EQ0045 (5.6)	105	■ 1.000 ● N +0	$\lambda^n = S(n, 1) \lambda^{(1)} + S(n, 2) \lambda^{(2)} + \dots + S(n, n) \lambda^{(n)}$	$\lambda^n = S(n, 1) \lambda^{(1)} + S(n, 2) \lambda^{(2)} + \dots + S(n, n) \lambda^{(n)}$	$\lambda^n = S(n, 1) \lambda^{(1)} + S(n, 2) \lambda^{(2)} + \dots + S(n, n) \lambda^{(n)}$
voloshin-hypergraph_ EQ0046	106	■ 1.000 ● K +0	$R(E_n) = (S(n, 1), S(n, 2), \dots, S(n, n)).$	$R(E_n) = (S(n, 1), S(n, 2), \dots, S(n, n)).$	$R(E_n) = (S(n, 1), S(n, 2), \dots, S(n, n)).$
voloshin-hypergraph_ EQ0047	106	■ 1.000 ● N +0	$\lambda^{(n)} = s(n, 1) \lambda + s(n, 2) \lambda^2 + \dots + s(n, n) \lambda^n$	$\lambda^{(n)} = s(n, 1) \lambda + s(n, 2) \lambda^2 + \dots + s(n, n) \lambda^n$	$\lambda^{(n)} = s(n, 1) \lambda + s(n, 2) \lambda^2 + \dots + s(n, n) \lambda^n$
voloshin-hypergraph_ EQ0048	106	■ 1.000 ● K +0	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda)$	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda)$	$P(G, \lambda) = P(G_1, \lambda) + P(G_2, \lambda)$
voloshin-hypergraph_ EQ0049 (5.7)	106	■ 1.000 ● K +0	$P(G_1, \lambda) = P(G, \lambda) - P(G_2, \lambda).$	$P(G_1, \lambda) = P(G, \lambda) - P(G_2, \lambda).$	$P(G_1, \lambda) = P(G, \lambda) - P(G_2, \lambda).$
voloshin-hypergraph_ EQ0053 (5.8)	108	■ 1.000 ● N +0	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(k)}}.$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(k)}}.$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(k)}}.$
voloshin-hypergraph_ EQ0055 (5.9)	109	■ 1.000 ● N +0	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_l, \lambda)}{[\lambda^{(k)}]^{l-1}}.$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_l, \lambda)}{[\lambda^{(k)}]^{l-1}}.$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \cdots P(G_l, \lambda)}{[\lambda^{(k)}]^{l-1}}.$
voloshin-hypergraph_ EQ0057 (5.10)	109	■ 1.000 ● K +0	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda).$	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda).$	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda).$
voloshin-hypergraph_ EQ0058	109	■ 1.000 ● N +0	$P(G, \lambda) = \frac{P(G_1, \lambda) P(K_{k+1}, \lambda)}{\lambda^{(k)}} =$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(K_{k+1}, \lambda)}{\lambda^{(k)}} =$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(K_{k+1}, \lambda)}{\lambda^{(k)}} =$
voloshin-hypergraph_ EQ0059	110	■ 1.000 ● N +0	$\frac{P(G - x, \lambda) \lambda^{(k+1)}}{\lambda^{(k)}} = (\lambda - k) P(G - x, \lambda).$	$\frac{P(G - x, \lambda) \lambda^{(k+1)}}{\lambda^{(k)}} = (\lambda - k) P(G - x, \lambda).$	$\frac{P(G - x, \lambda) \lambda^{(k+1)}}{\lambda^{(k)}} = (\lambda - k) P(G - x, \lambda).$
voloshin-hypergraph_ EQ0060	110	■ 1.000 ● K +0	$P(G, \lambda) = (\lambda - 2) P(G - x, \lambda) = \lambda(\lambda - 1)(\lambda - 2)(\lambda^2 - 3\lambda + 3).$	$P(G, \lambda) = (\lambda - 2) P(G - x, \lambda) = \lambda(\lambda - 1)(\lambda - 2)(\lambda^2 - 3\lambda + 3).$	$P(G, \lambda) = (\lambda - 2) P(G - x, \lambda) = \lambda(\lambda - 1)(\lambda - 2)(\lambda^2 - 3\lambda + 3).$
voloshin-hypergraph_ EQ0061	110	■ 1.000 ● N +0	$r_i(G) = (i - k) r_i(G - x) + r_{i-1}(G - x).$	$r_i(G) = (i - k) r_i(G - x) + r_{i-1}(G - x).$	$r_i(G) = (i - k) r_i(G - x) + r_{i-1}(G - x).$
voloshin-hypergraph_ EQ0062	110	■ 1.000 ● K +0	$S(n, i) = i S(n - 1, i) + S(n - 1, i - 1).$	$S(n, i) = i S(n - 1, i) + S(n - 1, i - 1).$	$S(n, i) = i S(n - 1, i) + S(n - 1, i - 1).$
voloshin-hypergraph_ EQ0063	110	■ 1.000 ● N +0	$S(5, 3) = 25 = 3 \cdot S(4, 3) + S(4, 2) = 3 \cdot 6 + 7 = 25.$	$S(5, 3) = 25 = 3 \cdot S(4, 3) + S(4, 2) = 3 \cdot 6 + 7 = 25.$	$S(5, 3) = 25 = 3 \cdot S(4, 3) + S(4, 2) = 3 \cdot 6 + 7 = 25.$
voloshin-hypergraph_ EQ0068 (5.13)	113	■ 1.000 ● N +1	$P(T_n, \lambda) = \lambda(\lambda - 1)^{n-1}$	$P(T_n, \lambda) = \lambda(\lambda - 1)^{n-1}$	$P(T_n, \lambda) = \lambda(\lambda - 1)^{n-1}.$
voloshin-hypergraph_ EQ0069 (5.14)	113	■ 1.000 ● N +0	$P(C_n, \lambda) = (\lambda - 1)^n + (-1)^n (\lambda - 1).$	$P(C_n, \lambda) = (\lambda - 1)^n + (-1)^n (\lambda - 1).$	$P(C_n, \lambda) = (\lambda - 1)^n + (-1)^n (\lambda - 1).$
voloshin-hypergraph_ EQ0070	113	■ 1.000 ● K +0	$P(C_n, \lambda) = P(T_n, \lambda) - P(C_{n-1}, \lambda) =$	$P(C_n, \lambda) = P(T_n, \lambda) - P(C_{n-1}, \lambda) =$	$P(C_n, \lambda) = P(T_n, \lambda) - P(C_{n-1}, \lambda) =$
voloshin-hypergraph_ EQ0073	114	■ 1.000 ● K +0	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0.$	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0.$	$G_{X_1 \cup A} = G_1, G_{X_2 \cup A} = G_2, \dots, G_{X_k \cup A} = G_k, G_A = G_0.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ E00074 (5.15)	114	■ 1.000 ● N +0	$P(\text{G}^{\backslash\prime}, \backslash\lambda\text{right})=\frac{P(\text{G}_{\text{1}}^{\backslash\prime}, \backslash\lambda\text{right})}{P(\text{G}_{\text{0}}^{\backslash\prime}, \backslash\lambda\text{right})} \dots P(\text{G}_{\text{k}}^{\backslash\prime}, \backslash\lambda\text{right})$	$P(G', \lambda) = \frac{P(G'_1, \lambda) P(G'_2, \lambda) \dots P(G'_k, \lambda)}{P(G'_0, \lambda)^{k-1}}.$	$P(G', \lambda) = \frac{P(G'_1, \lambda) P(G'_2, \lambda) \dots P(G'_k, \lambda)}{P(G'_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ E00075	114	■ 1.000 ● N +0	$P(\text{G}_{\text{1}}, \backslash\lambda\text{right})=(\backslash\lambda\text{p}) P(\text{G}_{\text{1}}-\text{x}_{\text{0}}, \backslash\lambda\text{right})$	$P(G_1, \lambda) = (\lambda - p) P(G_1 - x_0, \lambda).$	$P(G_1, \lambda) = (\lambda - p) P(G_1 - x_0, \lambda).$
voloshin-hypergraph_ E00076	114	■ 1.000 ● N +0	$P(\text{G}, \backslash\lambda\text{p})=(\backslash\lambda\text{p}) P(\text{G}-\text{x}_{\text{0}}, \backslash\lambda\text{right})$	$P(G, \lambda) = (\lambda - p) P(G - x_0, \lambda).$	$P(G, \lambda) = (\lambda - p) P(G - x_0, \lambda).$
voloshin-hypergraph_ E00077 (5.16)	114	■ 1.000 ● N +0	$P(\text{G}-\text{x}_{\text{0}}, \backslash\lambda\text{right})=\frac{P(\text{G}_{\text{1}}-\text{x}_{\text{0}}, \backslash\lambda\text{right})}{P(\text{G}_{\text{0}}, \backslash\lambda\text{right})} \dots P(\text{G}_{\text{k}}, \backslash\lambda\text{right})$	$P(G - x_0, \lambda) = \frac{P(G_1 - x_0, \lambda) P(G_2, \lambda) \dots P(G_k, \lambda)}{P(G_0, \lambda)^{k-1}}.$	$P(G - x_0, \lambda) = \frac{P(G_1 - x_0, \lambda) P(G_2, \lambda) \dots P(G_k, \lambda)}{P(G_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ E00078	115	■ 1.000 ● N +0	$P(\text{G}, \backslash\lambda\text{p})=\frac{P(\text{G}_{\text{1}}, \backslash\lambda\text{right})}{P(\text{G}_{\text{0}}, \backslash\lambda\text{right})} \dots P(\text{G}_{\text{k}}, \backslash\lambda\text{right})$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \dots P(G_k, \lambda)}{P(G_0, \lambda)^{k-1}}.$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \dots P(G_k, \lambda)}{P(G_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ E00080	115	■ 1.000 ● K +0	$P(\text{G}, \backslash\lambda\text{p})=(\backslash\lambda\text{p}) P(\text{G}-\text{x}_{\text{0}}, \backslash\lambda\text{right})$	$P(G, \lambda) = (\lambda - p) P(G - x_0, \lambda).$	$P(G, \lambda) = (\lambda - p) P(G - x_0, \lambda).$
voloshin-hypergraph_ E00081 (5.17)	115	■ 1.000 ● N +0	$P(\text{G}-\text{x}_{\text{0}}, \backslash\lambda\text{right})=\frac{P(\text{G}_{\text{1}}^{\backslash\prime}, \backslash\lambda\text{right})}{P(\text{G}_{\text{0}}^{\backslash\prime}, \backslash\lambda\text{right})} \dots P(\text{G}_{\text{k}}^{\backslash\prime}, \backslash\lambda\text{right})$	$P(G - x_0, \lambda) = \frac{P(G'_1, \lambda) P(G'_2, \lambda) \dots P(G'_k, \lambda)}{P(G'_0, \lambda)^{k-1}},$	$P(G - x_0, \lambda) = \frac{P(G'_1, \lambda) P(G'_2, \lambda) \dots P(G'_k, \lambda)}{P(G'_0, \lambda)^{k-1}},$
voloshin-hypergraph_ E00082	115	■ 1.000 ● N -1	$P(\text{G}_{\text{1}}, \backslash\lambda\text{right})=(\backslash\lambda\text{p}) P(\text{G}_{\text{1}}^{\backslash\prime}, \backslash\lambda\text{right}), \text{ and } P(\text{G}_{\text{i}}, \backslash\lambda\text{right})=(\backslash\lambda\text{p}_{\text{1}}) P(\text{G}_{\text{i}}^{\backslash\prime}, \backslash\lambda\text{right})$	$P(G_1, \lambda) = (\lambda - p) P(G'_1, \lambda), \text{ and } P(G_i, \lambda) = (\lambda - p_1) P(G'_i, \lambda)$	$P(G_1, \lambda) = (\lambda - p) P(G'_1, \lambda), \text{ and } P(G_i, \lambda) = (\lambda - p_1) P(G'_i, \lambda)$
voloshin-hypergraph_ E00083	116	■ 1.000 ● N +0	$P(\text{G}, \backslash\lambda\text{p})=\frac{P(\text{G}_{\text{1}}, \backslash\lambda\text{right})}{P(\text{G}_{\text{0}}, \backslash\lambda\text{right})} \dots P(\text{G}_{\text{k}}, \backslash\lambda\text{right})$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \dots P(G_k, \lambda)}{P(G_0, \lambda)^{k-1}}.$	$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda) \dots P(G_k, \lambda)}{P(G_0, \lambda)^{k-1}}.$
voloshin-hypergraph_ E00086 (5.18)	116	■ 1.000 ● K +0	$P(\text{G}^{\backslash\prime}, \backslash\lambda\text{right})=P(\text{G}^{\backslash\prime}-\text{x}, \backslash\lambda\text{right}) W(\text{G}_{\text{N}(\text{x})}^{\backslash\prime}, \backslash\lambda\text{right})$	$P(G', \lambda) = P(G' - x, \lambda) W(G'_{N(x)}, \lambda).$	$P(G', \lambda) = P(G' - x, \lambda) W(G'_{N(x)}, \lambda).$
voloshin-hypergraph_ E00088	116	■ 1.000 ● N +0	$P(\text{G}, \backslash\lambda\text{p})=\frac{P(\text{G}-\text{x}, \backslash\lambda\text{p})}{P(\text{G}_{\text{N}(\text{x})}, \backslash\lambda\text{right})} =$	$P(G, \lambda) = \frac{P(G - x, \lambda) P(G_{\{x\} \cup N(x)}, \lambda)}{P(G_{N(x)}, \lambda)} =$	$P(G, \lambda) = \frac{P(G - x, \lambda) P(G_{\{x\} \cup N(x)}, \lambda)}{P(G_{N(x)}, \lambda)} =$
voloshin-hypergraph_ E00089	117	■ 1.000 ● N -1	$P(\text{G}-\text{x}, \backslash\lambda\text{p}) \frac{\lambda P(\text{G}_{\text{N}(\text{x})}, \lambda - 1)}{P(\text{G}_{\text{N}(\text{x})}, \lambda)} = P(\text{G}-\text{x}, \backslash\lambda\text{p}) W(\text{G}_{\text{N}(\text{x})}, \lambda)$	$P(G - x, \lambda) \frac{\lambda P(G_{N(x)}, \lambda - 1)}{P(G_{N(x)}, \lambda)} = P(G - x, \lambda) W(G_{N(x)}, \lambda).$	$P(G - x, \lambda) \frac{\lambda P(G_{N(x)}, \lambda - 1)}{P(G_{N(x)}, \lambda)} = P(G - x, \lambda) W(G_{N(x)}, \lambda).$
voloshin-hypergraph_ E00091	117	■ 1.000 ● K +0	$P(\text{G}, \backslash\lambda\text{p})=(\backslash\lambda\text{p}-\text{k}) P(\text{G}-\text{x}, \backslash\lambda\text{p})$	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda).$	$P(G, \lambda) = (\lambda - k) P(G - x, \lambda).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ EQ0092	117	■ 1.000 ● N +0	$W\left(G_{N(x)}, \lambda\right)=W\left(K_{\{k\}}, \lambda\right)=\frac{\lambda(\lambda-1)^{\{k\}}}{\lambda^{\{k\}}}=(\lambda-k) .$	$W\left(G_{N(x)}, \lambda\right)=W\left(K_k, \lambda\right)=\frac{\lambda(\lambda-1)^{(k)}}{\lambda^{(k)}}=(\lambda-k) .$	$W\left(G_{N(x)}, \lambda\right)=W\left(K_k, \lambda\right)=\frac{\lambda(\lambda-1)^{(k)}}{\lambda^{(k)}}=(\lambda-k) .$
voloshin-hypergraph_ EQ0093	117	■ 1.000 ● K +0	$P(G, \lambda)=P(G-x, \lambda) W\left(G_{N(x)}, \lambda\right)=(\lambda-k) P(G-x, \lambda) .$	$P(G, \lambda)=P(G-x, \lambda) W\left(G_{N(x)}, \lambda\right)=(\lambda-k) P(G-x, \lambda) .$	$P(G, \lambda)=P(G-x, \lambda) W\left(G_{N(x)}, \lambda\right)=(\lambda-k) P(G-x, \lambda) .$
voloshin-hypergraph_ EQ0094	117	■ 1.000 ● N -1	$W\left(G_{N(4)}, \lambda\right)=\frac{\lambda P\left(T_3, \lambda-1\right)}{P\left(T_3, \lambda\right)}=\frac{\lambda(\lambda-1)(\lambda-2)^2}{\lambda(\lambda-1)^2}=\frac{(\lambda-2)^2}{\lambda-1} .$	$W\left(G_{N(4)}, \lambda\right)=\frac{\lambda P\left(T_3, \lambda-1\right)}{P\left(T_3, \lambda\right)}=\frac{\lambda(\lambda-1)(\lambda-2)^2}{\lambda(\lambda-1)^2}=\frac{(\lambda-2)^2}{\lambda-1} .$	$W\left(G_{N(4)}, \lambda\right)=\frac{\lambda P\left(T_3, \lambda-1\right)}{P\left(T_3, \lambda\right)}=\frac{\lambda(\lambda-1)(\lambda-2)^2}{\lambda(\lambda-1)^2}=\frac{(\lambda-2)^2}{\lambda-1} .$
voloshin-hypergraph_ EQ0095	117	■ 1.000 ● K +0	$P\left(G_1, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) .$	$P\left(G_1, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) .$	$P\left(G_1, \lambda\right)=\lambda(\lambda-1)^2(\lambda-2)^3(\lambda-3) .$
voloshin-hypergraph_ EQ0097	118	■ 1.000 ● W +0	$M(G)=\max _{X^{\prime} \subseteq X} \min _{x \in G^{\prime}} d(x)$	$M(G)=\max _{X^{\prime} \subseteq X} \min _{x \in G^{\prime}} d(x)$	$M(G)=\max _{X^{\prime} \subseteq X} \min _{x \in G^{\prime}} d(x)$
voloshin-hypergraph_ EQ0098 (5.19)	124	■ 1.000 ● N -1	$\chi(G) \geq \omega(G) .$	$\chi(G) \geq \omega(G) .$	$\chi(G) \geq \omega(G) .$
voloshin-hypergraph_ EQ0100	133	■ 1.000 ● K +0	$\bar{\chi}^{\prime}\left(G_i\right)=1, i=1, \ldots, \bar{\chi}^{\prime}(G) .$	$\bar{\chi}^{\prime}\left(G_i\right)=1, i=1, \ldots, \bar{\chi}^{\prime}(G) .$	$\bar{\chi}^{\prime}\left(G_i\right)=1, i=1, \ldots, \bar{\chi}^{\prime}(G) .$
voloshin-hypergraph_ EQ0101	134	■ 1.000 ● K +0	$\bar{\chi}^{\prime}(G)=c+m-n+p .$	$\bar{\chi}^{\prime}(G)=c+m-n+p .$	$\bar{\chi}^{\prime}(G)=c+m-n+p .$
voloshin-hypergraph_ EQ0102	135	■ 1.000 ● N +0	$n-v^{\prime}-p+r .$	$n-v^{\prime}-p+r .$	$n-v^{\prime}-p+r .$
voloshin-hypergraph_ EQ0103	135	■ 1.000 ● N -2	$m^{\prime}=v^{\prime}+\left(n-v^{\prime}-p+r\right)=n-p+r .$	$m^{\prime}=v^{\prime}+\left(n-v^{\prime}-p+r\right)=n-p+r .$	$m^{\prime}=v^{\prime}+\left(n-v^{\prime}-p+r\right)=n-p+r .$
voloshin-hypergraph_ EQ0104	135	■ 1.000 ● N -1	$\bar{\chi}^{\prime}(G)=c^{\prime}+r+\left(m-m^{\prime}\right)=c^{\prime}+m-n+p,$	$\bar{\chi}^{\prime}(G)=c^{\prime}+r+\left(m-m^{\prime}\right)=c^{\prime}+m-n+p,$	$\bar{\chi}^{\prime}(G)=c^{\prime}+r+\left(m-m^{\prime}\right)=c^{\prime}+m-n+p,$
voloshin-hypergraph_ EQ0105	135	■ 1.000 ● N +0	$\bar{\chi}^{\prime}(G)=c^{\prime}+m-n+p \leq c+m-n+p .$	$\bar{\chi}^{\prime}(G)=c^{\prime}+m-n+p \leq c+m-n+p .$	$\bar{\chi}^{\prime}(G)=c^{\prime}+m-n+p \leq c+m-n+p .$
voloshin-hypergraph_ EQ0108	136	■ 1.000 ● K +0	$\bar{\chi}^{\prime}(G)=p-1 .$	$\bar{\chi}^{\prime}(G)=p-1 .$	$\bar{\chi}^{\prime}(G)=p-1 .$
voloshin-hypergraph_ EQ0109	141	■ 1.000 ● N -1	$d(x)+d(y) \geq n,$	$d(x)+d(y) \geq n,$	$d(x)+d(y) \geq n,$
voloshin-hypergraph_ EQ0110	141	■ 1.000 ● N +0	$d\left(x_1\right)+d\left(x_n\right) \leq(n-2)+(n-2)-(n-3)=n-1$	$d\left(x_1\right)+d\left(x_n\right) \leq(n-2)+(n-2)-(n-3)=n-1$	$d\left(x_1\right)+d\left(x_n\right) \leq(n-2)+(n-2)-(n-3)=n-1$
voloshin-hypergraph_ EQ0111	142	■ 1.000 ● K +0	$x_1, x_{i+1}, x_{i+2}, \ldots, x_n, x_i, x_{i-1}, \ldots, x_1$	$x_1, x_{i+1}, x_{i+2}, \ldots, x_n, x_i, x_{i-1}, \ldots, x_1$	$x_1, x_{i+1}, x_{i+2}, \ldots, x_n, x_i, x_{i-1}, \ldots, x_1$
voloshin-hypergraph_ EQ0112	144	■ 1.000 ● N +0	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v$
voloshin-hypergraph_ EQ0113	144	■ 1.000 ● N +0	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v .$	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v .$	$u \rightarrow x_1 \rightarrow x_2 \rightarrow v .$
voloshin-hypergraph_ EQ0114	144	■ 1.000 ● N +0	$u \rightarrow x_4 \rightarrow x_3 \rightarrow v .$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow v .$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow v .$
voloshin-hypergraph_ EQ0115	144	■ 1.000 ● N +0	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v .$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v .$	$u \rightarrow x_4 \rightarrow x_3 \rightarrow x_1 \rightarrow x_2 \rightarrow v .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0116	152	■ 1.000 ● N +0	$\Delta(\mathcal{H})=\max_{x\in X} \mathcal{D}(x) $.	$\Delta(\mathcal{H}) = \max_{x \in X} \mathcal{D}(x) .$	$\Delta(\mathcal{H}) = \max_{x \in X} \mathcal{D}(x) .$
voloshin-hypergraph_ EQ0117	152	■ 1.000 ● N -1	$r(\mathcal{H})=\max_{D\in\mathcal{D}} \mathcal{D} $.	$r(\mathcal{H}) = \max_{D \in \mathcal{D}} D .$	$r(\mathcal{H}) = \max_{D \in \mathcal{D}} D .$
voloshin-hypergraph_ EQ0121 (7.1)	158	■ 1.000 ● S +0	$\sum_{i=1}^n \mathcal{D}(x_i) =\sum_{j=1}^m \mathcal{D}_j $.	$\sum_{i=1}^n \mathcal{D}(x_i) = \sum_{j=1}^m D_j .$	$\sum_{i=1}^n \mathcal{D}(x_i) = \sum_{j=1}^m D_j .$
voloshin-hypergraph_ EQ0122	158	■ 1.000 ● K +0	$3+3+2+2+0=12=1+2+3+3$.	$3+3+2+2+2+0=12=1+2+3+3+3.$	$3+3+2+2+2+0=12=1+2+3+3+3.$
voloshin-hypergraph_ EQ0123	159	■ 1.000 ● N +0	$L=\{\{1\},\{1,2\},\{1,2,4\},\{2,3,5\},\{3,4,5\}\}$.	$L = \{\{1\}, \{1, 2\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 5\}\}.$	$L = \{\{1\}, \{1, 2\}, \{1, 2, 4\}, \{2, 3, 5\}, \{3, 4, 5\}\}.$
voloshin-hypergraph_ EQ0126	160	■ 1.000 ● K +0	$\left \mathcal{D}\left(K_4^3\right)\right =\frac{\binom{4}{3}}{1 \cdot 2 \cdot 3}=4$.	$ \mathcal{D}\left(K_4^3\right) = \binom{4}{3} = \frac{4 \cdot 3 \cdot 2}{1 \cdot 2 \cdot 3} = 4.$	$ \mathcal{D}(K_4^3) = \binom{4}{3} = \frac{4 \cdot 3 \cdot 2}{1 \cdot 2 \cdot 3} = 4.$
voloshin-hypergraph_ EQ0127	160	■ 1.000 ● K +0	$\mu=x_0 D_0 x_1 D_1 x_2 \ldots x_{t-1} D_{t-1} x_t$	$\mu = x_0 D_0 x_1 D_1 x_2 \ldots x_{t-1} D_{t-1} x_t$	$\mu = x_0 D_0 x_1 D_1 x_2 \ldots x_{t-1} D_{t-1} x_t$
voloshin-hypergraph_ EQ0128	168	■ 1.000 ● N -2	$\alpha(\mathcal{H})+\tau(\mathcal{H})= X $.	$\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X .$	$\alpha(\mathcal{H}) + \tau(\mathcal{H}) = X .$
voloshin-hypergraph_ EQ0129 (7.2)	168	■ 1.000 ● N -1	$\tau(\mathcal{H}) \geq v(\mathcal{H})$.	$\tau(\mathcal{H}) \geq v(\mathcal{H}).$	$\tau(\mathcal{H}) \geq v(\mathcal{H}).$
voloshin-hypergraph_ EQ0131	169	■ 1.000 ● K +0	$\rho(\mathcal{H})=\tau\left(\mathcal{H}^*\right)$.	$\rho(\mathcal{H}) = \tau\left(\mathcal{H}^*\right).$	$\rho(\mathcal{H}) = \tau(\mathcal{H}^*).$
voloshin-hypergraph_ EQ0134	170	■ 1.000 ● K +0	$(\mathcal{H})_2=L\left(\mathcal{H}^*\right)$.	$(\mathcal{H})_2 = L\left(\mathcal{H}^*\right).$	$(\mathcal{H})_2 = L(\mathcal{H}^*).$
voloshin-hypergraph_ EQ0135	170	■ 1.000 ● N +0	$L(\mathcal{H})=\left(\mathcal{H}^*\right)_2$.	$L(\mathcal{H}) = (\mathcal{H}^*)_2.$	$L(\mathcal{H}) = (\mathcal{H}^*)_2.$
voloshin-hypergraph_ EQ0139	173	■ 1.000 ● K +0	$Y=\left(Y_1 \cap Y_2\right) \cup\left(Y_1 \cap Y_3\right) \cup\left(Y_2 \cap Y_3\right)$.	$Y = (Y_1 \cap Y_2) \cup (Y_1 \cap Y_3) \cup (Y_2 \cap Y_3).$	$Y = (Y_1 \cap Y_2) \cup (Y_1 \cap Y_3) \cup (Y_2 \cap Y_3).$
voloshin-hypergraph_ EQ0140	173	■ 1.000 ● K +0	$Y \subseteq\left(D_1 \cap D_2\right) \cup\left(D_1 \cap D_3\right) \cup\left(D_2 \cap D_3\right)$.	$Y \subseteq (D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3).$	$Y \subseteq (D_1 \cap D_2) \cup (D_1 \cap D_3) \cup (D_2 \cap D_3).$
voloshin-hypergraph_ EQ0142 (8.2)	190	■ 1.000 ● N +0	$w(T) \leq \sum_{D \in \mathcal{D}}(D -1)$.	$w(T) \leq \sum_{D \in \mathcal{D}} (D - 1).$	$w(T) \leq \sum_{D \in \mathcal{D}} (D - 1).$
voloshin-hypergraph_ EQ0147 (8.5)	192	■ 1.000 ● K +0	$c(T, D)= D -\left E_D\right $.	$c(T, D) = D - E_D .$	$c(T, D) = D - E_D .$
voloshin-hypergraph_ EQ0148	192	■ 1.000 ● K +0	$w(T)=\sum_{ D \geq 2}\left E_D\right $.	$w(T) = \sum_{ D \geq 2} E_D .$	$w(T) = \sum_{ D \geq 2} E_D .$
voloshin-hypergraph_ EQ0151	193	■ 1.000 ● N +0	$w(T)=\sum_{D \in \mathcal{D}}(D -1)$.	$w(T) = \sum_{D \in \mathcal{D}} (D - 1).$	$w(T) = \sum_{D \in \mathcal{D}} (D - 1).$
voloshin-hypergraph_ EQ0157	194	■ 1.000 ● N -1	$w\left(T_1\right)=w(T)-\left \mathcal{D}_T(x)\right $.	$w\left(T_1\right) = w(T) - \mathcal{D}_T(x) .$	$w(T_1) = w(T) - \mathcal{D}_T(x) .$
voloshin-hypergraph_ EQ0158	194	■ 1.000 ● N -2	$w\left(T^{\prime}\right)=w\left(T_1\right)+\left \mathcal{D}_T(x)\right \geq w\left(T_1\right)+1+\left \mathcal{D}_T(x)\right =w(T)+1$	$w\left(T^{\prime}\right) = w\left(T_1\right) + \mathcal{D}_T(x) \geq w\left(T_1\right) + 1 + \mathcal{D}_T(x) = w(T) + 1$	$w(T') = w(T_1) + \mathcal{D}_T(x) \geq w(T_1) + 1 + \mathcal{D}_T(x) = w(T) + 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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voloshin-hypergraph_ EQ0159	194	■ 1.000 ● N -1	$w(\mathcal{H})=\max_{\mathcal{T}} w(T)$.	$w(\mathcal{H}) = \max_T w(T).$	$w(\mathcal{H}) = \max_T w(T).$
voloshin-hypergraph_ EQ0163	197	■ 1.000 ● K +0	$\mu=x_{\{0\}} D_{\{0\}} x_{\{1\}} D_{\{1\}} x_{\{2\}} \ldots x_{\{t-1\}} D_{\{t-1\}} x_{\{t\}}$	$\mu = x_0 D_0 x_1 D_1 x_2 \dots x_{t-1} D_{t-1} x_t$	$\mu = x_0 D_0 x_1 D_1 x_2 \dots x_{t-1} D_{t-1} x_t$
voloshin-hypergraph_ EQ0164	201	■ 1.000 ● N -1	$\chi^{\prime}(\mathcal{H}) \geq \omega(L(\mathcal{H})) \geq \Delta(\mathcal{H})$.	$\chi'(\mathcal{H}) \geq \omega(L(\mathcal{H})) \geq \Delta(\mathcal{H}).$	$\chi'(\mathcal{H}) \geq \omega(L(\mathcal{H})) \geq \Delta(\mathcal{H}).$
voloshin-hypergraph_ EQ0165	201	■ 1.000 ● K +0	$\chi^{\prime}(\mathcal{H}) \left(\mathcal{H}^{\prime}\right)=\Delta \left(\mathcal{H}^{\prime}\right)$.	$\chi'(\mathcal{H}') = \Delta(\mathcal{H}').$	$\chi'(\mathcal{H}') = \Delta(\mathcal{H}').$
voloshin-hypergraph_ EQ0166	201	■ 1.000 ● N -1	$\chi^{\prime}(\mathcal{H})=\chi(L(\mathcal{H}))=\omega(L(\mathcal{H}))=\Delta(\mathcal{H})$.	$\chi'(\mathcal{H}) = \chi(L(\mathcal{H})) = \omega(L(\mathcal{H})) = \Delta(\mathcal{H}).$	$\chi'(\mathcal{H}) = \chi(L(\mathcal{H})) = \omega(L(\mathcal{H})) = \Delta(\mathcal{H}).$
voloshin-hypergraph_ EQ0167	204	■ 1.000 ● N +0	$n-\sum_{i=1}^m \left(\left E_i\right -1\right)+f=m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2$.	$n-\sum_{i=1}^m \left(\left E_i\right -1\right)+f=m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2.$	$n-\sum_{i=1}^m \left(\left E_i\right -1\right)+f=m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2.$
voloshin-hypergraph_ EQ0168 (9.1)	204	■ 1.000 ● N +0	$m^{\prime}=\sum_{j=1}^m\left D_j\right =\sum_{i=1}^n\left \mathcal{D}\left(x_i\right)\right $	$m' = \sum_{j=1}^m D_j = \sum_{i=1}^n \mathcal{D}(x_i) $	$m' = \sum_{j=1}^m D_j = \sum_{i=1}^n \mathcal{D}(x_i) $
voloshin-hypergraph_ EQ0169	205	■ 1.000 ● N +1	$n^{\prime}-m^{\prime}+f^{\prime}=2$	$n' - m' + f' = 2$	$n' - m' + f' = 2.$
voloshin-hypergraph_ EQ0170 (9.2)	205	■ 1.000 ● N +1	$n-\sum_{i=1}^m \left(\left D_i\right -1\right)+f=2$	$n-\sum_{i=1}^m \left(\left D_i\right -1\right)+f=2$	$n-\sum_{i=1}^m \left(\left D_i\right -1\right)+f=2,$
voloshin-hypergraph_ EQ0171 (9.3)	205	■ 1.000 ● N +1	$m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2$	$m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2$	$m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2.$
voloshin-hypergraph_ EQ0172	205	■ 1.000 ● N +0	$n-\sum_{i=1}^m \left(\left E_i\right -1\right)+f=4-(2+2)+2=2$	$n-\sum_{i=1}^m \left(\left E_i\right -1\right)+f=4-(2+2)+2=2$	$n-\sum_{i=1}^m \left(\left E_i\right -1\right)+f=4-(2+2)+2=2$
voloshin-hypergraph_ EQ0173	205	■ 1.000 ● N -1	$\left.m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2-(0+1+0+1)+2\right)=2$.	$m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2-(0+1+0+1)+2\right)=2.$	$m-\sum_{j=1}^n \left(\left \mathcal{D}\left(x_j\right)\right -1\right)+f=2-(0+1+0+1)+2)=2.$
voloshin-hypergraph_ EQ0174 (10.1)	209	■ 1.000 ● K +0	$X=\cup_{i=1}^k S_{\{i\}}, \quad S_{\{i\}} \neq \emptyset, \quad S_{\{i\}} \cap S_{\{j\}}=\emptyset, \quad i \neq j$.	$X = \cup_{i=1}^k S_i, \quad S_i \neq \emptyset, \quad S_i \cap S_j = \emptyset, i \neq j.$	$X = \cup_{i=1}^k S_i, \quad S_i \neq \emptyset, \quad S_i \cap S_j = \emptyset, i \neq j.$
voloshin-hypergraph_ EQ0175	210	■ 1.000 ● W -2	$\begin{gathered} \alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n; \\ \chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1 = n - \alpha(\mathcal{H}) + 1. \end{gathered}$	$\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n; \\ \chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1 = n - \alpha(\mathcal{H}) + 1.$	$\alpha(\mathcal{H}) \chi(\mathcal{H}) \geq n; \\ \chi(\mathcal{H}) \leq \tau(\mathcal{H}) + 1 = n - \alpha(\mathcal{H}) + 1.$
voloshin-hypergraph_ EQ0177	211	■ 1.000 ● K +0	$\min \left\{ D , \lambda \right\}$	$\min\{ D , \lambda\}$	$\min\{ D , \lambda\}$
voloshin-hypergraph_ EQ0178	211	■ 1.000 ● N +0	$\lambda \leq \min_{D \in \mathcal{D}} D ,$	$\lambda \leq \min_{D \in \mathcal{D}} D ,$	$\lambda \leq \min_{D \in \mathcal{D}} D ,$
voloshin-hypergraph_ EQ0181	211	■ 1.000 ● K +0	$a_j \leq \left D_j \cap S_i\right \leq b_j$.	$a_j \leq D_j \cap S_i \leq b_j.$	$a_j \leq D_j \cap S_i \leq b_j.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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voloshin-hypergraph_ EQ0183	213	■ 1.000 ● K +0	$D_{\{i\}}, D_{\{j\}} \in \mathcal{D}_1(x) \Rightarrow D_{\{i\}} \cap D_{\{j\}} = \{x\}.$	$D_i, D_j \in \mathcal{D}_1(x) \Rightarrow D_i \cap D_j = \{x\}.$	$D_i, D_j \in \mathcal{D}_1(x) \Rightarrow D_i \cap D_j = \{x\}.$
voloshin-hypergraph_ EQ0184	214	■ 1.000 ● N -2	$M(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} m(x, \mathcal{H}_Y).$	$M(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} m(x, \mathcal{H}_Y).$	$M(\mathcal{H}) = \max_{Y \subseteq X} \min_{x \in Y} m(x, \mathcal{H}_Y).$
voloshin-hypergraph_ EQ0187	215	■ 1.000 ● N +0	$\chi(\mathcal{H}) \leq M(\mathcal{H}) + 1.$	$\chi(\mathcal{H}) \leq M(\mathcal{H}) + 1.$	$\chi(\mathcal{H}) \leq M(\mathcal{H}) + 1.$
voloshin-hypergraph_ EQ0188	215	■ 1.000 ● N +0	$m(1, \mathcal{H})=2, m(2, \mathcal{H})=2, m(3, \mathcal{H})=2, m(4, \mathcal{H})=1, m(5, \mathcal{H})=1.$	$m(1, \mathcal{H}) = 2, m(2, \mathcal{H}) = 2, m(3, \mathcal{H}) = 2, m(4, \mathcal{H}) = 1, m(5, \mathcal{H}) = 1.$	$m(1, \mathcal{H}) = 2, m(2, \mathcal{H}) = 2, m(3, \mathcal{H}) = 2, m(4, \mathcal{H}) = 1, m(5, \mathcal{H}) = 1.$
voloshin-hypergraph_ EQ0191	216	■ 1.000 ● K +0	$c=(2,1,2,1,1).$	$c = (2, 1, 2, 1, 1).$	$c = (2, 1, 2, 1, 1).$
voloshin-hypergraph_ EQ0192	218	■ 1.000 ● K +0	$c: X \rightarrow \{1, 2, \dots, \lambda\}.$	$c: X \rightarrow \{1, 2, \dots, \lambda\}.$	$c: X \rightarrow \{1, 2, \dots, \lambda\}.$
voloshin-hypergraph_ EQ0193	219	■ 1.000 ● N +0	$R(\mathcal{H}) = (r_1, r_2, \dots, r_n)$	$R(\mathcal{H}) = (r_1, r_2, \dots, r_n)$	$R(\mathcal{H}) = (r_1, r_2, \dots, r_n)$
voloshin-hypergraph_ EQ0196	220	■ 1.000 ● N +0	$r_{\bar{\chi}} \lambda (\lambda - 1) (\lambda - 2) \cdots (\lambda - \bar{\chi} + 1)$	$r_{\bar{\chi}} \lambda (\lambda - 1) (\lambda - 2) \cdots (\lambda - \bar{\chi} + 1)$	$r_{\bar{\chi}} \lambda (\lambda - 1) (\lambda - 2) \cdots (\lambda - \bar{\chi} + 1)$
voloshin-hypergraph_ EQ0205	230	■ 1.000 ● N -1	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) + P(\mathcal{H}_2, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1) + R(\mathcal{H}_2),$	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) + P(\mathcal{H}_2, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1) + R(\mathcal{H}_2),$	$P(\mathcal{H}, \lambda) = P(\mathcal{H}_1, \lambda) + P(\mathcal{H}_2, \lambda), \quad R(\mathcal{H}) = R(\mathcal{H}_1) + R(\mathcal{H}_2),$
voloshin-hypergraph_ EQ0209	231	■ 1.000 ● W +0	$\begin{gathered} Z = \{K_3, K_3, K_2\}, \\ R(\mathcal{H}) = (0, 1, 2, 0), \\ P(\mathcal{H}, \lambda) = 2\lambda^3 + \lambda^2 = 2\lambda^3 - 5\lambda^2 + 3\lambda, \end{gathered}$	$Z = \{K_3, K_3, K_2\}, R(\mathcal{H}) = (0, 1, 2, 0), \\ P(\mathcal{H}, \lambda) = 2\lambda^3 + \lambda^2 = 2\lambda^3 - 5\lambda^2 + 3\lambda,$	$Z = \{K_3, K_3, K_2\}, R(\mathcal{H}) = (0, 1, 2, 0), \\ P(\mathcal{H}, \lambda) = 2\lambda^3 + \lambda^2 = 2\lambda^3 - 5\lambda^2 + 3\lambda,$
voloshin-hypergraph_ EQ0210	231	■ 1.000 ● K +0	$L = \{(1, 2, 3, 2), (1, 2, 1, 3), (1, 2, 1, 2)\}.$	$L = \{(1, 2, 3, 2), (1, 2, 1, 3), (1, 2, 1, 2)\}.$	$L = \{(1, 2, 3, 2), (1, 2, 1, 3), (1, 2, 1, 2)\}.$
voloshin-hypergraph_ EQ0211	233	■ 1.000 ● K +0	$R(\mathcal{H}) = R(\mathcal{H} - C) (R(\mathcal{H}) = R(\mathcal{H} - D)).$	$R(\mathcal{H}) = R(\mathcal{H} - C) (R(\mathcal{H}) = R(\mathcal{H} - D)).$	$R(\mathcal{H}) = R(\mathcal{H} - C) (R(\mathcal{H}) = R(\mathcal{H} - D)).$
voloshin-hypergraph_ EQ0212	233	■ 1.000 ● N +0	$R(\mathcal{H}) = R(\mathcal{H} - C_j) = (1, 7, 0, 0), j = 1, 2, 3, 4.$	$R(\mathcal{H}) = R(\mathcal{H} - C_j) = (1, 7, 0, 0), j = 1, 2, 3, 4.$	$R(\mathcal{H}) = R(\mathcal{H} - C_j) = (1, 7, 0, 0), j = 1, 2, 3, 4.$
voloshin-hypergraph_ EQ0213	233	■ 1.000 ● K +0	$\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$	$\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$	$\mathcal{D} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}\}$
voloshin-hypergraph_ EQ0215	237	■ 1.000 ● K +0	$v(k) = k + 4.$	$v(k) = k + 4.$	$v(k) = k + 4.$
voloshin-hypergraph_ EQ0217	240	■ 1.000 ● N +0	$n \geq (l - 1)(m - 1) + 1.$	$n \geq (l - 1)(m - 1) + 1.$	$n \geq (l - 1)(m - 1) + 1.$
voloshin-hypergraph_ EQ0219	240	■ 1.000 ● N +0	$N_n = \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor \simeq n \log n$	$N_n = \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor \simeq n \log n$	$N_n = \sum_{k=1}^n \left\lfloor \frac{n}{k} \right\rfloor \simeq n \log n$
voloshin-hypergraph_ EQ0220	240	■ 1.000 ● W -2	$\lim_{n \rightarrow \infty} \frac{N_n}{n^2} = \lim_{n \rightarrow \infty} \frac{\log n}{n} = 0,$	$\lim_{n \rightarrow \infty} \frac{N_n}{n^2} = \lim_{n \rightarrow \infty} \frac{\log n}{n} = 0,$	$\lim_{n \rightarrow \infty} \frac{N_n}{n^2} = \lim_{n \rightarrow \infty} \frac{\log n}{n} = 0,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
voloshin-hypergraph_ EQ0221	243	■ 1.000 ● N +0	$\chi(\mathcal{H})=\bar{\chi}(\mathcal{H})=\chi, \ r_{\chi}(\mathcal{H})=1,$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = \chi, r_{\chi}(\mathcal{H}) = 1,$	$\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = \chi, \ r_{\chi}(\mathcal{H}) = 1,$
voloshin-hypergraph_ EQ0222	243	■ 1.000 ● K +0	$R(\mathcal{H})=(0, \ldots, 0, 1, 0, \ldots, 0) \ .$	$R(\mathcal{H}) = (0, \dots, 0, 1, 0, \dots, 0).$	$R(\mathcal{H}) = (0, \dots, 0, 1, 0, \dots, 0).$
voloshin-hypergraph_ EQ0225	244	■ 1.000 ● N +0	$R\left(\mathcal{H}_{\{1\}}\right)=R\left(\mathcal{H}_{\{2\}}\right.\\ \left.\right)=(0,1,0,0), \ P\left(\mathcal{H}_{\{1\}}, \ \lambda\right.\\ \left.\right)=P\left(\mathcal{H}_{\{2\}}, \ \lambda\right)=\lambda(\lambda-1) \ .$	$R(\mathcal{H}_1) = R(\mathcal{H}_2) = (0, 1, 0, 0), P(\mathcal{H}_1, \lambda) = P(\mathcal{H}_2, \lambda) = \lambda(\lambda - 1).$	$R(\mathcal{H}_1) = R(\mathcal{H}_2) = (0, 1, 0, 0), P(\mathcal{H}_1, \lambda) = P(\mathcal{H}_2, \lambda) = \lambda(\lambda - 1).$
voloshin-hypergraph_ EQ0227	248	■ 1.000 ● K +0	$L_{\{1\}}=\left(X_{\{1\}}, \ E_{\{1\}}\right), \ L_{\{2\}}=\left(X_{\{2\}}, \ E_{\{2\}}\right.\\ \left.\right), \ \ldots, \ L_{\{\chi\}}=\left(X_{\{\chi\}}, \ E_{\{\chi\}}\right) \ .$	$L_1 = (X_1, E_1), L_2 = (X_2, E_2), \dots, L_{\chi} = (X_{\chi}, E_{\chi}).$	$L_1 = (X_1, E_1), L_2 = (X_2, E_2), \dots, L_{\chi} = (X_{\chi}, E_{\chi}).$
voloshin-hypergraph_ EQ0230	249	■ 1.000 ● K +0	$\left\{x_{\{1\}}, \ x_{\{2\}}, \ \ldots, \ x_{\{i\}}\right\}, \ i=1,2, \ \ldots, \ n$	$\{x_1, x_2, \dots, x_i\}, i = 1, 2, \dots, n$	$\{x_1, x_2, \dots, x_i\}, i = 1, 2, \dots, n$
voloshin-hypergraph_ EQ0234	253	■ 1.000 ● K +0	$C_{\{i\}} \ \cap \ C_{\{j\}}=\{1\}, \ \text{quad } i, \ j \ \text{in } I=\{1,2, \ \ldots \ k+2\}, \ \text{quad } i \neq j \ .$	$C_i \cap C_j = \{1\}, \quad i, j \in I = \{1, 2, \dots k + 2\}, \quad i \neq j.$	$C_i \cap C_j = \{1\}, \quad i, j \in I = \{1, 2, \dots k + 2\}, \quad i \neq j.$
voloshin-hypergraph_ EQ0236	255	■ 1.000 ● N +0	$\begin{aligned} &\& n_{\{12\}}+n_{\{23\}}+n_{\{34\}}+2<r, \ \& \\ &n_{\{12\}}+n_{\{14\}}+n_{\{34\}}+2<r \ . \end{aligned}$	$n_{12} + n_{23} + n_{34} + 2 < r,$ $n_{12} + n_{14} + n_{34} + 2 < r.$	$n_{12} + n_{23} + n_{34} + 2 < r,$ $n_{12} + n_{14} + n_{34} + 2 < r.$
voloshin-hypergraph_ EQ0238	256	■ 1.000 ● N +0	$n_{\{31\}}+1+n_{\{12\}}<r \ .$	$n_{31} + 1 + n_{12} < r.$	$n_{31} + 1 + n_{12} < r.$
voloshin-hypergraph_ EQ0239	256	■ 1.000 ● N +0	$X=\{1,2,3, \ \ldots, \ 2 \ r\},$	$X = \{1, 2, 3, \dots, 2r\},$	$X = \{1, 2, 3, \dots, 2r\},$
voloshin-hypergraph_ EQ0240	256	■ 1.000 ● K +0	$\mathcal{C}=\left\{C_{\{1\}}, \ C_{\{2\}}, \ \ldots, \ C_{\{2 \ r\}}, \ C_{\{o\}}, \ C_{\{e\}}\right\}$	$\mathcal{C} = \{C_1, C_2, \dots, C_{2r}, C_o, C_e\}$	$\mathcal{C} = \{C_1, C_2, \dots, C_{2r}, C_o, C_e\}$
voloshin-hypergraph_ EQ0241	256	■ 1.000 ● N +0	$C_{\{1\}}=\{1,2,4,6,8, \ \ldots, \ 2 \ r-2\} \ .$	$C_1 = \{1, 2, 4, 6, 8, \dots, 2r - 2\}.$	$C_1 = \{1, 2, 4, 6, 8, \dots, 2r - 2\}.$
voloshin-hypergraph_ EQ0247	260	■ 1.000 ● K +0	$R(\mathcal{H})=(0,7,0,0)$	$R(\mathcal{H}) = (0, 7, 0, 0)$	$R(\mathcal{H}) = (0, 7, 0, 0)$
voloshin-hypergraph_ EQ0249	261	■ 1.000 ● N -1	$R(\mathcal{H})=\left(0, \ r_{\{2\}}, \ 0,1,0,0,0,0\right) \ .$	$R(\mathcal{H}) = (0, r_2, 0, 1, 0, 0, 0).$	$R(\mathcal{H}) = (0, r_2, 0, 1, 0, 0, 0).$
voloshin-hypergraph_ EQ0250	263	■ 1.000 ● K +0	$\left\{x_{\{1\}}, \ x_{\{2\}}, \ a_{\{1\}}, \ \ldots, \ a_{\{t-2\}}, \ b_{\{1\}}, \ \ldots, \ b_{\{t-2\}}\right\} \ .$	$\{x_1, x_2, a_1, \dots, a_{t-2}, b_1, \dots, b_{t-2}\}.$	$\{x_1, x_2, a_1, \dots, a_{t-2}, b_1, \dots, b_{t-2}\}.$
voloshin-hypergraph_ EQ0251	264	■ 1.000 ● K +0	$R\left(\mathcal{H}_{\{2,4\}}\right)=(0,4,0,1,0,0),$	$R(\mathcal{H}_{2,4}) = (0, 4, 0, 1, 0, 0),$	$R(\mathcal{H}_{2,4}) = (0, 4, 0, 1, 0, 0),$
voloshin-hypergraph_ EQ0252	264	■ 1.000 ● K +0	$P(\mathcal{H}, \ \lambda)=\lambda(\lambda-1)\left(\lambda^2-5\lambda+10\right) \ .$	$P(\mathcal{H}, \lambda) = \lambda(\lambda - 1) (\lambda^2 - 5\lambda + 10) .$	$P(\mathcal{H}, \lambda) = \lambda(\lambda - 1) (\lambda^2 - 5\lambda + 10).$
voloshin-hypergraph_ EQ0253	266	■ 1.000 ● N +0	$R(\mathcal{H})=\left(0, \ \ldots, \ 0, \ r_{\{\chi\}}, \ \ldots, \ r_{\{\bar{\chi}\}}, \ 0, \ \ldots, \ 0\right) \ .$	$R(\mathcal{H}) = (0, \dots, 0, r_{\chi}, \dots, r_{\bar{\chi}}, 0, \dots, 0) .$	$R(\mathcal{H}) = (0, \dots, 0, r_{\chi}, \dots, r_{\bar{\chi}}, 0, \dots, 0).$
voloshin-hypergraph_ EQ0255	268	■ 1.000 ● K +0	$(A, \ A, \ B, \ B, \ C, \ C, \ D) \ .$	$(A, A, B, B, C, C, D).$	$(A, A, B, B, C, C, D).$
voloshin-hypergraph_ EQ0256	268	■ 1.000 ● K +0	$\{1,2,7\}, \{3,4,7\}, \{5,6,7\} \ \text{in } \mathcal{E} \ .$	$\{1, 2, 7\}, \{3, 4, 7\}, \{5, 6, 7\} \in \mathcal{E}.$	$\{1, 2, 7\}, \{3, 4, 7\}, \{5, 6, 7\} \in \mathcal{E}.$
voloshin-hypergraph_ EQ0260	274	■ 1.000 ● K +0	$\mathcal{H}=\left(X, \ \binom{X}{3}, \ \binom{X}{3}\right)$	$\mathcal{H} = \left(X, \binom{X}{3}, \binom{X}{3}\right)$	$\mathcal{H} = (X, \binom{X}{3}, \binom{X}{3})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

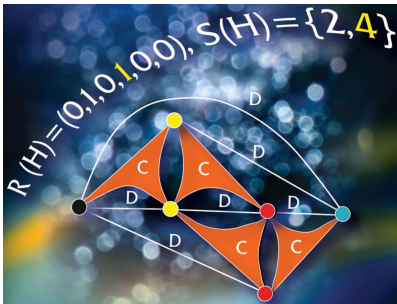
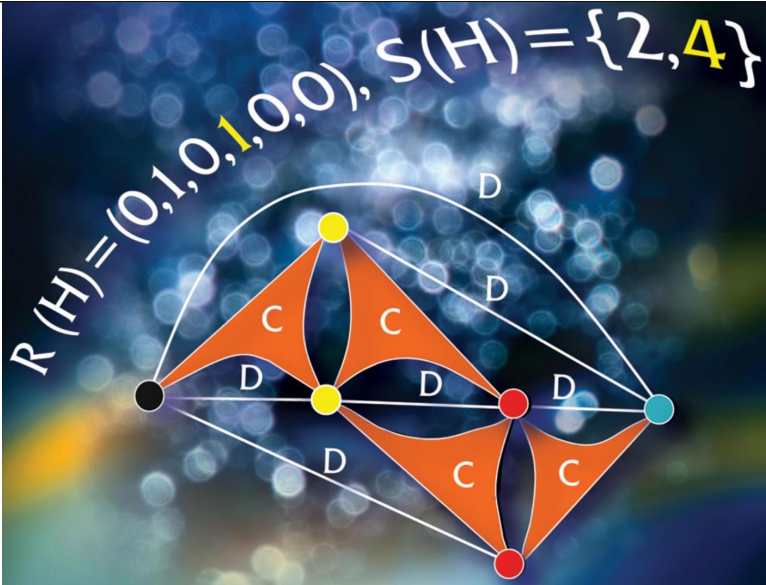
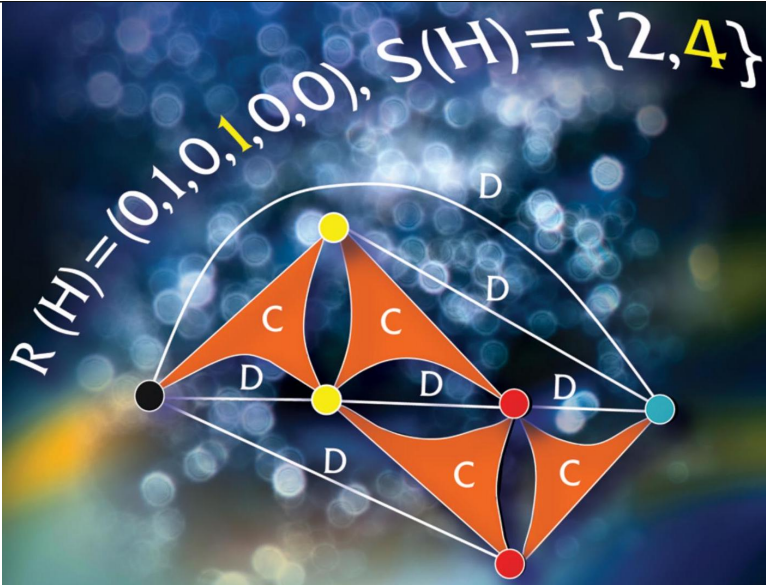
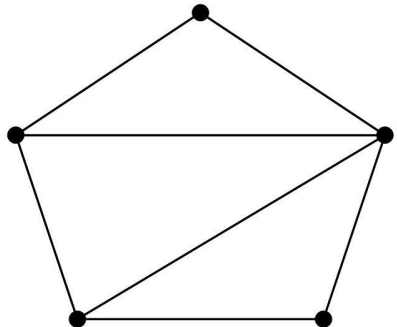
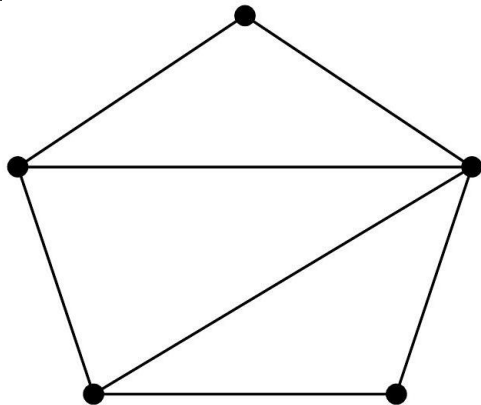
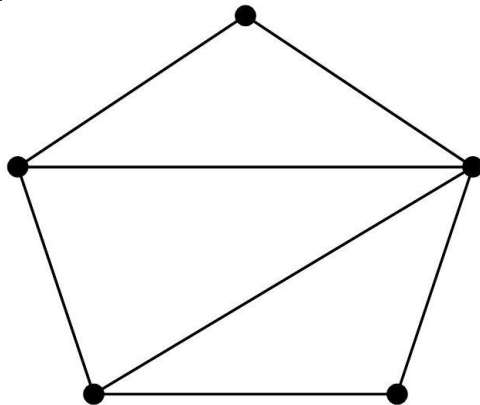
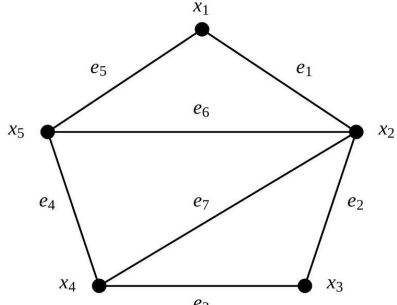
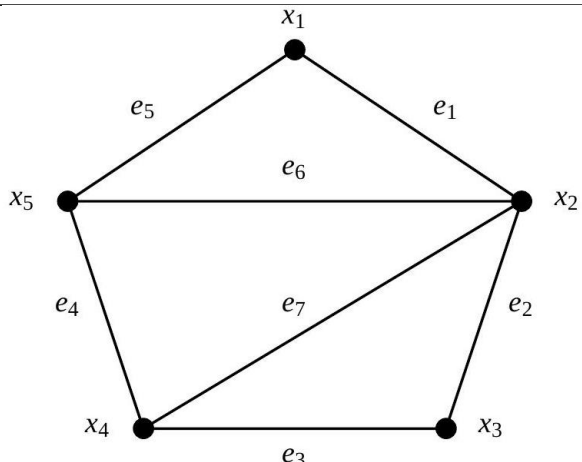
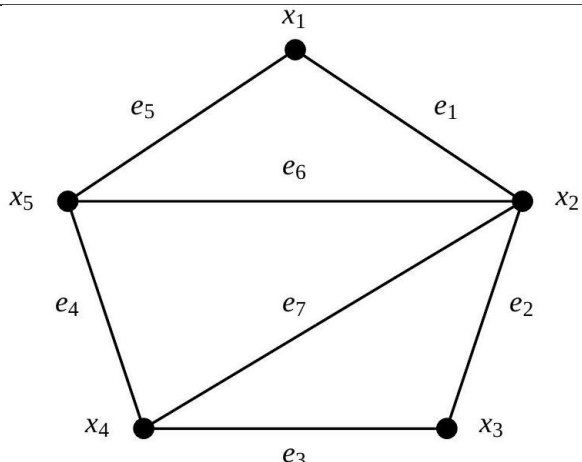
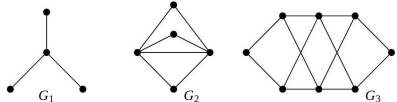
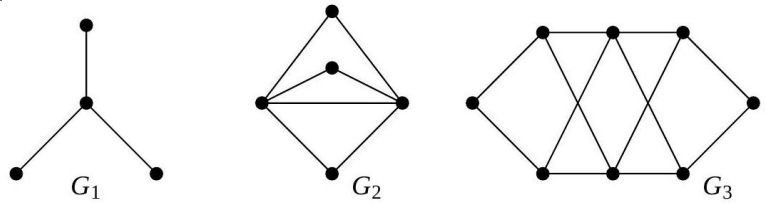
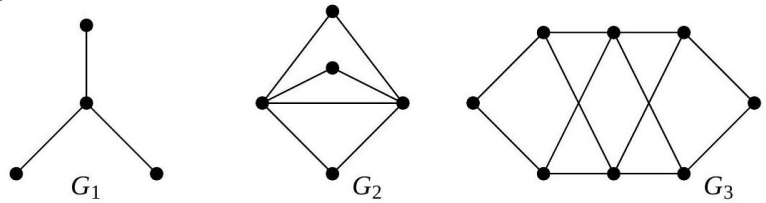
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voloshin-hypergraph_ EQ0262	276	■ 1.000 ● K +0	$R\left(\mathrm{mathcal{H}}_{2,4}^{\prime}\right)=(0,1,0,1,0,0),$	$R\left(\mathcal{H}_{2,4}^{\prime}\right)=(0,1,0,1,0,0),$	$R\left(\mathcal{H}_{2,4}^{\prime}\right)=(0,1,0,1,0,0),$
voloshin-hypergraph_ EQ0263	276	■ 1.000 ● K +0	$P\left(\mathrm{mathcal{H}}_{2,4}^{\prime},\lambda\right)=\lambda\left(\lambda-1\right)\left(\lambda^2-5\lambda+7\right)$	$P\left(\mathcal{H}_{2,4}^{\prime},\lambda\right)=\lambda\left(\lambda-1\right)\left(\lambda^2-5\lambda+7\right)$	$P\left(\mathcal{H}_{2,4}^{\prime},\lambda\right)=\lambda\left(\lambda-1\right)\left(\lambda^2-5\lambda+7\right)$
voloshin-hypergraph_ EQ0264	285	■ 1.000 ● N +0	$f(n)\leq Cg(n)$	$f(n)\leq Cg(n)$	$f(n)\leq Cg(n)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Tables

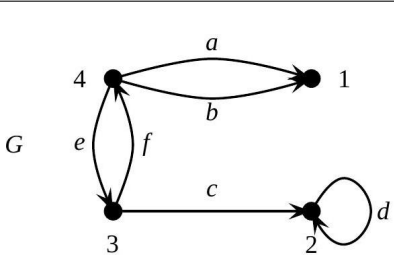
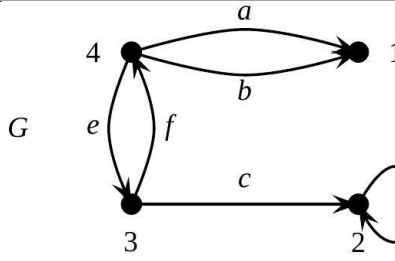
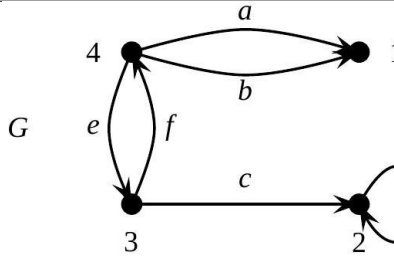
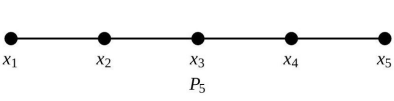
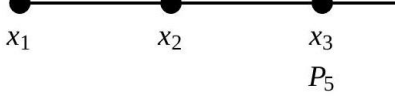
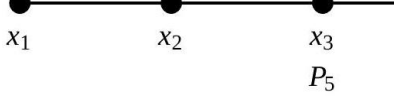
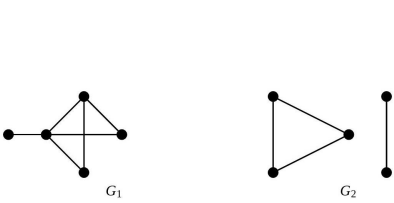
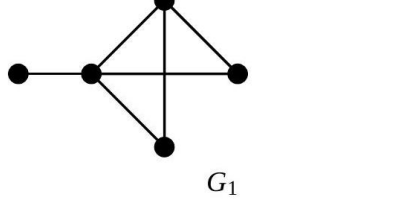
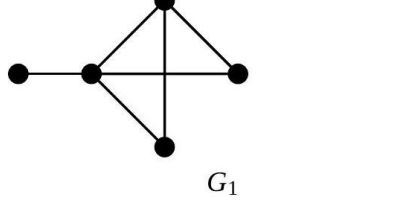
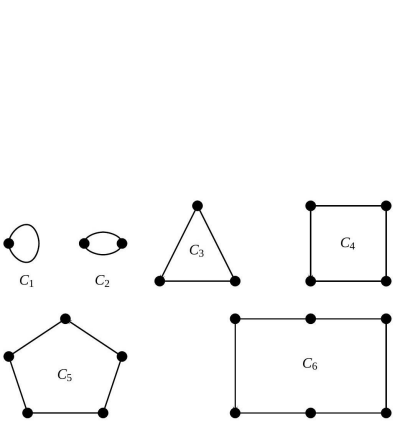
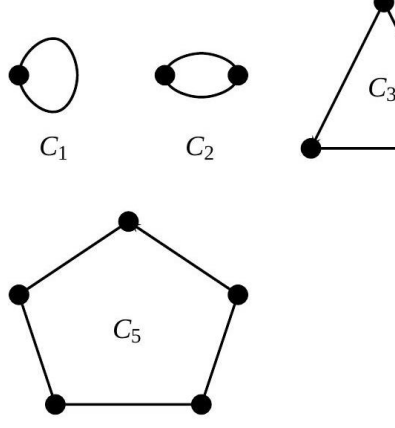
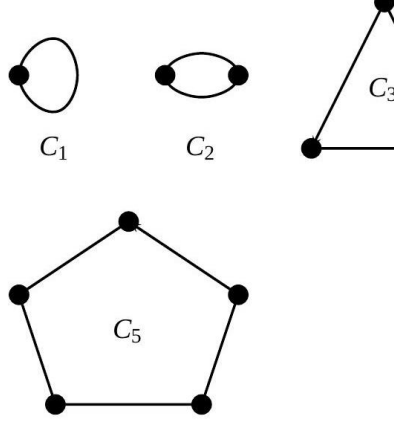
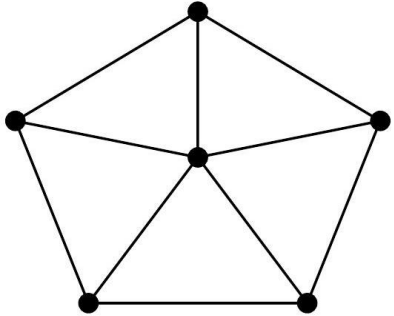
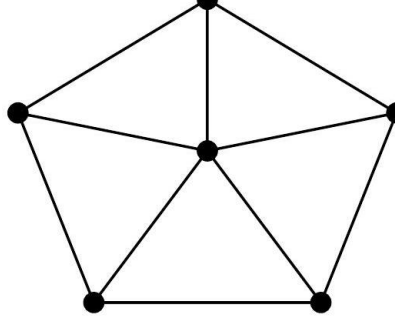
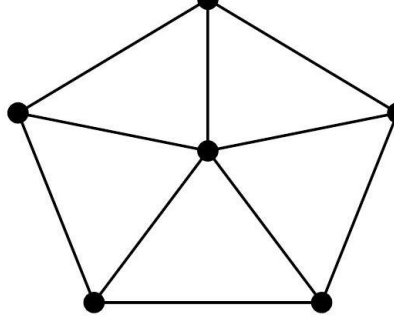
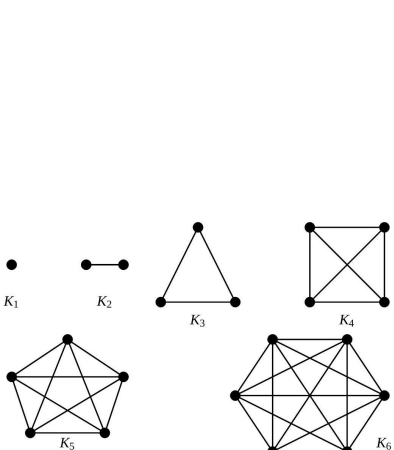
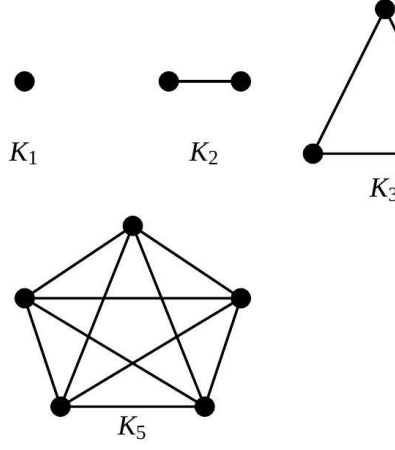
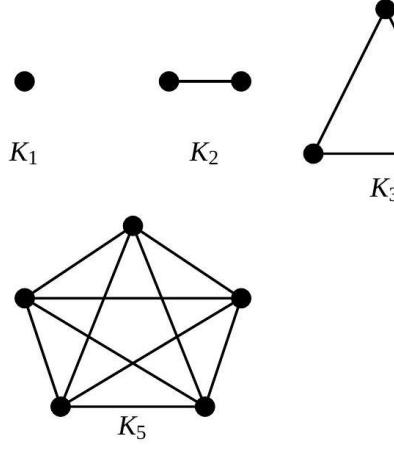
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voloshin-hypergraph_TAB_041	241	—	(no LaTeX source; 564 × 391 px region)	—	12345678910 <table><tr><th>$l \backslash m$</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>2</td><td>−</td><td>−</td><td>−</td><td>−</td><td>−</td><td>−</td><td>−</td></tr><tr><td>3</td><td>−</td><td>−</td><td>−</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>4</td><td>−</td><td>−</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>5</td><td>−</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>6</td><td>−</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>7</td><td>−</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>8</td><td>−</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr></table> 12345678910 Figure 10.17. The irreducibility of $g \in G$ for $l \leq 8$.	$l \backslash m$	2	3	4	5	6	7	8	2	−	−	−	−	−	−	−	3	−	−	−	+	+	+	+	4	−	−	+	+	+	+	+	5	−	+	+	+	+	+	+	6	−	+	+	+	+	+	+	7	−	+	+	+	+	+	+	8	−	+	+	+	+	+	+
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8	−	+	+	+	+	+	+																																																														
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																																					

Image regions — rendered against scan

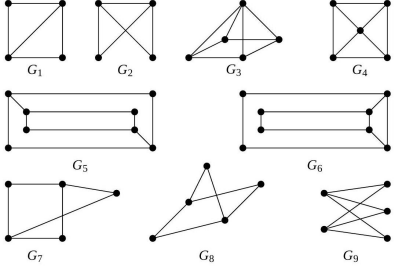
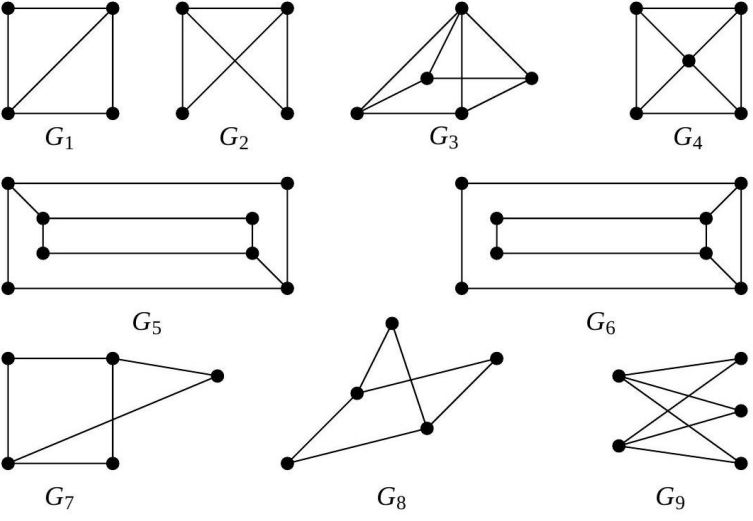
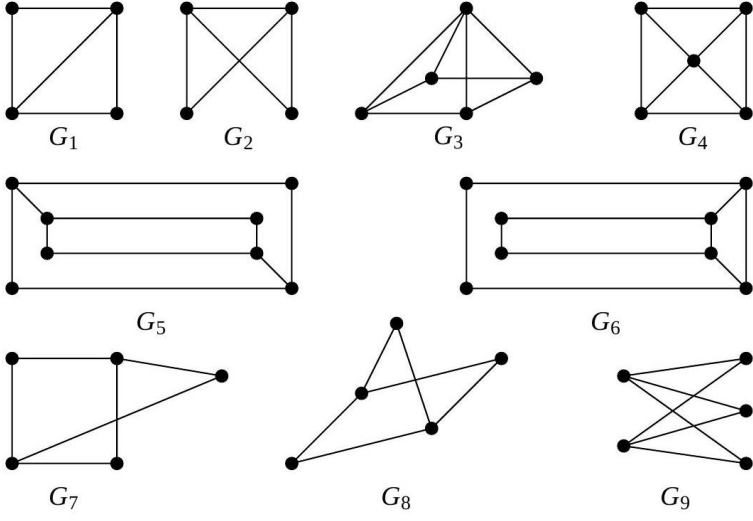
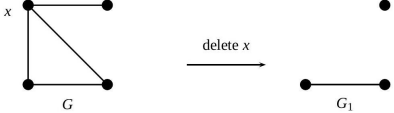
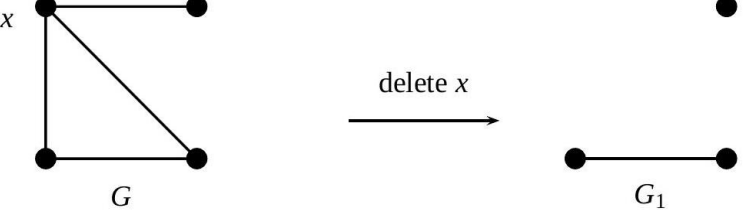
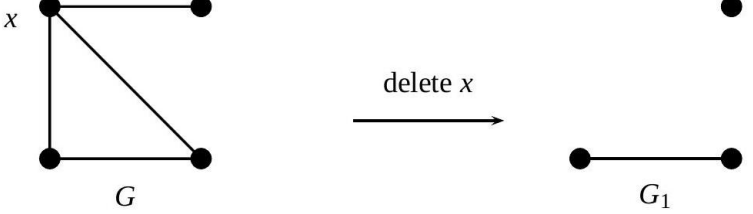
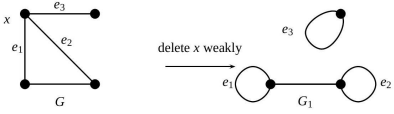
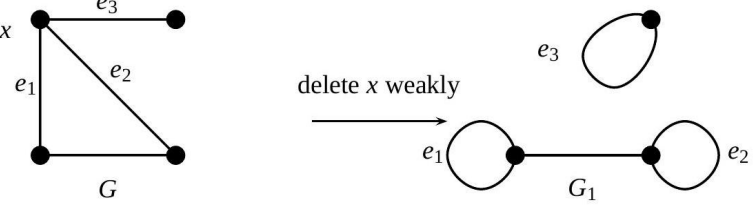
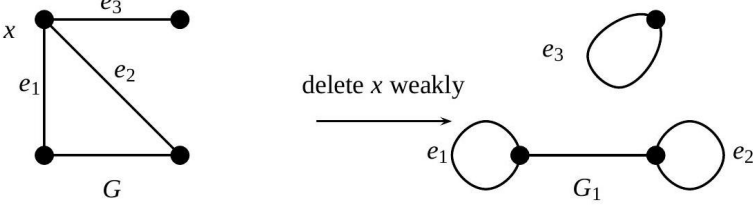
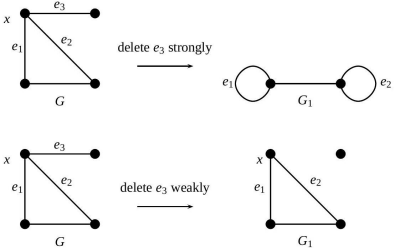
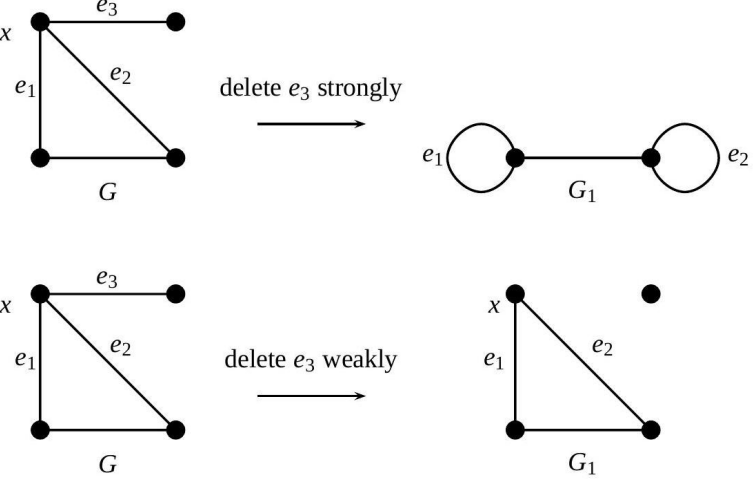
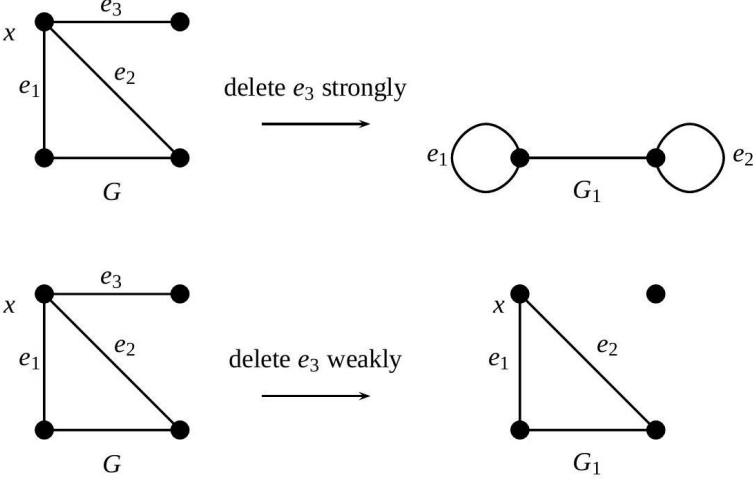
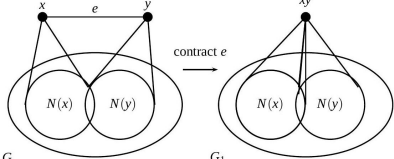
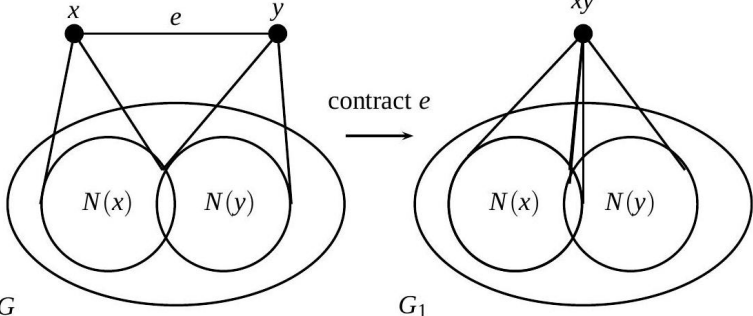
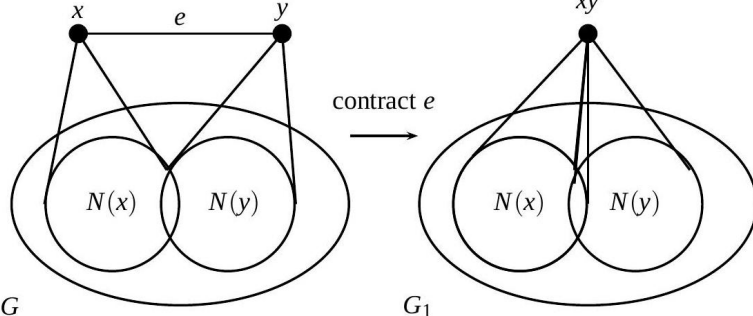
Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

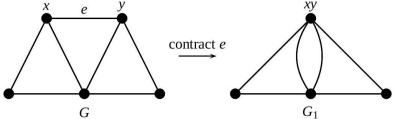
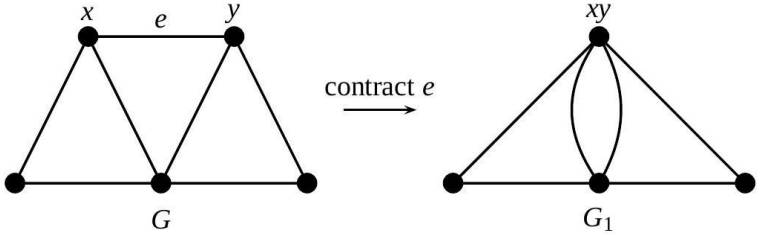
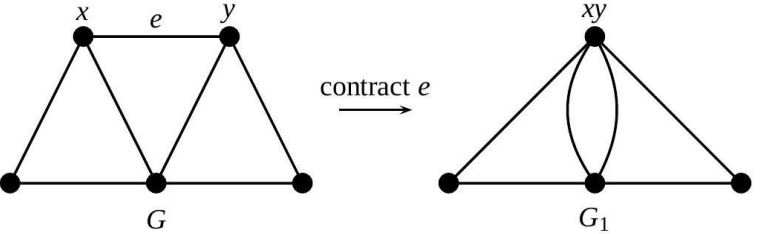
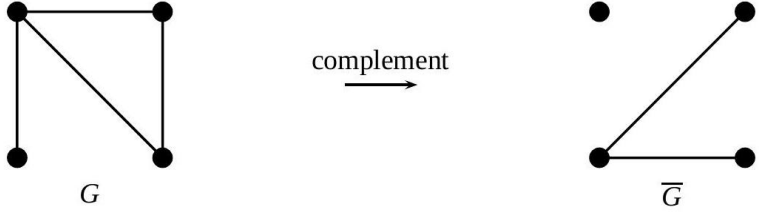
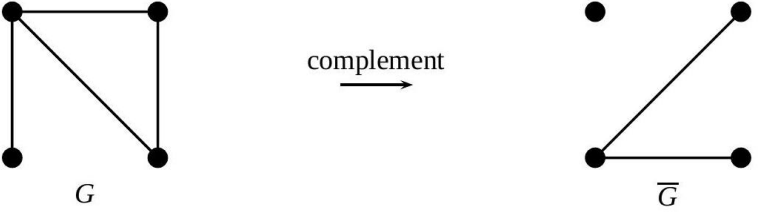
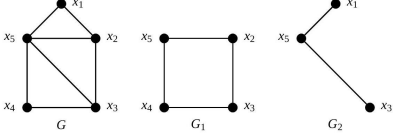
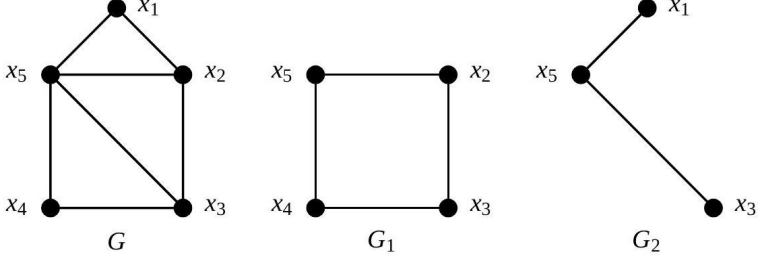
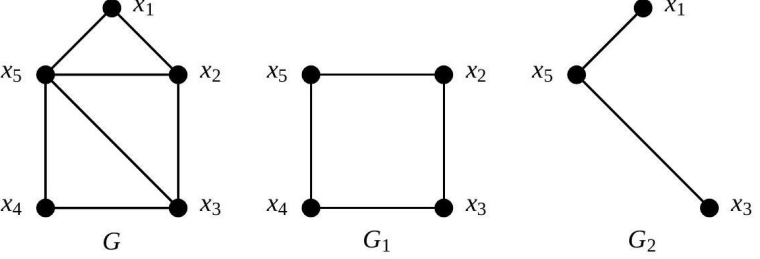
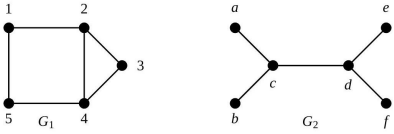
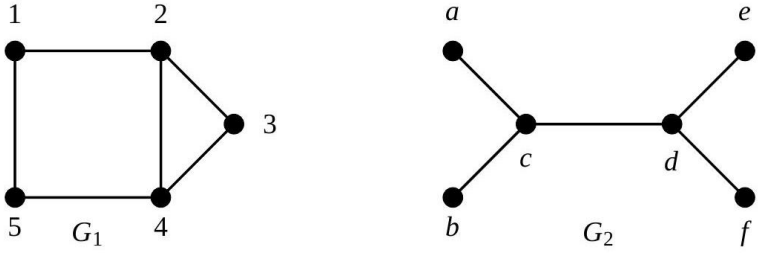
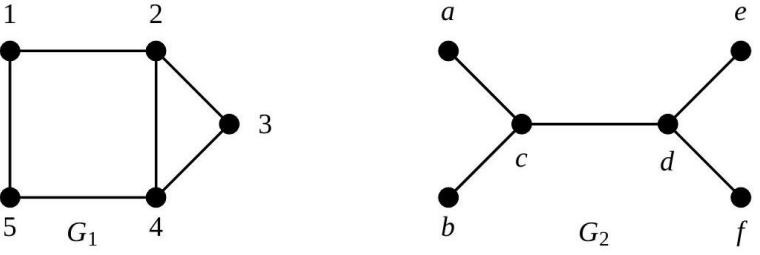
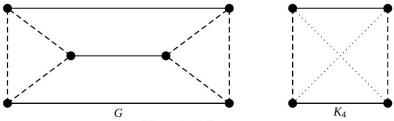
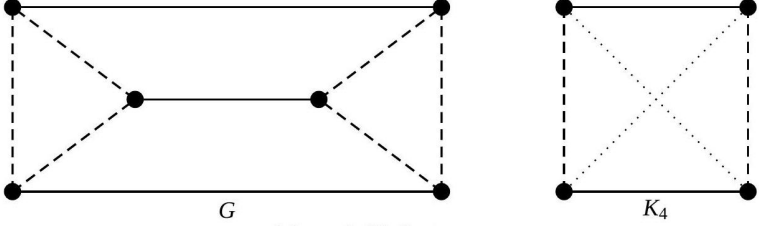
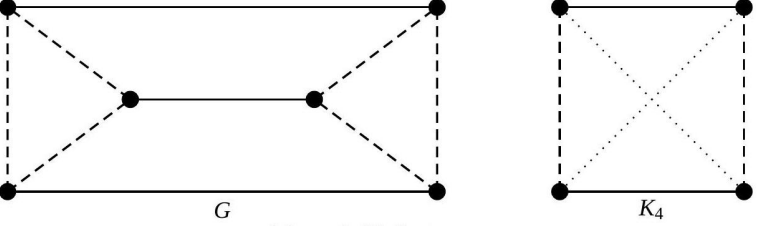
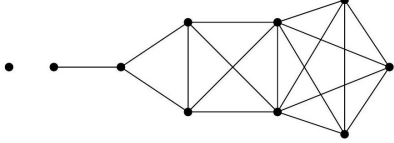
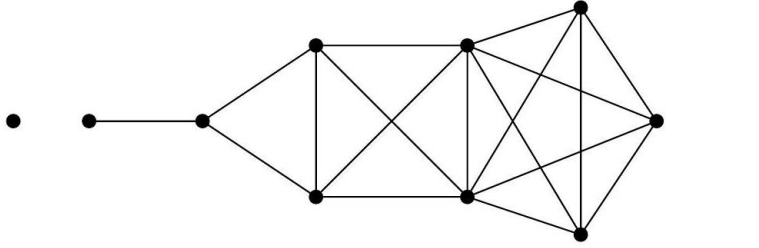
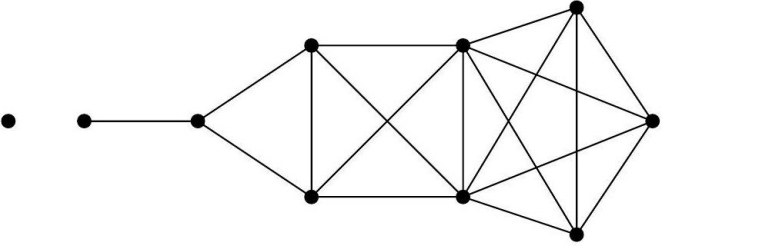
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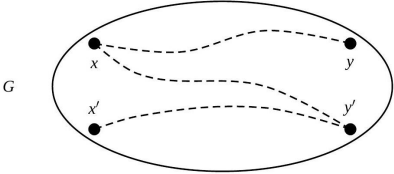
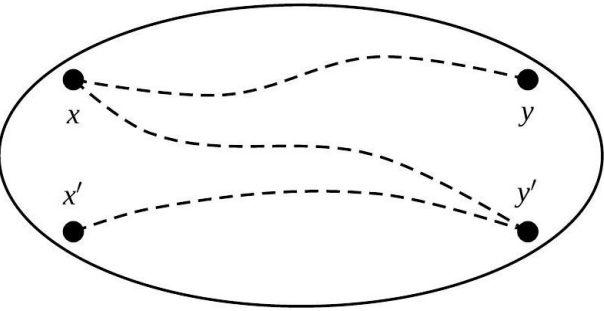
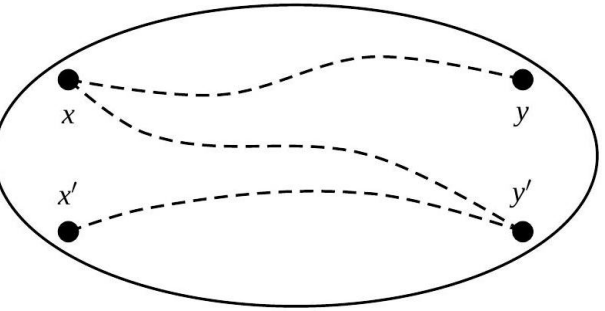
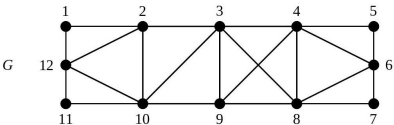
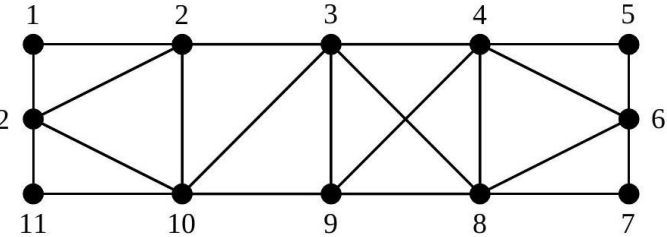
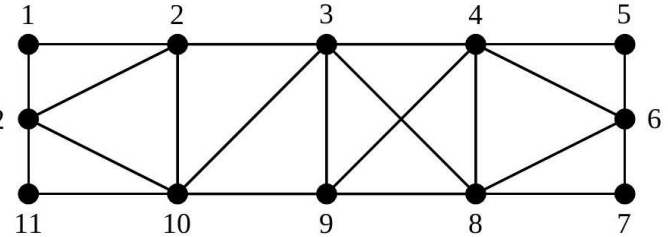
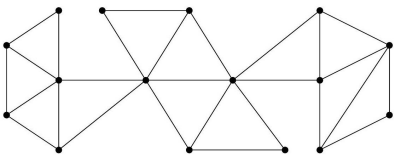
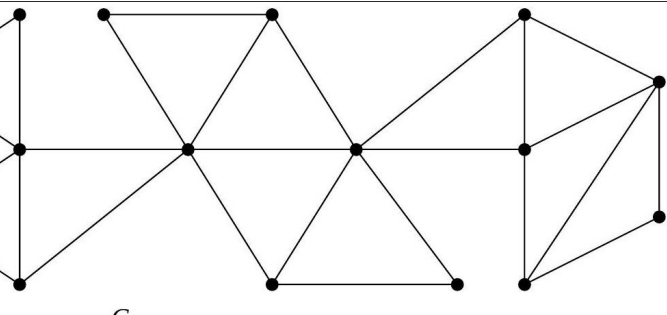
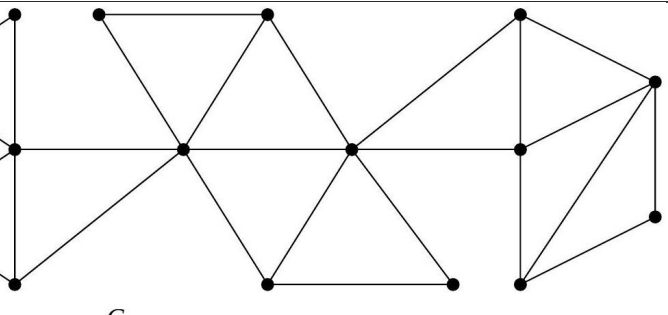
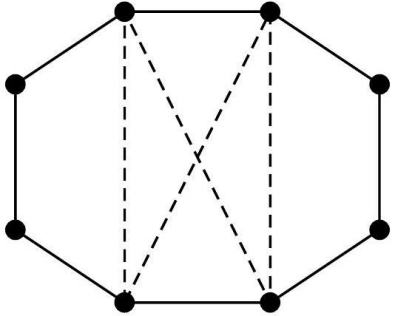
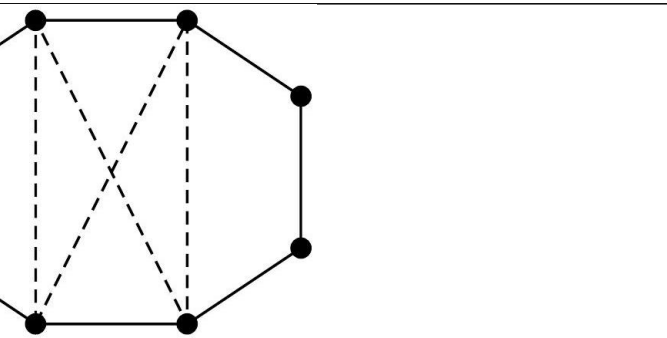
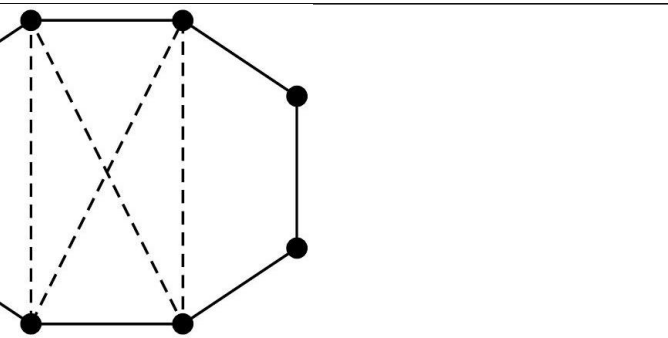
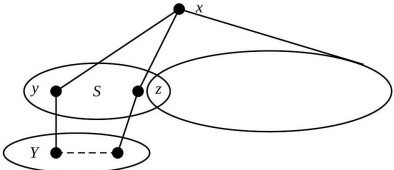
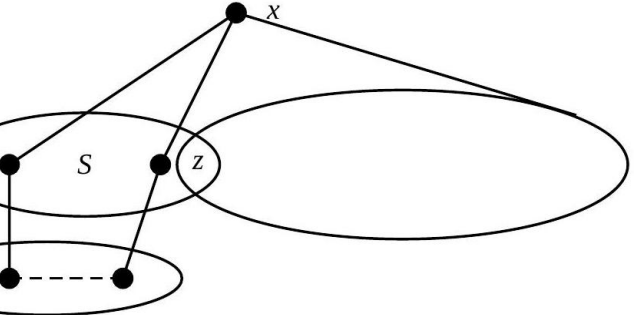
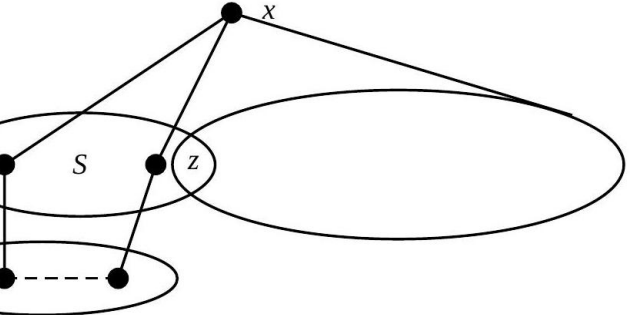
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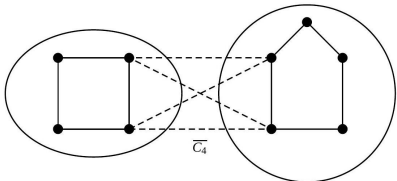
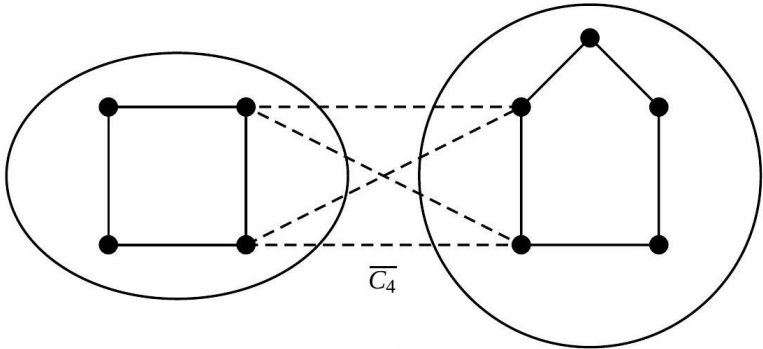
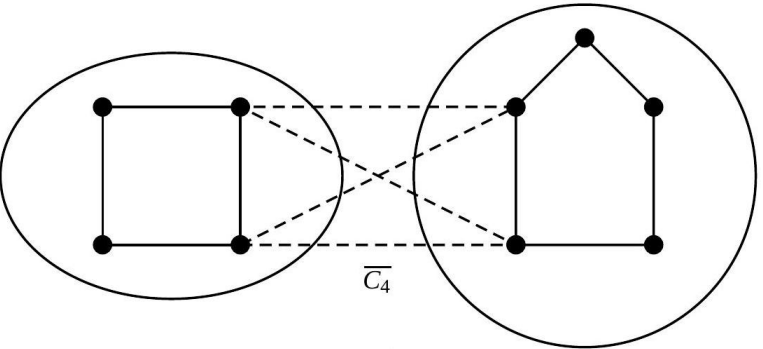
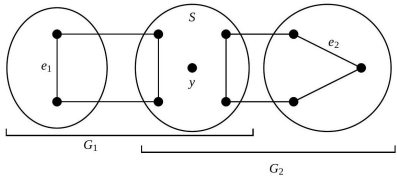
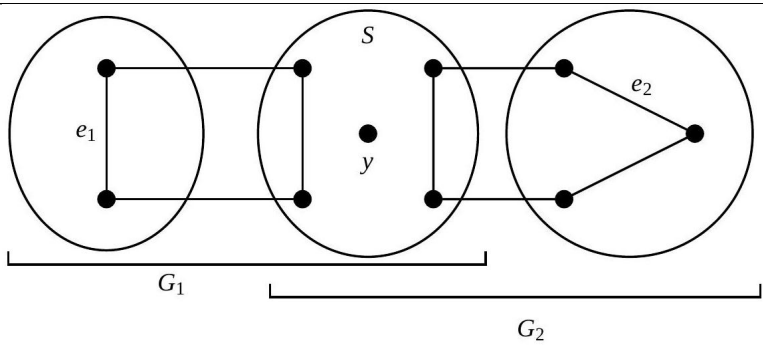
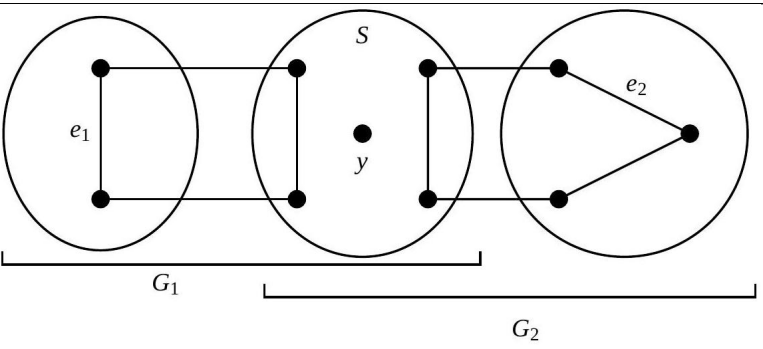
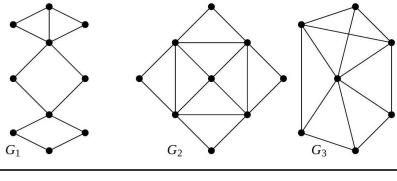
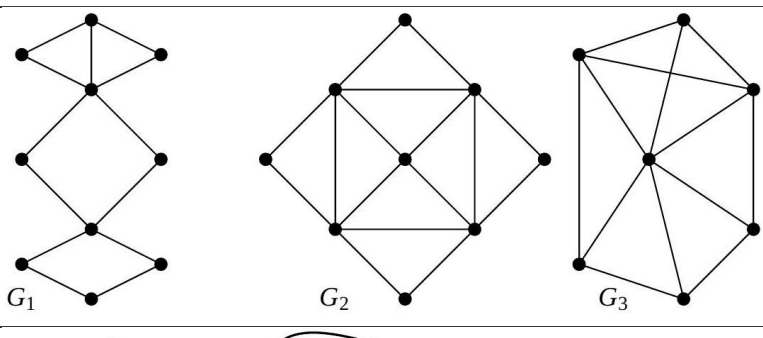
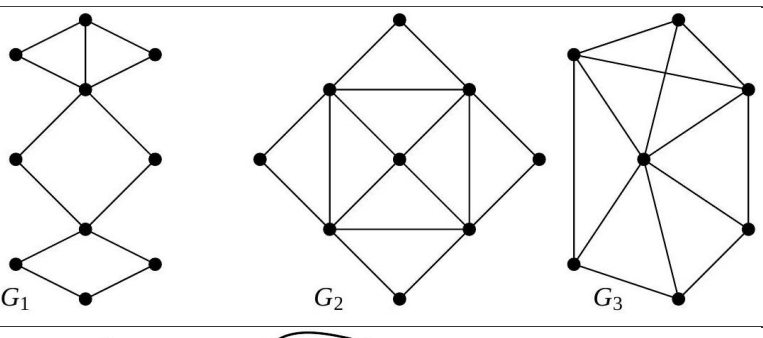
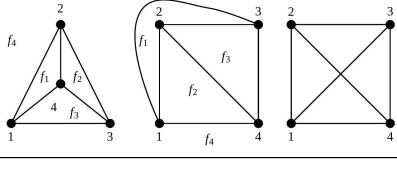
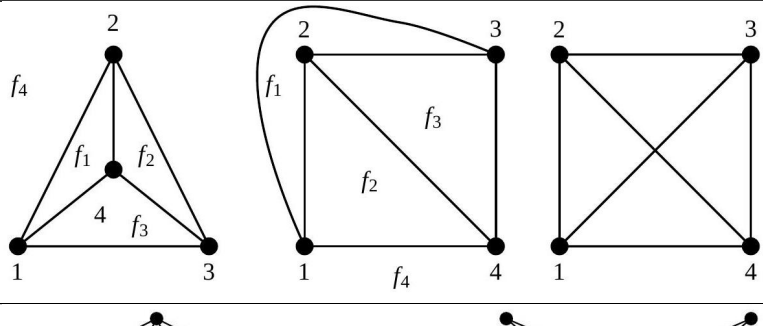
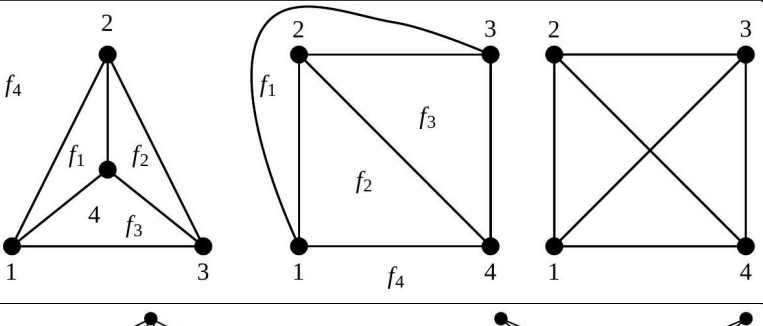
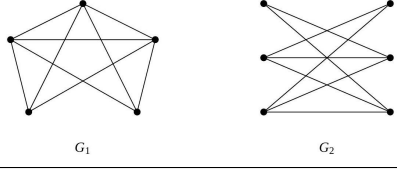
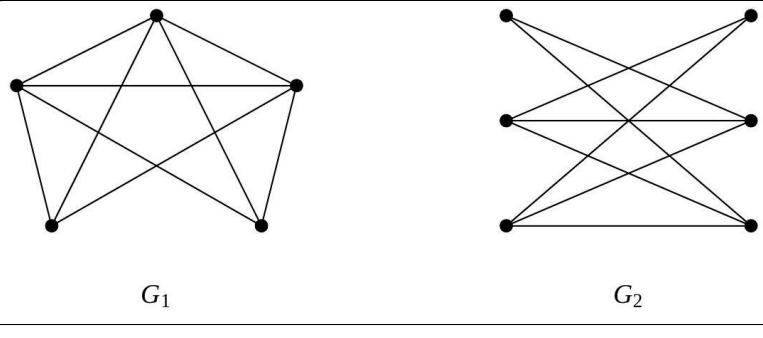
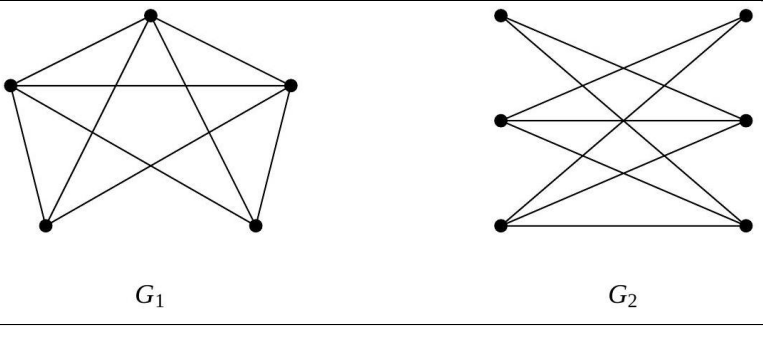
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voloshin-hypergraph_DIA_068	068	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-068_471_1227_210_269.jpg			
voloshin-hypergraph_DIA_068	068	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-068_277_1191_1722_287.jpg			
voloshin-hypergraph_DIA_070	070	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-070_316_935_1421_362.jpg			
voloshin-hypergraph_DIA_072	072	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-072_437_1121_1156_321.jpg			
voloshin-hypergraph_DIA_074	074	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-074_206_602_204_581.jpg			
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voloshin-hypergraph_DIA_0057	076	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-076_404_911_761_370.jpg			
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voloshin-hypergraph_DIA_0057	079	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-079_419_948_691_453.jpg			

voloshin-hypergraph_DIA_0030	080	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-080_505_1093_761_337.jpg			
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voloshin-hypergraph_DIA_0030	081	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-081_437_1079_1358_363.jpg			
voloshin-hypergraph_DIA_0030	084	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-084_451_1180_225_317.jpg			
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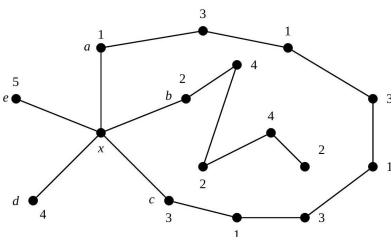
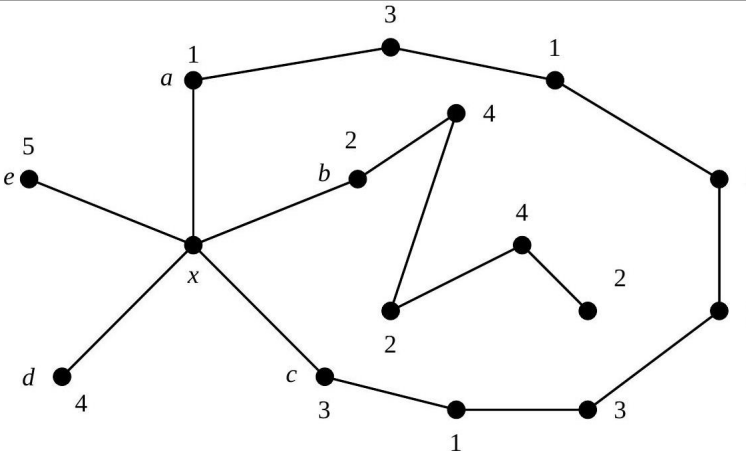
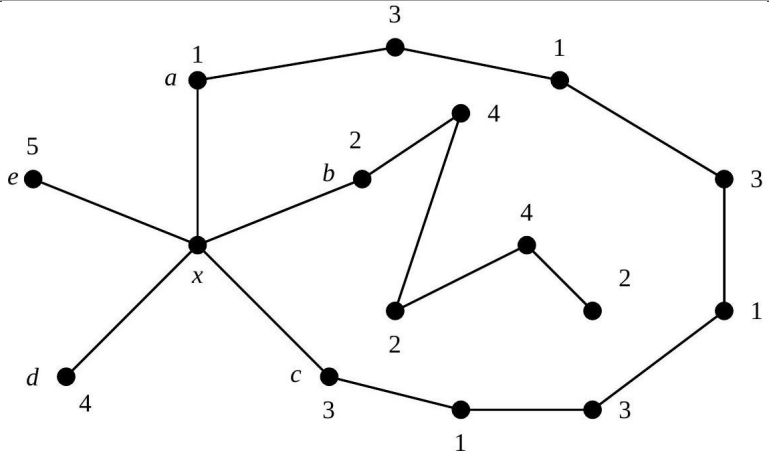
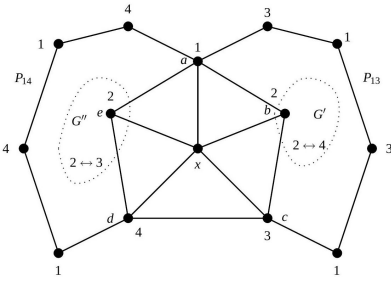
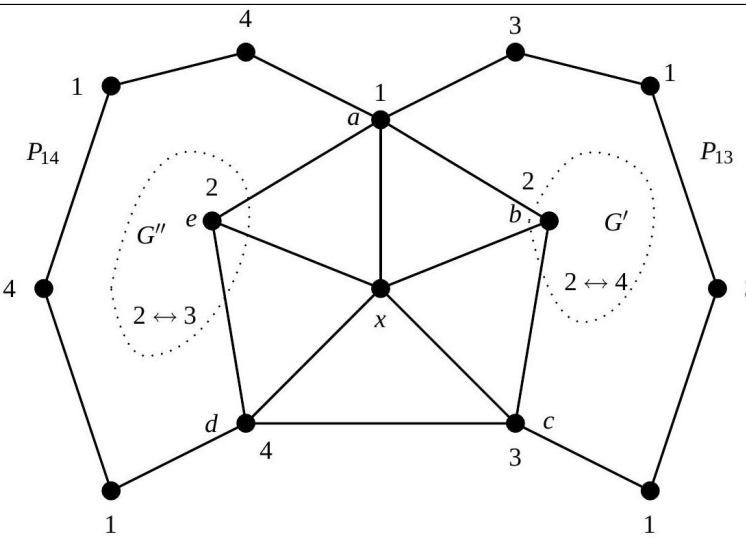
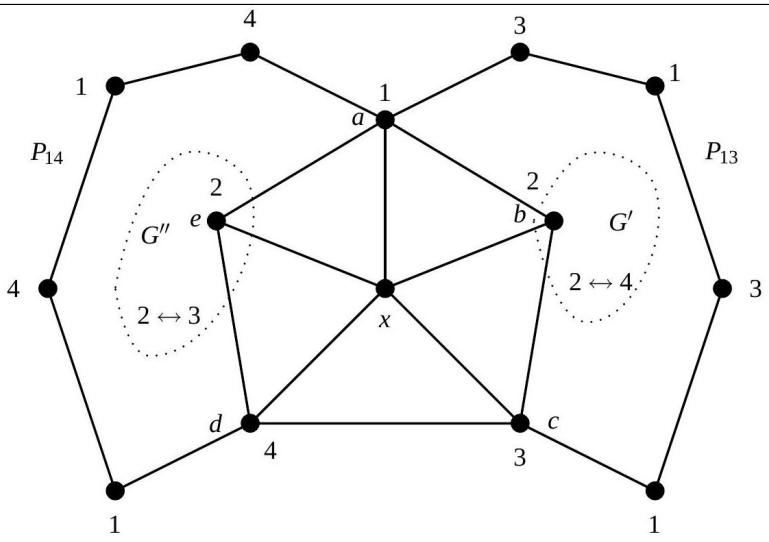
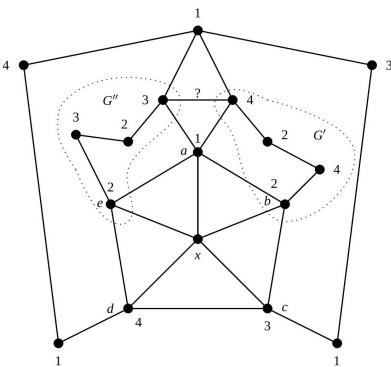
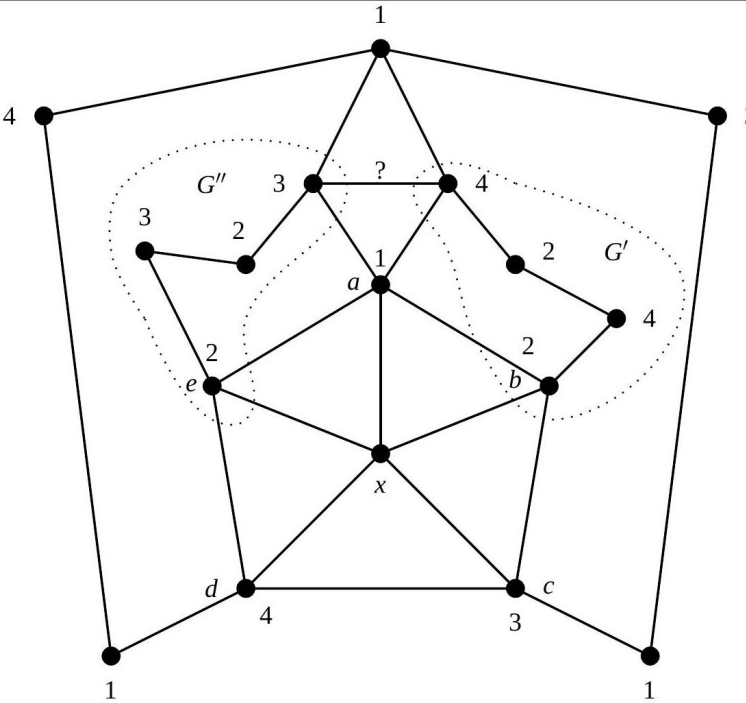
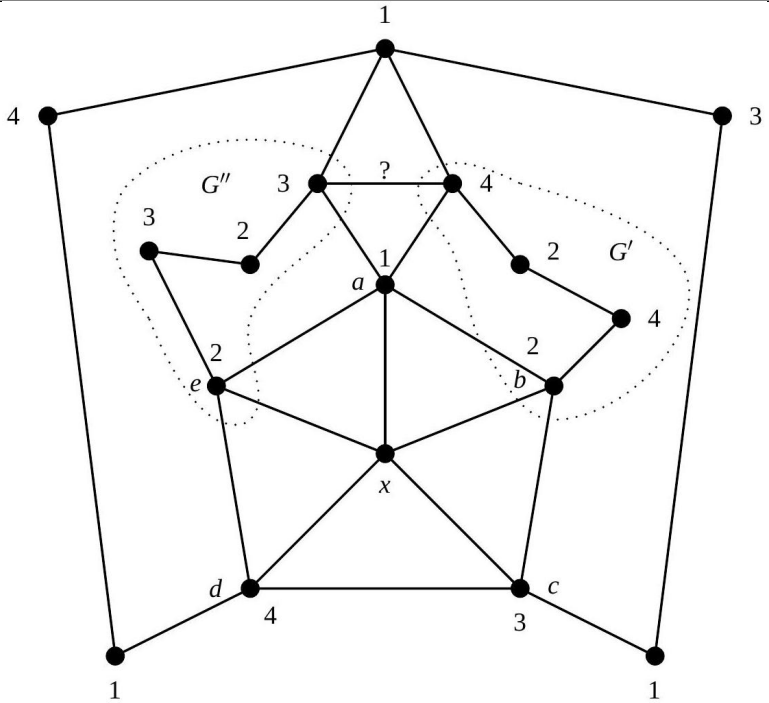
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voloshin-hypergraph_DIA_0088	088	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-088_1543_1245_233_241.jpg	 	 	
voloshin-hypergraph_DIA_0091	091	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-091_437_1071_1232_321.jpg	 	 	

voloshin-hypergraph_DIA_0092	0092	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-092_363_606_1343_581.jpg			
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voloshin-hypergraph_DIA_0097	0097	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-097_624_1227_230_269.jpg			
voloshin-hypergraph_DIA_0099	0099	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-099_338_933_237_406.jpg			
voloshin-hypergraph_DIA_100	100	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-100_399_1093_233_365.jpg			

voloshin-hypergraph_DIA_011	101	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-101_474_1103_205_331.jpg			
voloshin-hypergraph_DIA_012	102	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-102_368_1082_984_365.jpg			
voloshin-hypergraph_DIA_013	103	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-103_686_1227_238_269.jpg			
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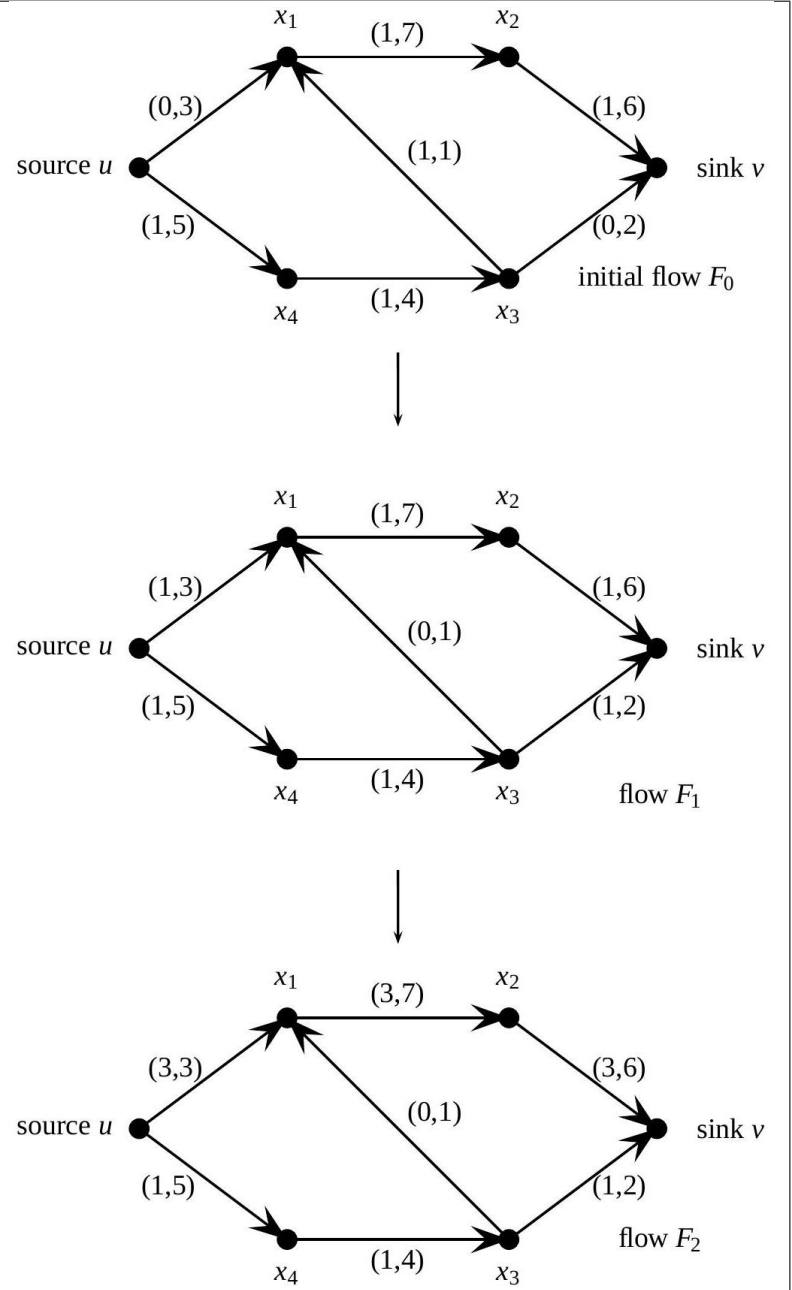
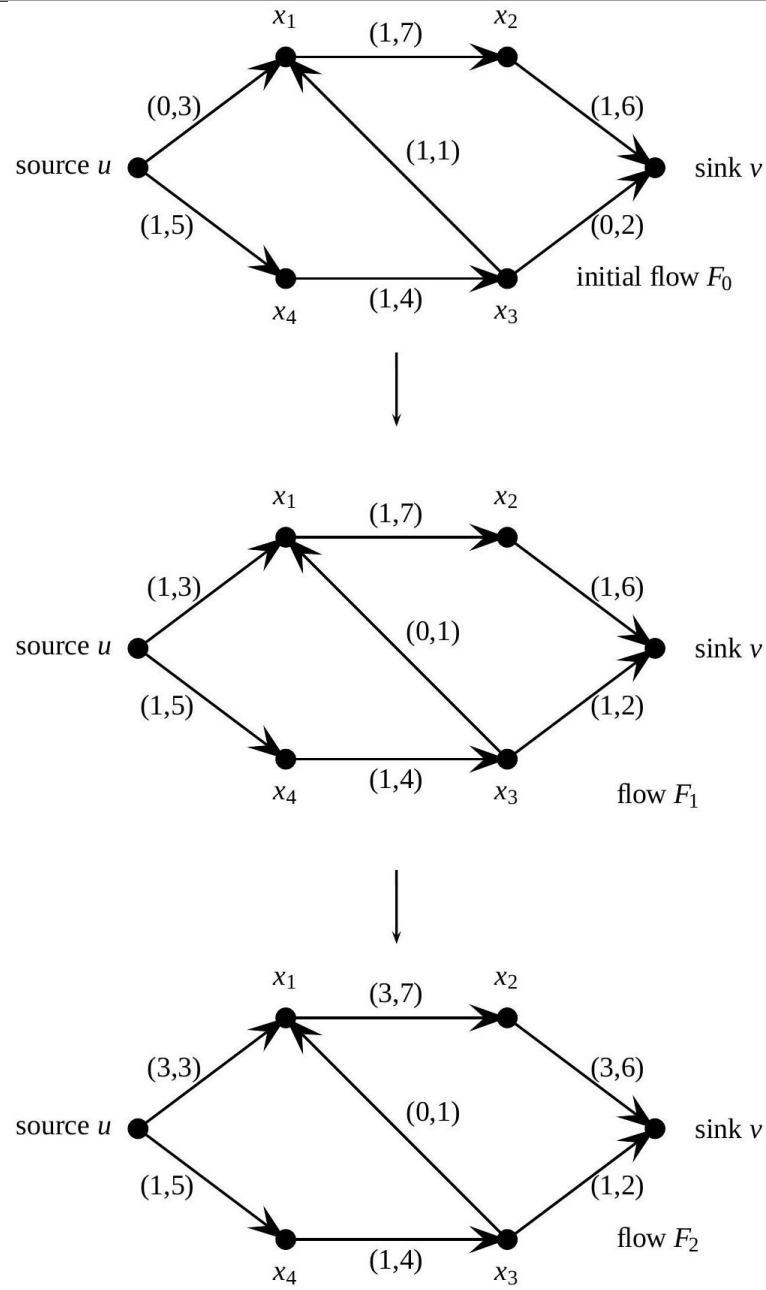
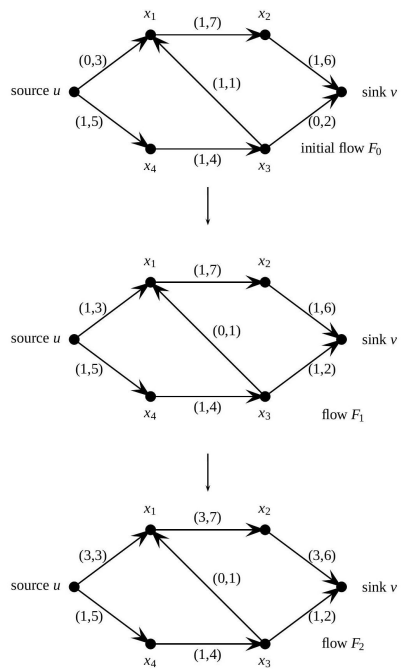
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voloshin-hypergraph_DIA_0111	107	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-107_790_1175_236_269.jpg	 $E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$	 $E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$	 $E_3 - 2E_2 + E_1 \Rightarrow P(G, \lambda) = \lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda - 1)^2$
voloshin-hypergraph_DIA_0112	108	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-108_378_1196_1315_285.jpg		 $G_1 = G_{X_1 \cup S}$ $G_2 = G_{X_2 \cup S}$	 $G_1 = G_{X_1 \cup S}$ $G_2 = G_{X_2 \cup S}$
voloshin-hypergraph_DIA_0113	109	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-109_557_1085_816_313.jpg			
voloshin-hypergraph_DIA_0114	111	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-111_414_1075_214_319.jpg			
voloshin-hypergraph_DIA_0115	112	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-112_316_935_1473_362.jpg			

voloshin-hypergraph_DIA_0113	113	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-113_316_935_536_362.jpg			
voloshin-hypergraph_DIA_0115	115	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-115_782_999_917_383.jpg			
voloshin-hypergraph_DIA_0117	117	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-117_316_924_1372_370.jpg			
voloshin-hypergraph_DIA_0119	119	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-119_469_911_205_430.jpg			

voloshin-hypergraph_DIA_0182	120	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-120_681_1144_238_295.jpg			
voloshin-hypergraph_DIA_0182	121	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-121_779_1116_539_324.jpg			
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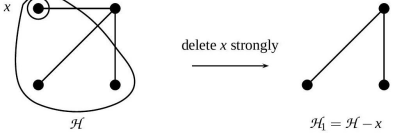
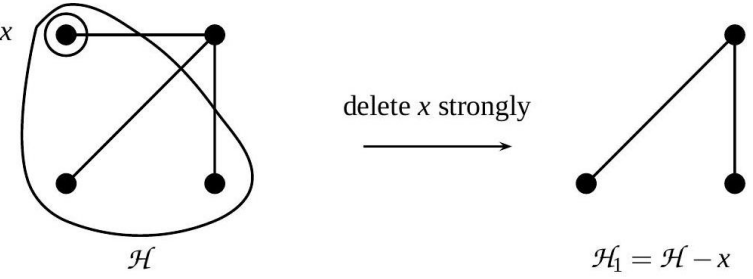
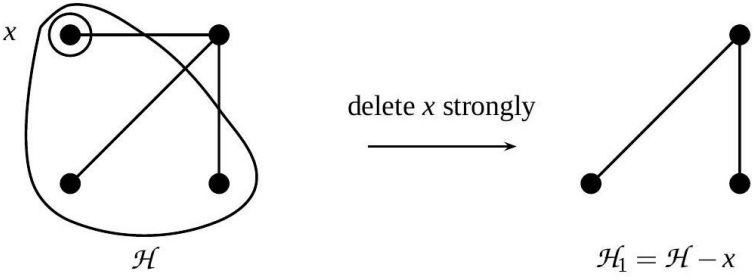
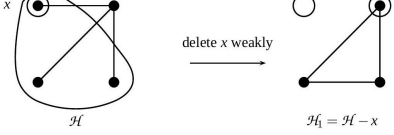
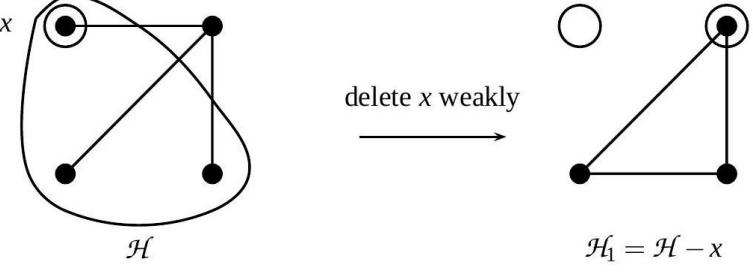
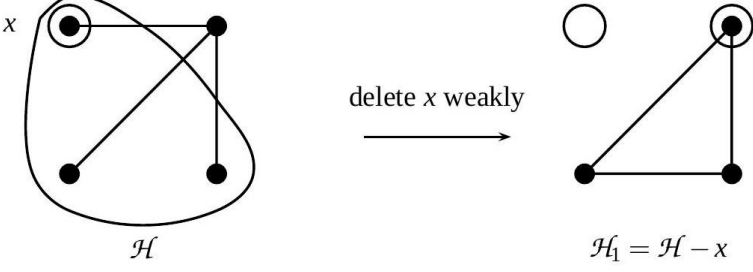
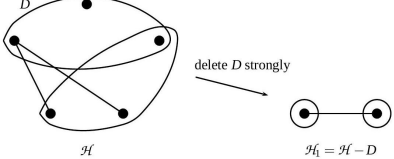
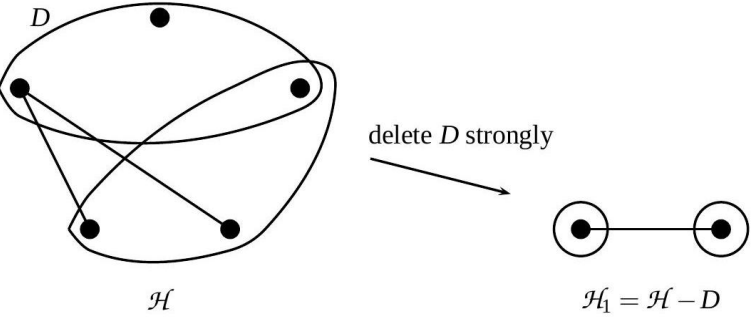
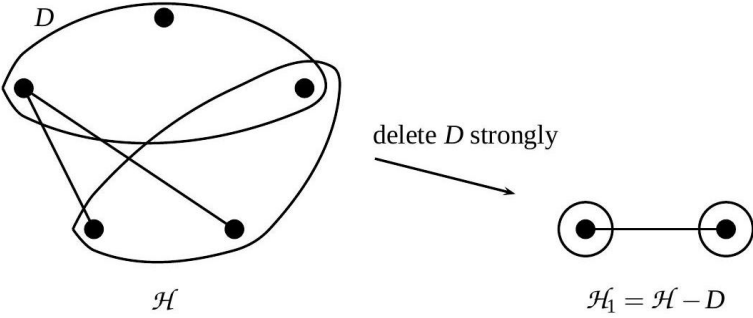
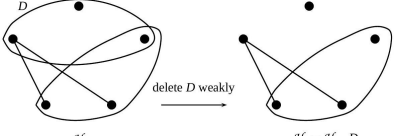
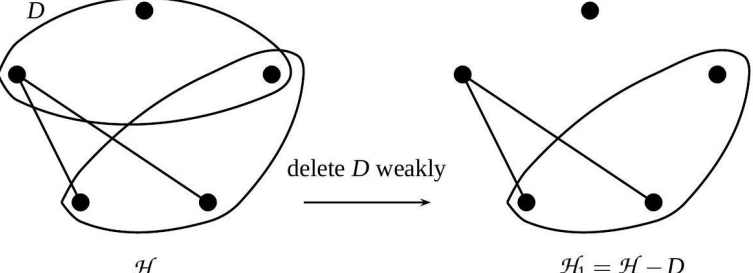
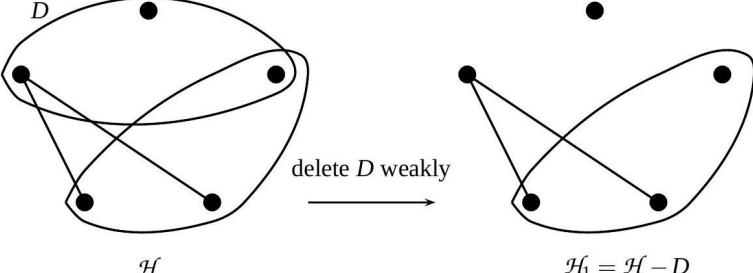
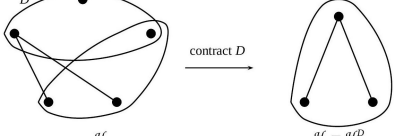
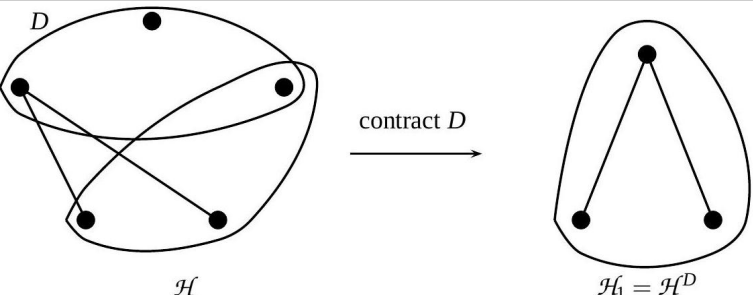
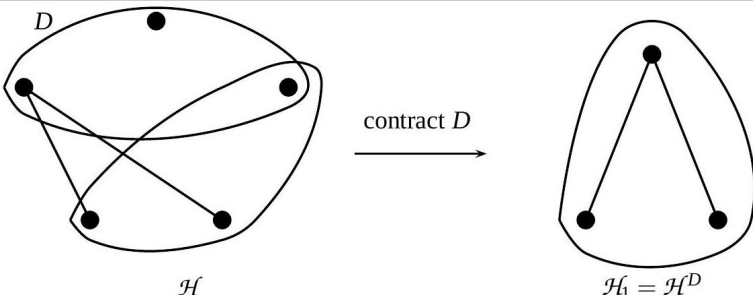
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voloshin-hypergraph_DIA_0192	126	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-126_557_1116_909_318.jpg			

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voloshin-hypergraph_DIA_0141	141	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-141_533_1033_1315_365.jpg			



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voloshin-hypergraph_DIA_0164	164	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-164_451_1072_228_352.jpg			
voloshin-hypergraph_DIA_0164	164	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-164_451_1181_1271_303.jpg			
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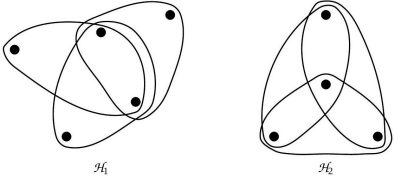
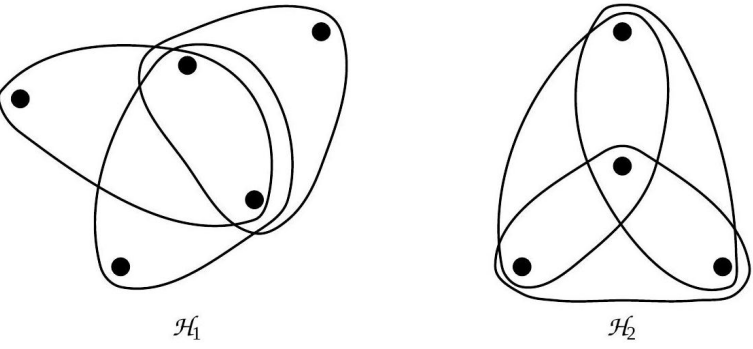
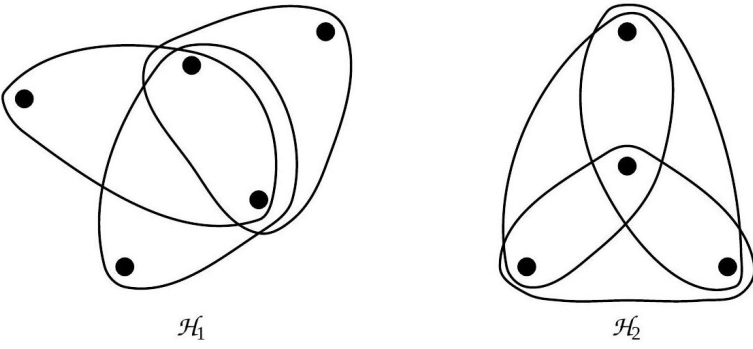
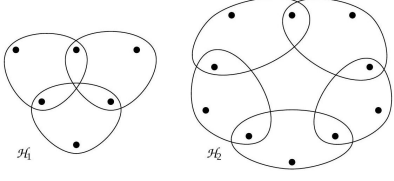
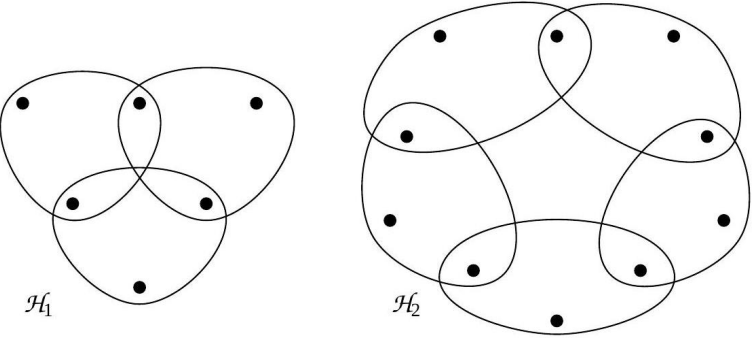
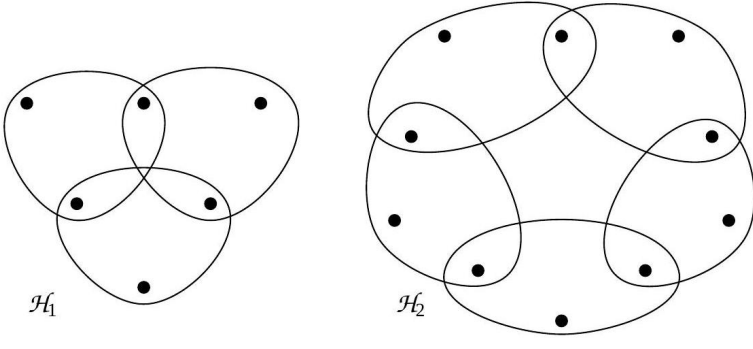
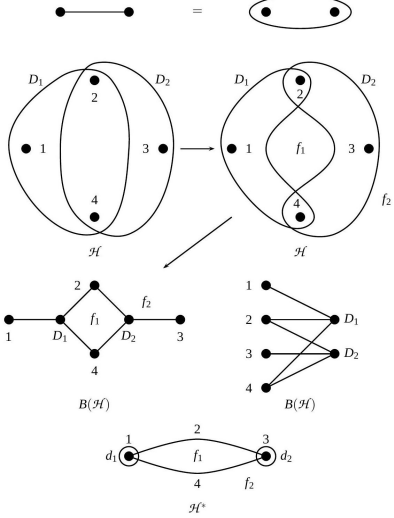
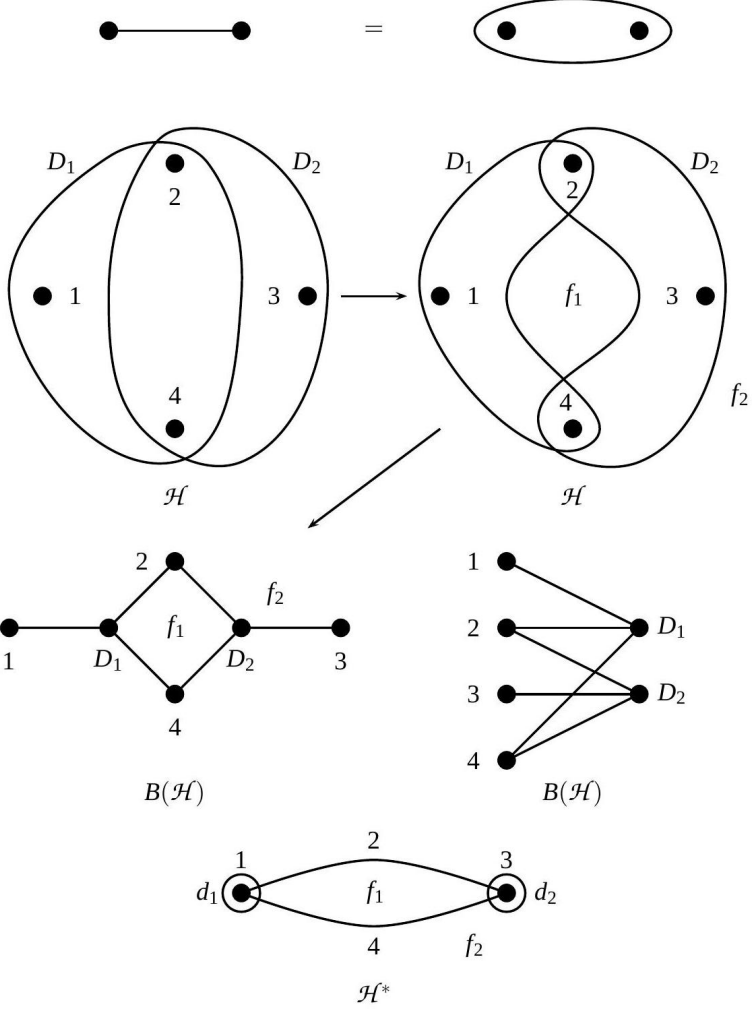
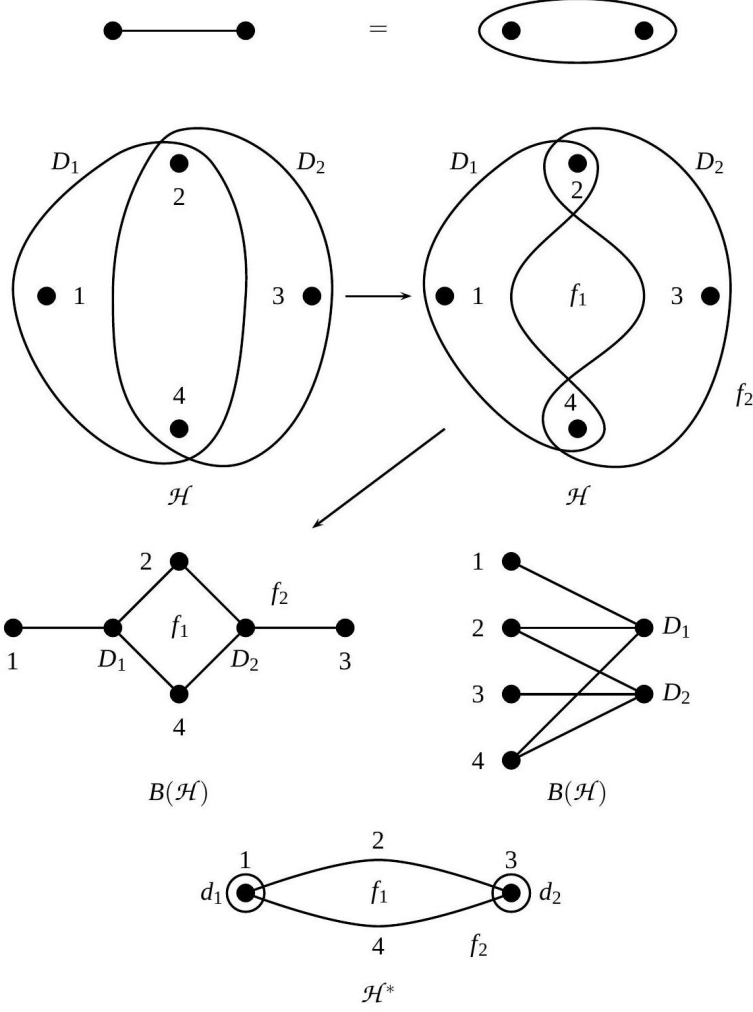
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voloshin-hypergraph_DIA_01171	171	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-171_466_1126_590_298.jpg		
voloshin-hypergraph_DIA_01172	172	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-172_469_1142_217_282.jpg		

voloshin-hypergraph_DIA_0174	174	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-174_1266_1313_220_249.jpg	$\mathcal{H} = K_3$ not conformal, not Helly $\mathcal{H}_1$ - conformal, not Helly $\mathcal{H}_2 \cong \mathcal{H}_1^*$ - Helly, not conformal $\mathcal{H}_3 \cong \mathcal{H}_3^*$ - Helly, conformal	$\mathcal{H} = K_3$ not conformal, not Helly $\mathcal{H}_1$ - conformal, not Helly $\mathcal{H}_2 \cong \mathcal{H}_1^*$ - Helly, not conformal $\mathcal{H}_3 \cong \mathcal{H}_3^*$ - Helly, conformal
voloshin-hypergraph_DIA_0178	178	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-178_746_1245_256_267.jpg	T C_4	T C_4
voloshin-hypergraph_DIA_0179	179	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-179_502_1126_246_347.jpg	$\mathcal{H}_1$ $\mathcal{H}_2$	$\mathcal{H}_1$ $\mathcal{H}_2$
voloshin-hypergraph_DIA_0180	180	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-180_406_1139_388_274.jpg	$\mathcal{H}$ $\mathcal{H}_0$	$\mathcal{H}$ $\mathcal{H}_0$

voloshin-hypergraph_DIA_0181	181	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-181_406_1033_207_381.jpg			
voloshin-hypergraph_DIA_0181	181	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-181_406_1033_1271_381.jpg			
voloshin-hypergraph_DIA_0182	182	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-182_404_1240_225_267.jpg			
voloshin-hypergraph_DIA_0183	183	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-183_269_1186_841_272.jpg			
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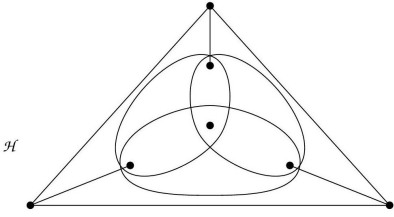
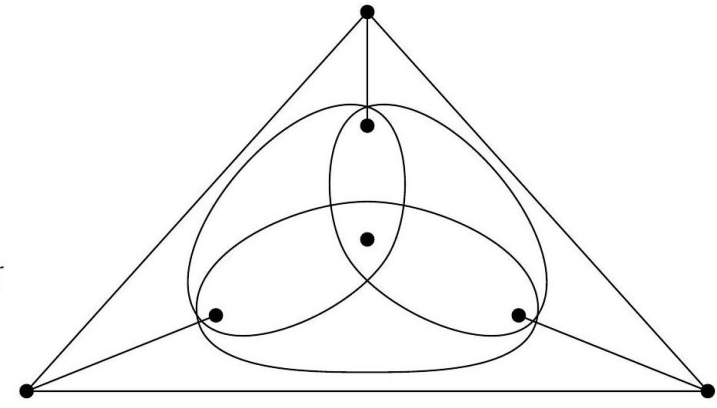
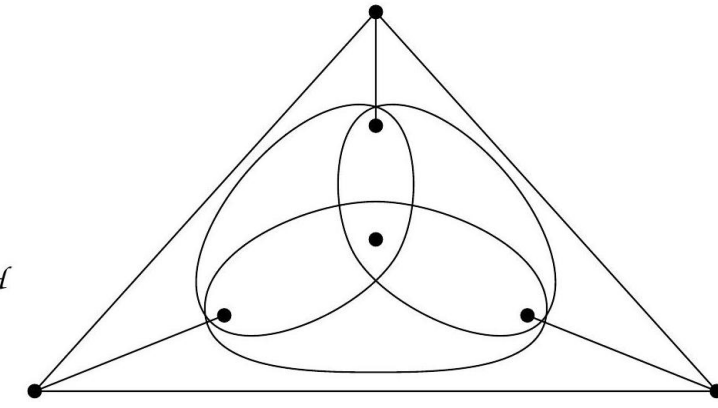
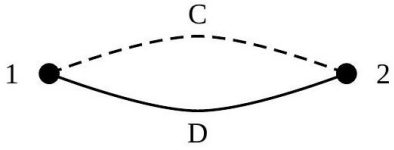
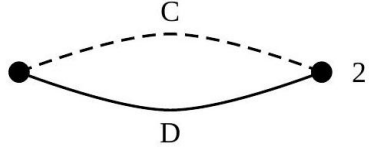
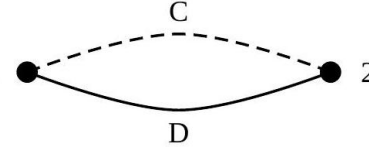
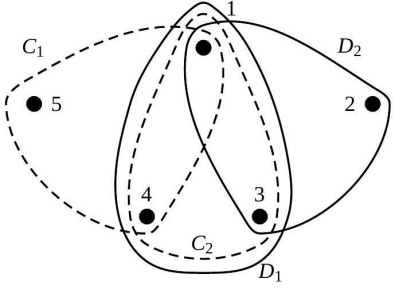
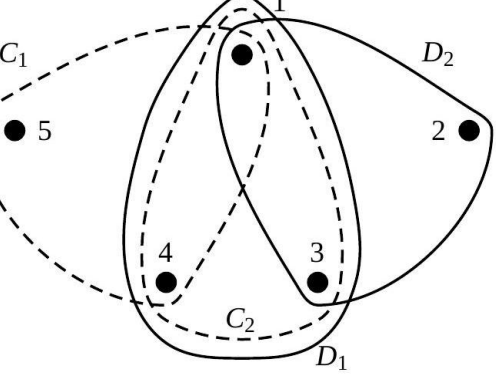
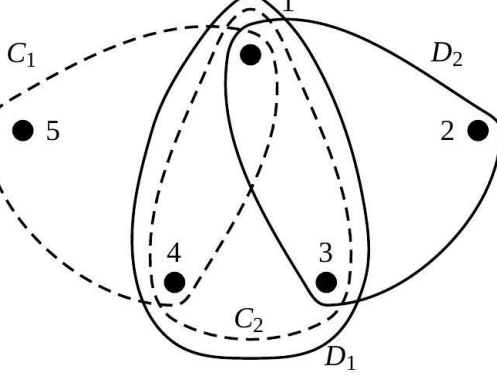
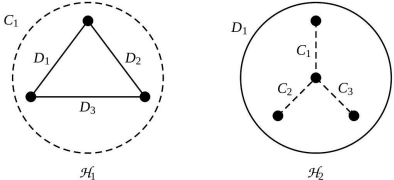
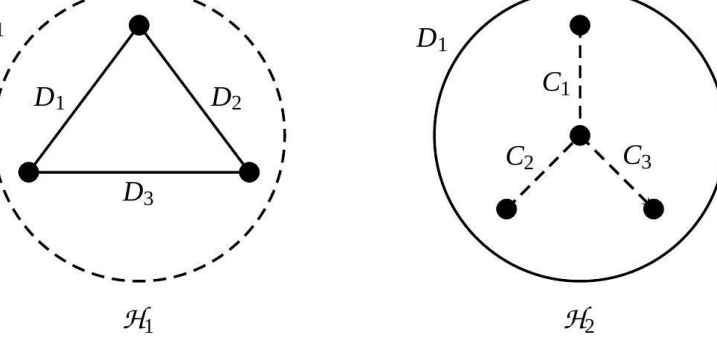
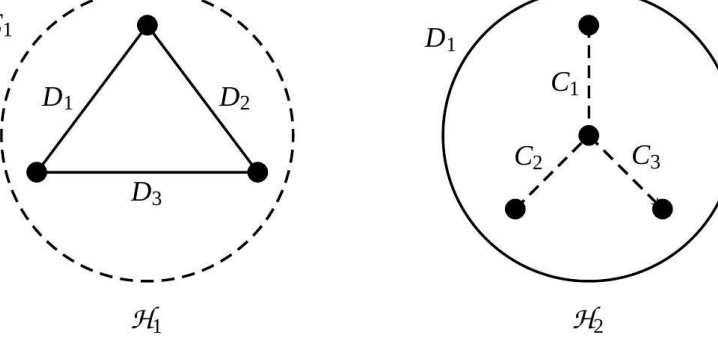
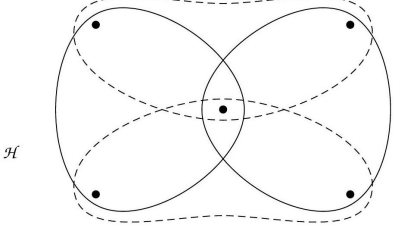
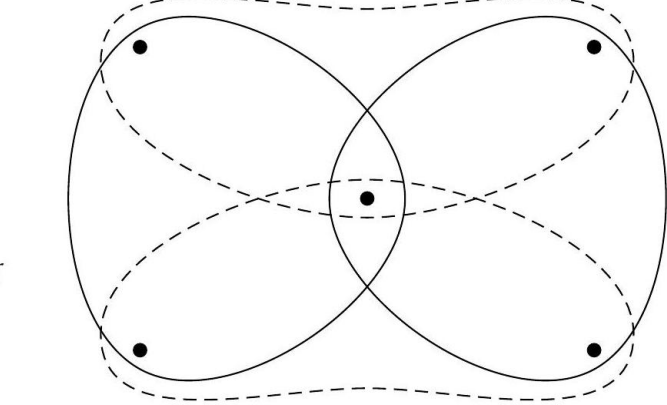
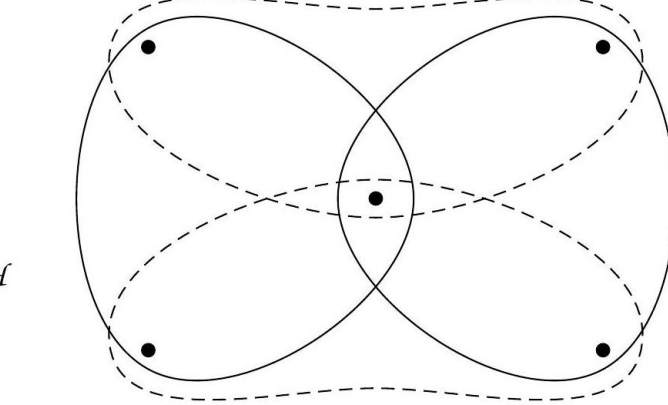
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voloshin-hypergraph_DIA_0188	188	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-188_803_1258_217_264.jpg			
voloshin-hypergraph_DIA_0189	189	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-189_479_1113_751_342.jpg			
voloshin-hypergraph_DIA_0191	191	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-191_927_1227_197_269.jpg			

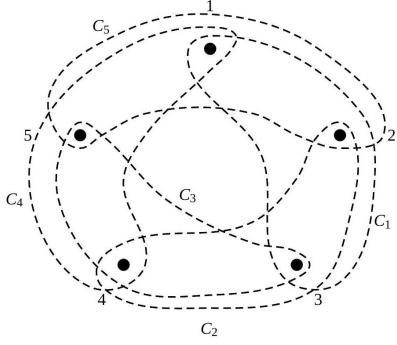
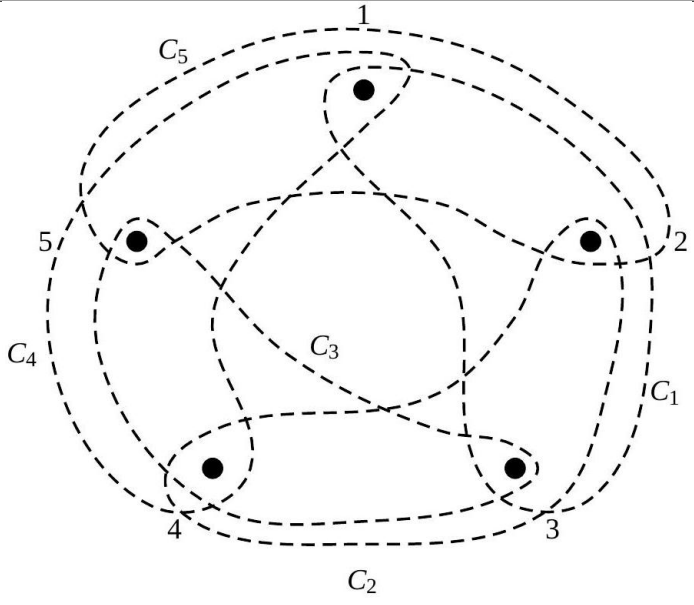
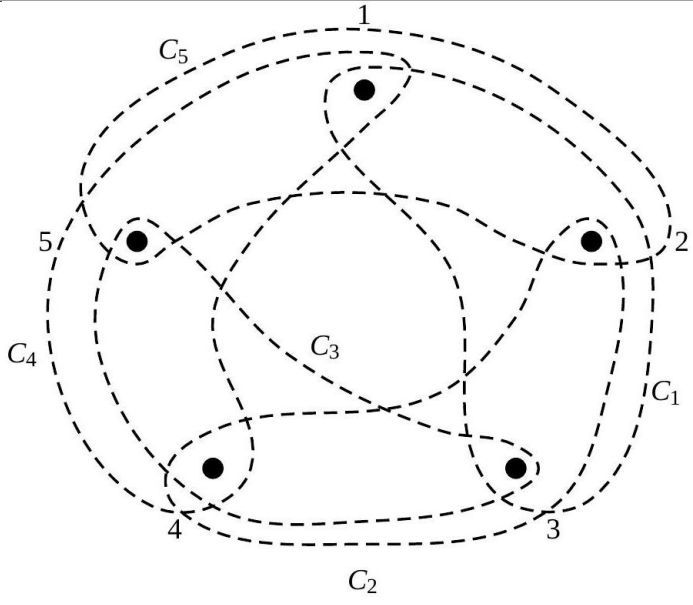
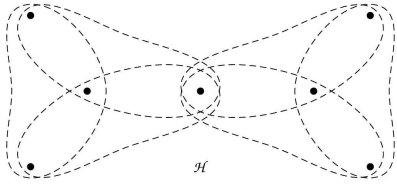
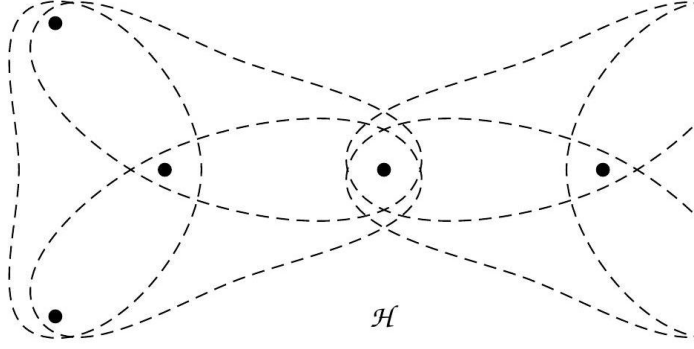
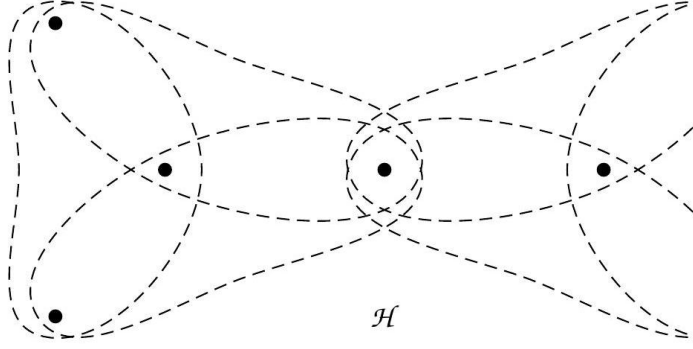
voloshin-hypergraph_DIA_013	194	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-194_945_1194_197_277.jpg			
voloshin-hypergraph_DIA_013	196	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-196_432_1129_792_321.jpg			
voloshin-hypergraph_DIA_013	198	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-198_624_743_212_505.jpg			
voloshin-hypergraph_DIA_013	199	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-199_316_1230_1465_277.jpg			

voloshin-hypergraph_DIA_0200	200	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-200_507_1124_981_300.jpg			
voloshin-hypergraph_DIA_0202	202	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-202_507_1126_948_321.jpg			
voloshin-hypergraph_DIA_0203	203	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-203_1515_1134_562_316.jpg			

voloshin-hypergraph_DIA_0230	206	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-206_1623_1054_236_347.jpg	The diagram shows a hypergraph with 4 vertices labeled 1, 2, 3, and 4. There are six hyperedges: f_1 (vertices 1, 2), f_2 (vertices 1, 4), f_3 (vertices 1, 2, 3, 4), f_4 (vertices 2, 3), f_5 (vertices 3, 4), and f_6 (vertices 4, 1). Below the hypergraph is a graph with the same 4 vertices and edges connecting 1-2, 1-4, 2-3, 3-4, and 1-3.	The diagram shows a hypergraph with 4 vertices labeled 1, 2, 3, and 4. There are six hyperedges: f_1 (vertices 1, 2), f_2 (vertices 1, 4), f_3 (vertices 1, 2, 3, 4), f_4 (vertices 2, 3), f_5 (vertices 3, 4), and f_6 (vertices 4, 1). Below the hypergraph is a graph with the same 4 vertices and edges connecting 1-2, 1-4, 2-3, 3-4, and 1-3.	The diagram shows a hypergraph with 4 vertices labeled 1, 2, 3, and 4. There are six hyperedges: f_1 (vertices 1, 2), f_2 (vertices 1, 4), f_3 (vertices 1, 2, 3, 4), f_4 (vertices 2, 3), f_5 (vertices 3, 4), and f_6 (vertices 4, 1). Below the hypergraph is a graph with the same 4 vertices and edges connecting 1-2, 1-4, 2-3, 3-4, and 1-3.
voloshin-hypergraph_DIA_0230	207	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-207_507_1121_422_324.jpg	The diagram shows two hypergraphs, $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ has 4 vertices and 3 hyperedges. $\mathcal{H}_2$ has 6 vertices and 5 hyperedges.	The diagram shows two hypergraphs, $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ has 4 vertices and 3 hyperedges. $\mathcal{H}_2$ has 6 vertices and 5 hyperedges.	The diagram shows two hypergraphs, $\mathcal{H}_1$ and $\mathcal{H}_2$. $\mathcal{H}_1$ has 4 vertices and 3 hyperedges. $\mathcal{H}_2$ has 6 vertices and 5 hyperedges.

voloshin-hypergraph_DIA_0233	210	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-210_598_709_259_531.jpg			
voloshin-hypergraph_DIA_0241	213	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-213_458_924_225_417.jpg			
voloshin-hypergraph_DIA_0244	216	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-216_1323_1230_194_267.jpg			

voloshin-hypergraph_DIA_02417	217	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-217_523_974_492_365.jpg			
voloshin-hypergraph_DIA_02418	218	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-218_194_523_1686_621.jpg			
voloshin-hypergraph_DIA_02421	221	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-221_507_689_217_528.jpg			
voloshin-hypergraph_DIA_02422	222	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-222_482_1023_199_360.jpg			
voloshin-hypergraph_DIA_02423	223	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-223_536_917_236_365.jpg			

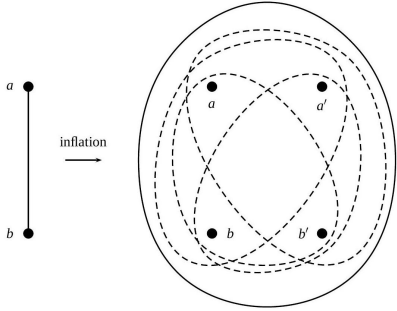
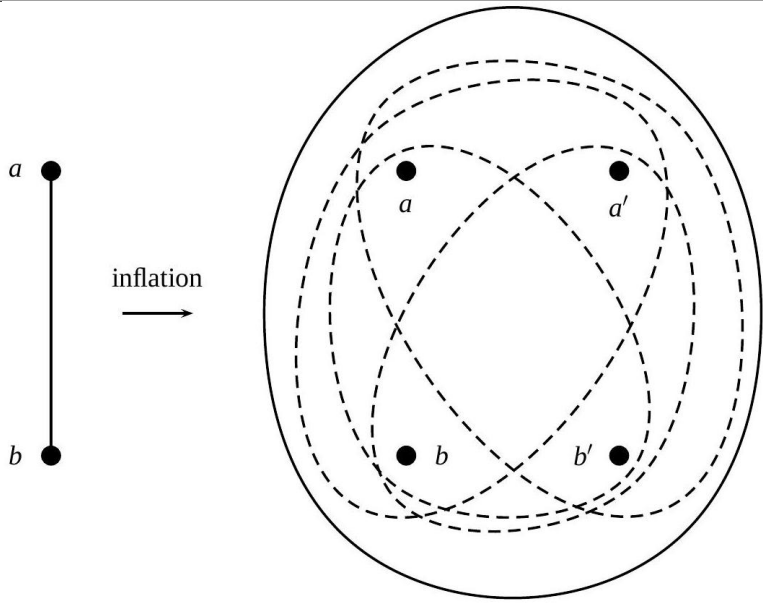
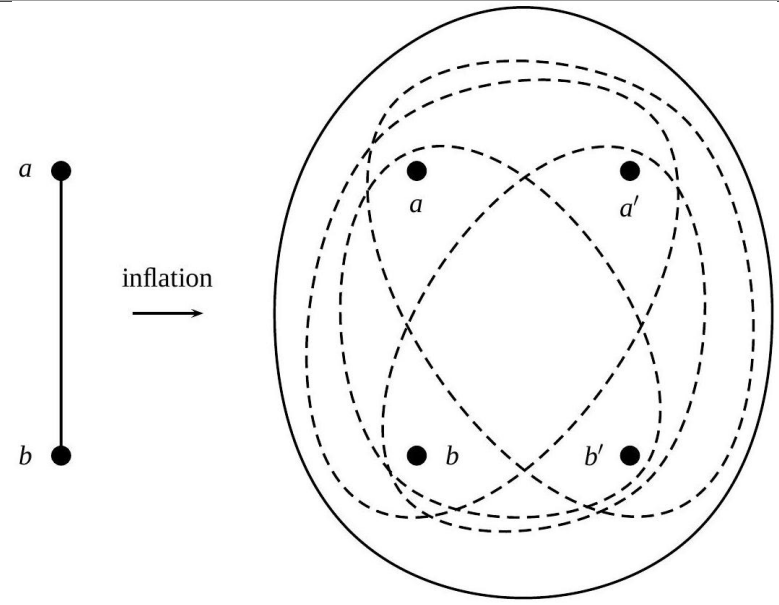
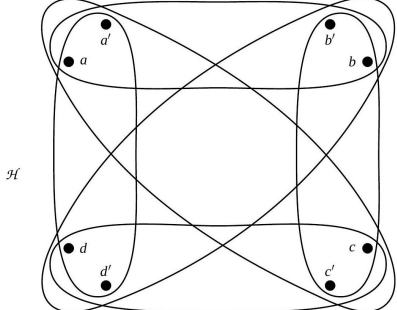
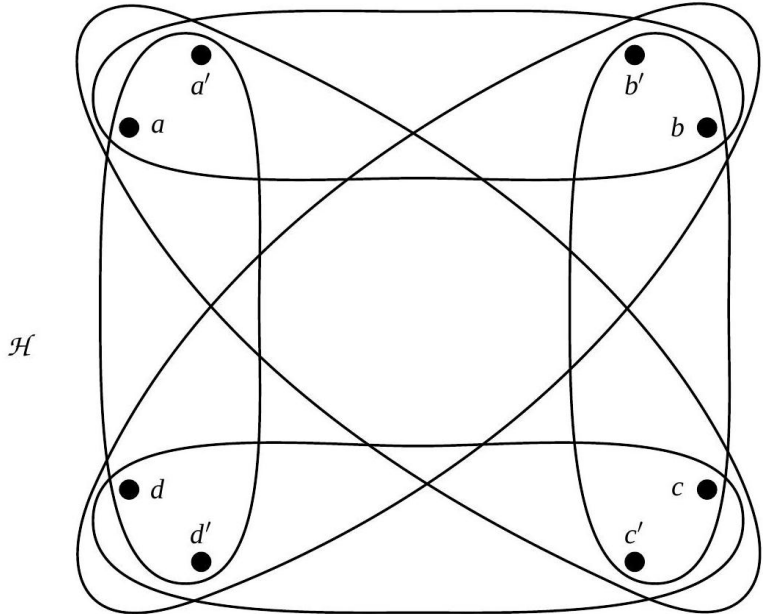
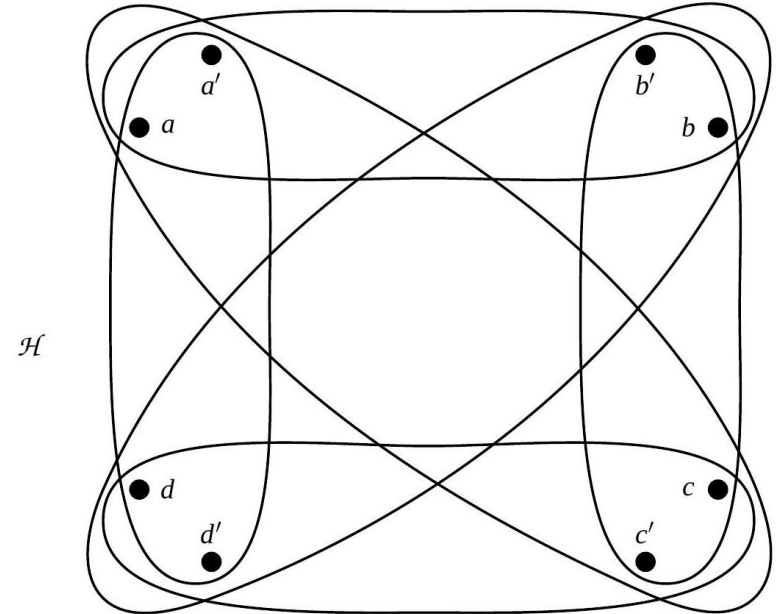
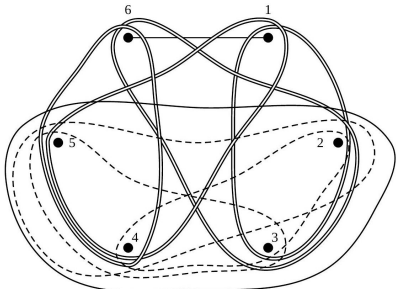
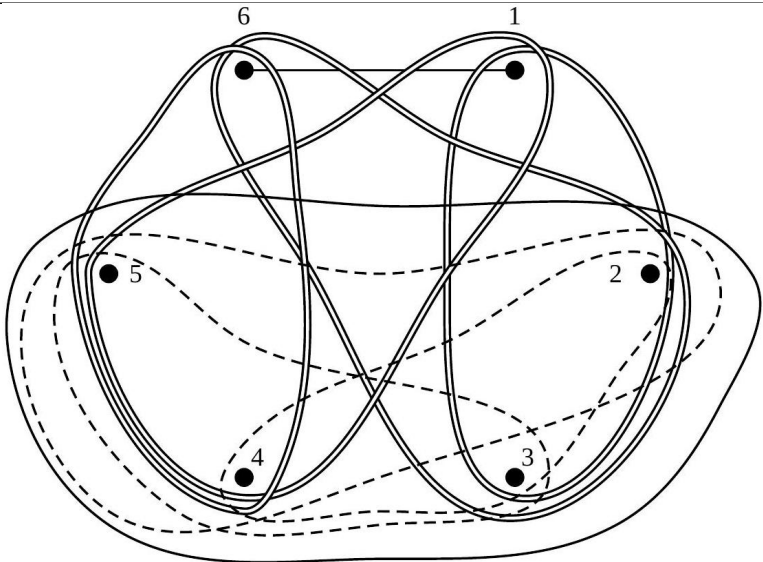
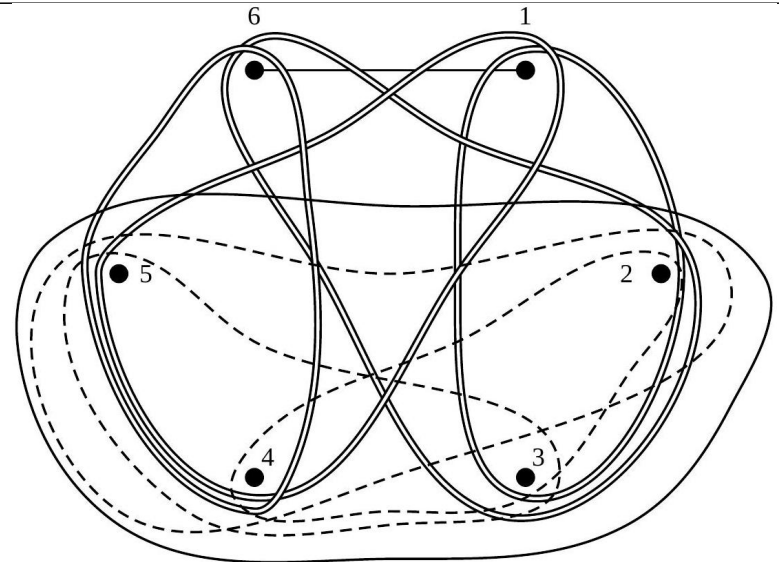
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voloshin-hypergraph_DIA_0442	229	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-229_466_1030_223_368.jpg			

voloshin-hypergraph_DIA_0240	232	(dup)	cde1d261-30b2-451d-a9dd-108d5796775a-232_1709_1165_249_313.jpg		

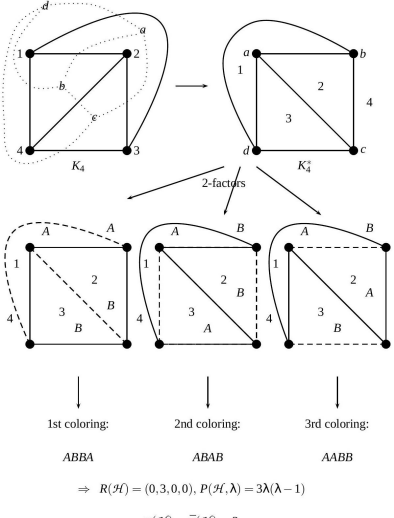
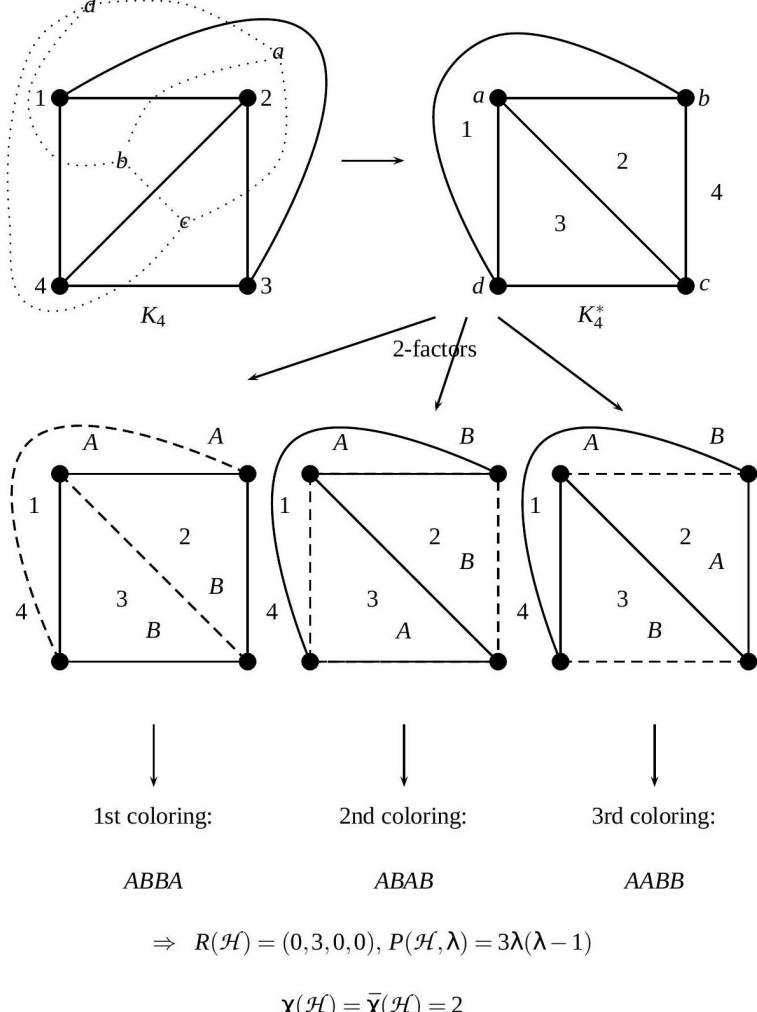
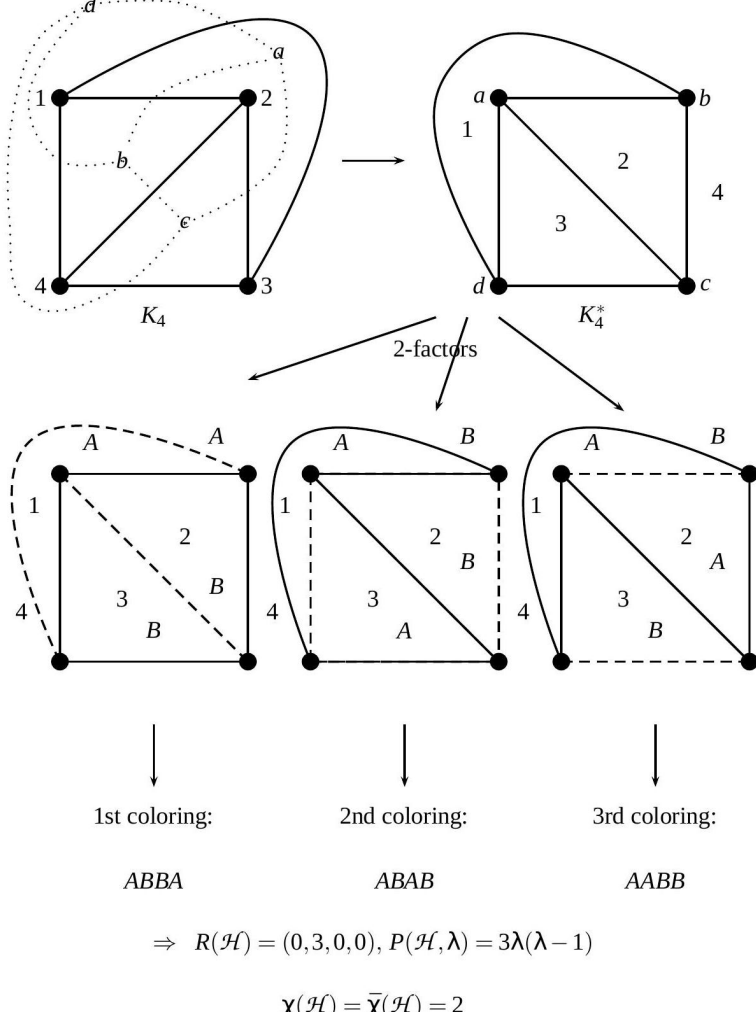
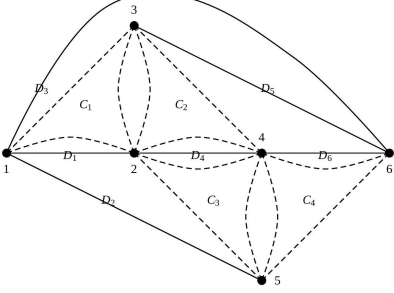
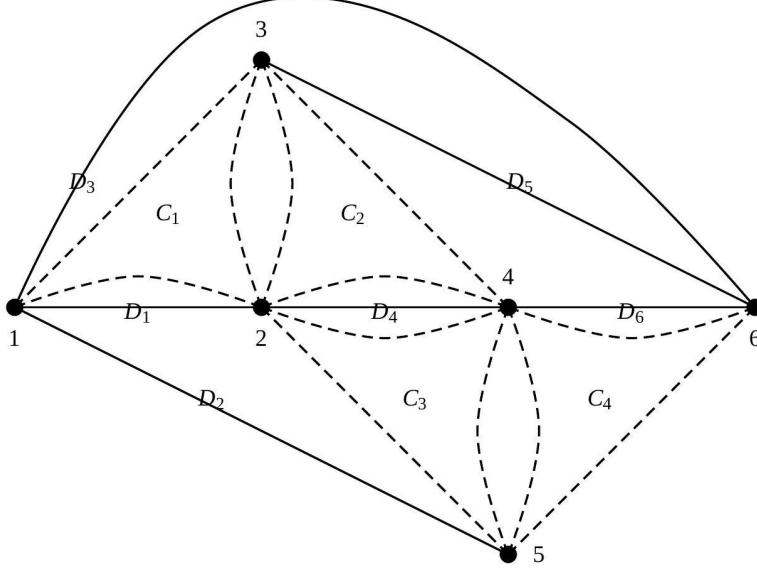
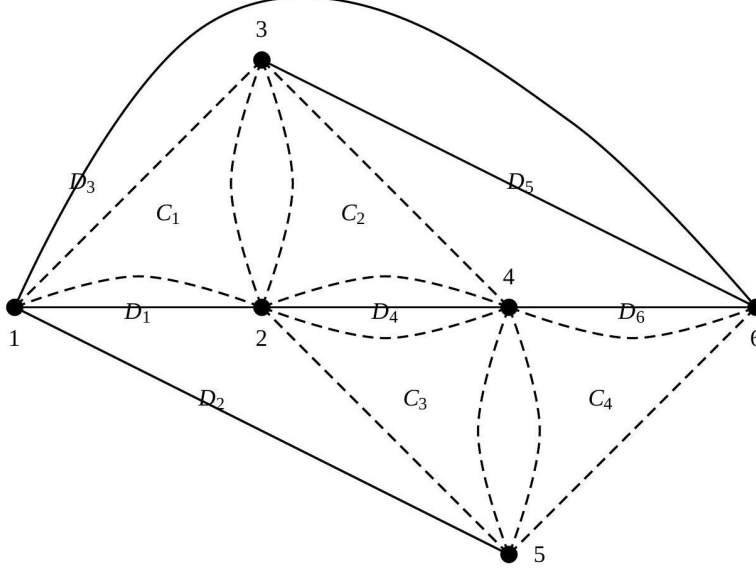
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voloshin-hypergraph_DIA_0253	237	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-237_466_466_269_647.jpg			
voloshin-hypergraph_DIA_0253	238	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-238_893_1121_241_321.jpg			
voloshin-hypergraph_DIA_0253	239	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-239_362_368_1828_699.jpg			

voloshin-hypergraph_DIA_0254	242	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-242_440_958_269_394.jpg			
voloshin-hypergraph_DIA_0254	242	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-242_518_836_1282_414.jpg			
voloshin-hypergraph_DIA_0254	244	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-244_386_1139_919_280.jpg			
voloshin-hypergraph_DIA_0255	250	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-250_606_1121_205_329.jpg			
voloshin-hypergraph_DIA_0255	251	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-251_492_1170_1129_311.jpg			

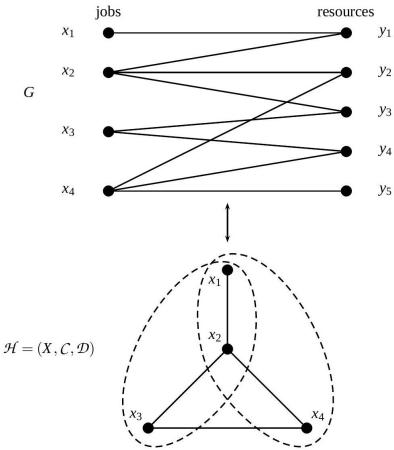
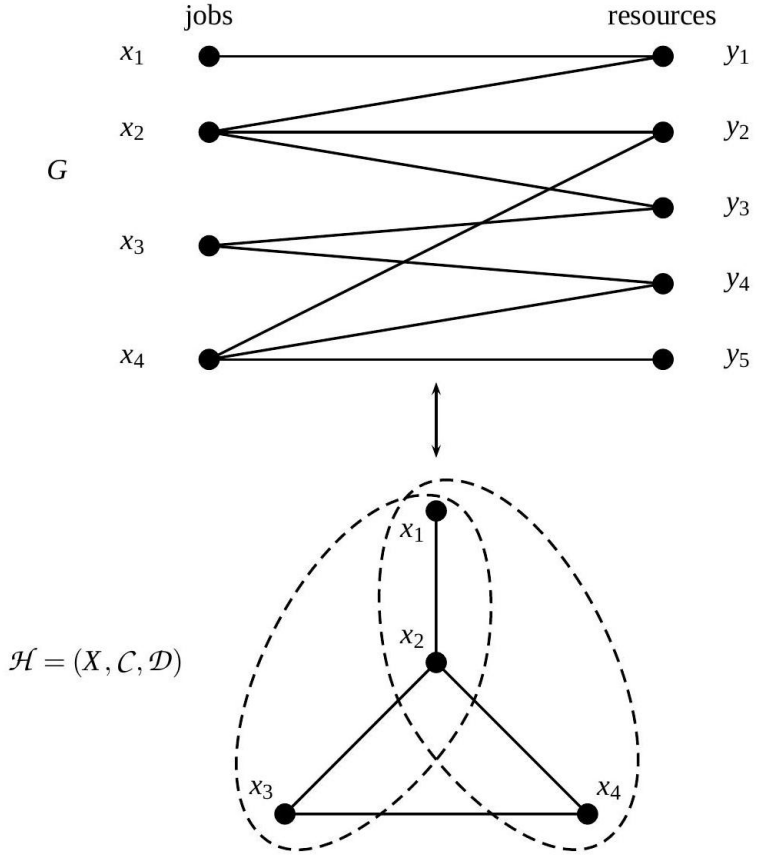
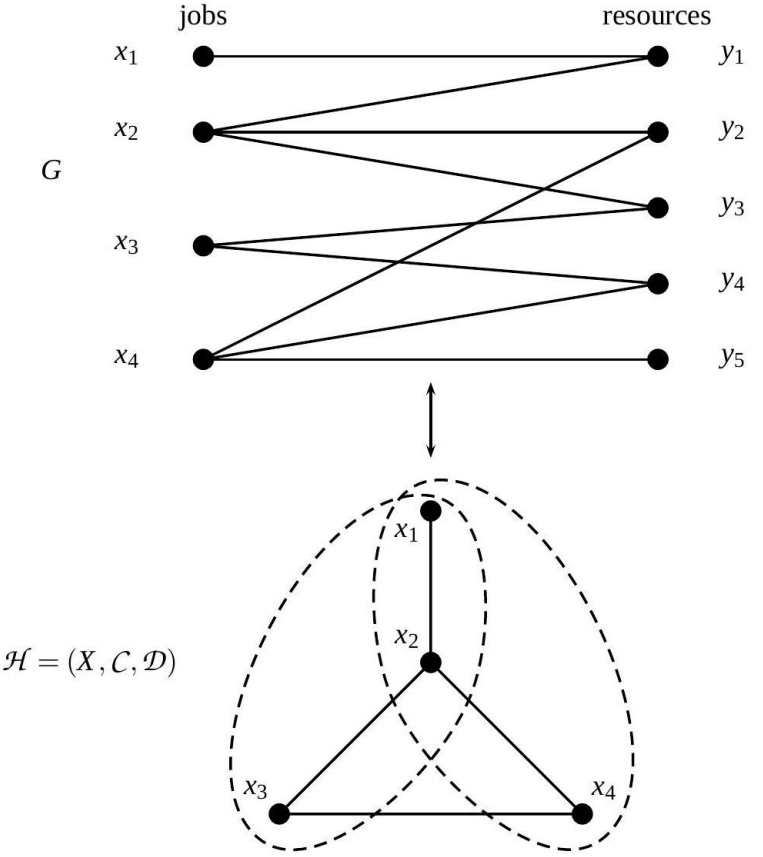
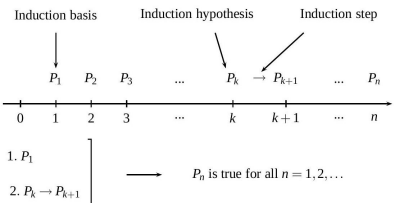
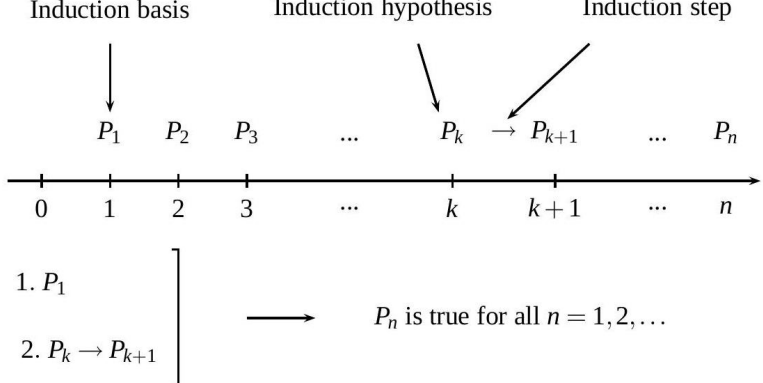
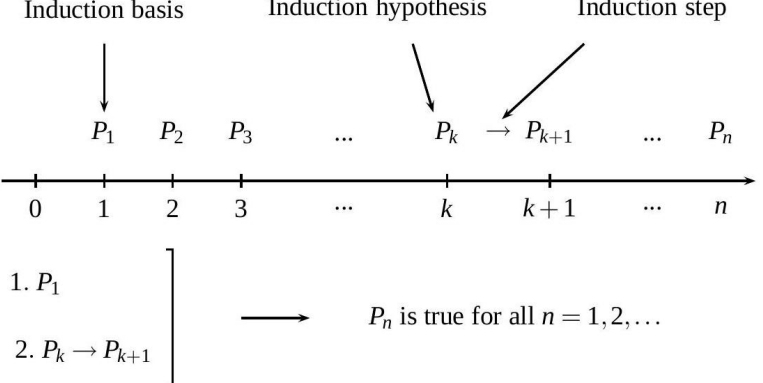
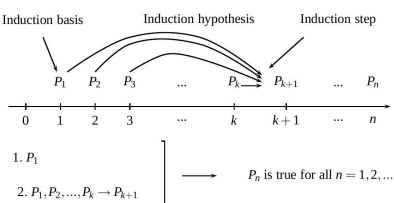
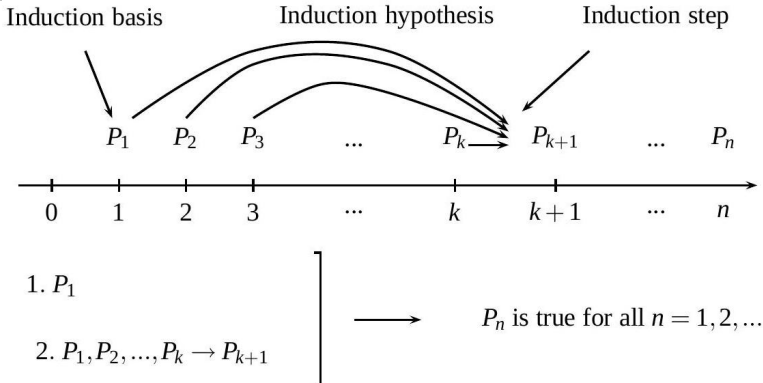
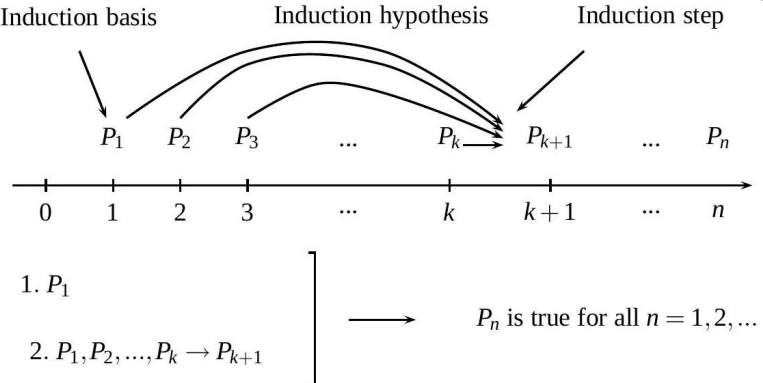
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voloshin-hypergraph_DIA_0257	257	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-257_640_1098_233_321.jpg			
voloshin-hypergraph_DIA_0258	258	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-258_557_1175_267_295.jpg			
voloshin-hypergraph_DIA_0259	259	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-259_484_1139_986_295.jpg			

voloshin-hypergraph_DIA_0261	261	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-261_834_1059_220_324.jpg			
voloshin-hypergraph_DIA_0262	262	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-262_844_1041_230_316.jpg			
voloshin-hypergraph_DIA_0265	265	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-265_818_1111_210_334.jpg			

voloshin-hypergraph_DIA_267	267	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-267_702_1152_207_303.jpg			
voloshin-hypergraph_DIA_269	269	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-269_541_1020_1059_365.jpg			
voloshin-hypergraph_DIA_271	271	— (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-271_544_702_199_531.jpg			

voloshin-hypergraph_DIA_0275	275 (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-275_1631_1201_233_300.jpg	 Diagram illustrating the decomposition of a K_4 hypergraph into three 2-factors and their corresponding colorings. The top part shows the decomposition of K_4 into three 2-factors. The bottom part shows the colorings: 1st coloring (ABBA), 2nd coloring (ABAB), and 3rd coloring (AABB). The resulting coloring is $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$.	 Diagram illustrating the decomposition of a K_4^* hypergraph into three 2-factors and their corresponding colorings. The top part shows the decomposition of K_4^* into three 2-factors. The bottom part shows the colorings: 1st coloring (ABBA), 2nd coloring (ABAB), and 3rd coloring (AABB). The resulting coloring is $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$.	 Diagram illustrating the decomposition of a K_4^* hypergraph into three 2-factors and their corresponding colorings. The top part shows the decomposition of K_4^* into three 2-factors. The bottom part shows the colorings: 1st coloring (ABBA), 2nd coloring (ABAB), and 3rd coloring (AABB). The resulting coloring is $\chi(\mathcal{H}) = \bar{\chi}(\mathcal{H}) = 2$.
voloshin-hypergraph_DIA_0276	276 (dup)	cde1d261-30b2-451d-a9dd-108d5796775a-276_917_1219_999_272.jpg	 Diagram illustrating a hypergraph structure with vertices 1, 2, 3, 4, 5, 6 and edges $D_1, D_2, D_3, D_4, D_5, D_6$.	 Diagram illustrating a hypergraph structure with vertices 1, 2, 3, 4, 5, 6 and edges $D_1, D_2, D_3, D_4, D_5, D_6$.	 Diagram illustrating a hypergraph structure with vertices 1, 2, 3, 4, 5, 6 and edges $D_1, D_2, D_3, D_4, D_5, D_6$.

voloshin-hypergraph_DIA_277	(dup)	cde1d261-30b2-451d-a9dd-108d5796775a-277_430_707_1491_528.jpg			
voloshin-hypergraph_DIA_280	(dup)	cde1d261-30b2-451d-a9dd-108d5796775a-280_1142_914_228_427.jpg	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$	$\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$

						
voloshin-hypergraph_DIA_281	281	(dup)	cde1d261-30b2-451d-a9dd-108d5796775a-281_1116_981_233_318.jpg		Induction basis Induction hypothesis Induction step  1. P_1 2. $P_k \rightarrow P_{k+1}$ $\longrightarrow$ P_n is true for all $n = 1, 2, \dots$	Induction basis Induction hypothesis Induction step  1. P_1 2. $P_k \rightarrow P_{k+1}$ $\longrightarrow$ P_n is true for all $n = 1, 2, \dots$
voloshin-hypergraph_DIA_284	284	(dup)	cde1d261-30b2-451d-a9dd-108d5796775a-284_580_1098_230_334.jpg		Induction basis Induction hypothesis Induction step  1. P_1 2. $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$ $\longrightarrow$ P_n is true for all $n = 1, 2, \dots$	Induction basis Induction hypothesis Induction step  1. P_1 2. $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$ $\longrightarrow$ P_n is true for all $n = 1, 2, \dots$